



Supporting Online Material for

Proteome Organization in a Genome-Reduced Bacterium

Sebastian Kühner, Vera van Noort, Matthew J. Betts, Alejandra Leo-Macias, Claire Batisse, Michaela Rode, Takuji Yamada, Tobias Maier, Samuel Bader, Pedro Beltran-Alvarez, Daniel Castaño-Diez, Wei-Hua Chen, Damien Devos, Marc Güell, Tomas Norambuena, Ines Racke, Vladimir Rybin, Alexander Schmidt, Eva Yus, Ruedi Aebersold, Richard Herrmann, Bettina Böttcher, Achilleas S. Frangakis, Robert B. Russell, Luis Serrano, Peer Bork,* Anne-Claude Gavin*

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Proteome organization in a genome-reduced bacterium

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Materials and methods

Construction of a collection of *Mycoplasma pneumoniae* strains expressing a TAP-fusion.

TAP-fusions were expressed in *M. pneumoniae* M129 (ATTC29342 broth passage No. 31) strains. Primers were designed for all 689 ORFs in *M. pneumoniae*. Each ORF was amplified by PCR using cosmid DNA as template (S1, S2). PCR products were digested (*NotI*/ *SfiI*) and ligated into the pMT85-based pMT_ClpB-TAPtag vector (S3), which carries the *Staphylococcus aureus* transposon Tn4001 and the TAP cassette downstream of the *SfiI* site. Transformation of *M. pneumoniae* was done by electroporation, following published protocols (S4). Following transformation and recovery, cells were plated out in modified Hayflick medium (S5), containing gentamycin (78 mg/l) and incubated at 37°C as described earlier (S6). Cell culture was tested for expression by Western blot with a Peroxidase-Anti-Peroxidase antibody developed in rabbit (Sigma, P1291).

TAP purification of protein complexes and mass spectrometric analysis

Two-liter cultures were incubated in ten 300 cm² cell culture flasks (Sarstedt) and harvested 96 h after inoculation. Cells were washed twice with ice cold PBS and centrifuged at 9,860xg. Pellets were resuspended in 2 ml lysis buffer (50 mM Tris pH 7.5, 5% glycerol, 1.5 mM MgCl₂, 100 mM NaCl, 0.2% NP40, 1 mM DTT, 1 mM AEBSF, 1 mM PMSF, 1 µg/ml pepstatin A, 1 µg/ml antipain, 2 µg/ml aprotinin, 1 µg/ml leupeptin and 16 µg/ml benzamidin) and lysed mechanically using a douncer. TAP purification was done following established protocols (S7, S8). Protein samples were then analyzed by SDS-PAGE (4-12% NuPage, Invitrogen). Sample preparation for mass spectrometry was done essentially as described earlier (S9, S10). Briefly, whole gel lanes were cut into 32 equal slices; the proteins were then destained and digested (trypsin, 12.5 mg/ml). Peptides were extracted in 1% formic acid in acetonitrile, resuspended in 0.1% TFA in water, mixed (1:1) with matrix solution (0.1% TFA in 50% acetonitrile and 50% methanol) and spotted on MALDI-TOF plates (Micromass,

Waters). The mass spectrometry (MS) analyses were carried out on a MALDI Micro MXTM (Waters, Milford, MA, USA). All data were acquired in positive ion mode over m/z range of 900-3,500 in reflectron mode. Peak masses were extracted from the spectra and used for protein identification (table S1).

Sample preparation for PeptideAtlas data collection

M. pneumoniae M129 (ATTC29342 broth passage No. 31) strain cells were washed twice in ice cold PBS, harvested and pelleted by centrifugation at 9,860 $\times g$. Cells were resuspended in 100 μ l denaturation buffer (100 mM ammoniumbicarbonate, 8 M urea, 0.1% RapiGestTM(Waters)), sonicated for 5 minutes and spun down for 5 minutes at 20,000 $\times g$ at 4°C. A small aliquot of the supernatant was taken to determine the protein concentration using a commercially available bicinchonic acid-(BCA) based protein assay kit (Pierce #23227).

Protein digestion for PeptideAtlas data collection

The proteins were reduced with 5 mM Tris(2-carboxyethyl)phosphine (TCEP) for 60 minutes at 37°C and alkylated with 10 mM iodoacetamide for 30 minutes in the dark before diluting the sample with 100 mM ammoniumbicarbonate to a final urea concentration below 2 M. Proteins were digested by incubation with trypsin (1/50, w/w) overnight at 37°C. The peptides were cleaned up by C18 reversed-phase spin columns according to the manufacturer's instructions (Harvard Apparatus).

Directed mass spectrometry for PeptideAtlas data collection

The setup of the μ RPLC-MS system was as described previously (S11). The hybrid LTQ-FT-ICR mass spectrometer was interfaced to a nanoelectrospray ion source (both Thermo Electron, Bremen, Germany) coupled online to a Tempo 1D-plus nanoLC (Applied Biosystems/MDS Sciex, Foster City, CA). Peptides were separated on a RP-LC column (75

μm x 15 cm) packed in-house with C18 resin (Magic C18 AQ 3 μm; Michrom BioResources, Auburn, CA, USA) using a linear gradient from 98% solvent A (98% water, 2% acetonitrile, 0.15% formic acid) and 2% solvent B (98% acetonitrile, 2% water, 0.15% formic acid) to 30% solvent B over 120 minutes at a flow rate of 0.3 μl/minutes. Each survey scan acquired in the ICR-cell at 100,000 FWHM was followed by MS/MS scans of the three most intense precursor ions in the linear ion trap with enabled dynamic exclusion for 30 seconds. Charge state screening was employed to select for ions with at least two charges and rejecting ions with undetermined charge state. The normalized collision energy was set to 32%, and one microscan was acquired for each spectrum.

For directed LC-MS/MS, two data-dependent LC-MS/MS runs, the SuperHirn peak extraction and alignment algorithm (S12) was used to extract all MS1 peaks. After removing all features for which MS2-data was obtained, the remaining 24,461 masses were divided in four lists and subsequently sequenced using directed LC-MS/MS analysis as recently specified (S11).

Data processing and compilation of PeptideAtlas

MS/MS spectra were searched using the SEQUEST search tool (S13) against a decoy database (consisting of forward and reverse protein sequences) of the predicted proteome from *M. pneumoniae* strain M129 (ATTC29342), complete genome NCBI genome number NC_000912 (<http://www.ncbi.nlm.nih.gov/entrez>), consisting of 689 proteins as well as known contaminants such as porcine trypsin and human keratins (Non-Redundant Protein Database, National Cancer Institute Advanced Biomedical Computing Center, 2004, <ftp://ftp.ncifcrf.gov/pub/nonredundant>). The search was performed with semi-tryptic cleavage specificity, mass tolerance of 15 ppm, methionine oxidation as variable modification and cysteine carbamidomethylation as fixed modification. The database search results were further processed using the PeptideProphet program (S14) with decoy option. The peptide false discovery rate (FDR) was set to 1% (p=0.7).

Peptide Mass Fingerprinting

Mass intensity files were analyzed with MASCOT (S15) and AlDente (S16, S17) to identify proteins (fig. S1, A and B). Parameter settings were 80 ppm and 0 missed cleavages for both programs. For Aldente, shift was set at 0.20, slope at 200 and minimum number of peptides at two. For MASCOT, identifications above PMF score 45 were always taken and above 30 only if the consolidated score was at least 80 percent of the maximum consolidated score for that protein. For AlDente the same files were used for PMF against a random database. All identifications scoring lower than the random database were excluded. From these PMF identifications, proteins scoring higher than 1.5 were always taken and higher than 0.5 only if the consolidated score was at least 80 percent of the maximum consolidated score for that protein.

Reproducibility

To assess the quality of the entire approach, we first measured the reproducibility of the overall experimental process, i.e. growth of adherent *M. pneumoniae* strains expressing TAP-fusions, lysis of the cell pellets, TAP purification of protein complexes and mass spectrometric identification of protein complex components. For this purpose, we ran this entire procedure in duplicate for 18 strains expressing different TAP-fusions. We measure a reproducibility rate of 73% for proteins involved in the high confidence interactions, which are the ones that cluster in complexes in the dataset. This is comparable to most published studies, including that of Sowa *et al.* that recently reported 80% reproducibility (derived from four duplicated experiments) for high confidence interactions (S18).

We also more specifically assess the quality of the MS and protein identification strategy. For this purpose, 72 MS samples (see procedure above) were processed for MALDI-TOF analysis in duplicate; the reproducibility for protein identification is 97%.

Profilogs

Proteins that have overlapping peptide mass profiles (we coined the term Profilogs) could lead to spurious identifications. To detect spurious identifications of this sort we use the results from the PeptideAtlas. The masses of sequenced peptides by FT-MS (PeptideAtlas) were entered into MASCOT and AlDente to identify proteins by Peptide Mass Fingerprinting (PMF). False positives (Profilogs) are in this case identifications of proteins by PMF that were not the proteins sequenced in MS2 (PeptideAtlas). Mascot scores of false positive identifications (Profilogs) were recorded (fig. S1, C and D). For each protein identified by FT-MS, we identify all possible Profilogs as false positive identifications by PMF. In the PMF results of the TAP purifications, all identifications that were not the top-ranking proteins were considered as background if they are Profilogs and they were not further considered.

Consolidated scores

MASCOT and AlDente scores of overlapping identifications were used to re-scale all the PMF scores, such that they have the same distribution. For PMF result, the maximum of the two rescaled scores was taken as the consolidated score. These consolidated scores were used in the subsequent analyses. Altogether, the set of successful redundant purifications comprises 259 purifications (table S2) from 212 unique baits. In total 411 proteins were identified in the consolidated set of identified proteins (table S2; fig. S2, A, B, C, D, E; see also fig. S3).

Socio-affinity scoring and integration of quantitative prey profiling across the entire dataset

The socio-affinity scoring was adapted from (S19, S20). To further discriminate contaminants from cognate complexes components we also integrated systematic quantification of the preys across the entire dataset using spectral counting. Our approach is very similar to what has

been reported by Sowa *et al.* (S18) who originally integrated averaged spectral counts from technical duplicates to the interaction scores. Their datasets, focused on a limited set of purifications, contained only rare instances of reciprocal data (reverse tagging). In contrast, in genome-wide datasets run to saturation (fig. S3), reciprocal interactions are frequent. Thus, for each protein in the dataset, we could derive quantitative profiles across different purifications where proteins were present as bait or preys (fig. S4A). We ranked all interactions according to the quality of the MS data. This ranking is based on the PMF scores from MASCOT and AlDente, which explicitly take into account spectral counts, further normalized for length of the peptides and number of possible tryptic peptides per protein. Therefore, the PMF score represents a more accurate measure than spectral counts alone.

More specifically, the approach is based on the principle that proteins present as contaminants in a given purification often have low PMF scores (i.e. they are less abundant and they are thus identified by a smaller number of peptides; they have lower spectral counts). Reciprocally, the presence of such contaminant proteins with a high PMF score in a given purification is usually indicative of a true interaction. Thus, we wanted to give a higher socio-affinity, if the PMF score was close to the maximum for that prey. We then sorted all identified preys by PMF score across different purifications (using different baits), and use the rank divided by the total number of times the prey was identified instead of the count divided by the background frequency. In this way, we ensure that preys when identified with a higher spectral count (reflected in the PMF score) receive a higher interaction score with this bait than when they are identified with a lower spectral count (usually indicative of spurious interactions).

Socio-affinity (SA) is build up of the Spokes component (S) where direct bait-prey relations are counted and a Matrix component (M) where preys are counted that are seen together in a purification (equation 1). S is a summation over the scores (s) of all purifications (k) where the bait (i) was used and prey (j) was identified (equation 3) or prey (j) was not identified

$$SA = S_{ij} + S_{ji} + M_{ij} \quad (1)$$

$$S_{ij} = \sum_k s_{ijk} \quad (2)$$

$$s_{ijk} = \log_{10} \frac{r + (1 - r)p_{ijk}}{p_{ijk}} \quad (3)$$

$$s_{ijk} = \log_{10} \frac{(1 - r)(1 - p_{ijk})}{(1 - p_{ijk})} = \log_{10}(1 - r) \quad (4)$$

$$p_{ijk} = 1 - \exp(-f_i f_j n_{\text{tot}}^{\text{prey-prey}}) \quad (5)$$

$$f_j = \frac{n_j^{\text{prey}_{\text{obs}}} + n_{\text{pseudo}}}{n_{\text{tot}}^{\text{prey}_{\text{obs}}} + (n_{\text{distinct}_{\text{preys}}} n_{\text{pseudo}})} \quad (6)$$

$$M_{ijk} = \sum_k m_{ijk} \quad (7)$$

$$m_{ijk} = \log_{10} \frac{r + (1 - r)p_{ijk}}{p_{ijk}} \quad (8)$$

$$p_{ijk} = 1 - \exp(-f_j n_{ik}^{\text{prey}} n_i^{\text{bait}}) \quad (9)$$

$$MI = \frac{\frac{\sum P_{pc1} \sum P_{pc2}}{N_{pc1} N_{pc2}}}{\frac{\sum P_{c1c2}}{N_{c1c2}}} \quad 10 \quad (10)$$

(equation 4). The retrieval rate (r) is estimated to be 0.25 and the probability (p_{ijk}) of finding a bait-prey interaction is estimated from a Poisson distribution (S20) where n_{ik}^{prey} is the number of preys identified in purification k with bait i , n_i^{bait} is the number of times protein i was used as bait, and f_j is an estimate of the nonspecific frequency of occurrence of prey j in the dataset, where we took a value of 1 for n_{pseudo} . We computed f_j as Bayesian posterior estimates based on the observed frequency of occurrence of preys in the dataset and the prior hypothesis that all preys occur nonspecifically with equal frequency, for f_j , we take only the prey frequency until the PMF score of the prey- identification with bait i (fig. S4A). Similarly, for the Matrix component (M), m was summed over all purifications where both prey (i) and prey (j) occur together. The value for p_{ijk} is calculated using the Poisson distribution as in equation 9 where f_i and f_j are computed as in equation 6, and $n_{\text{tot}}^{\text{prey-prey}}$ is the total number of prey-prey pairs observed in the dataset.

Socio-affinity score benchmarking

A curated list of known complexes was inferred by homology to known complexes from the literature (table S5). The interactions between proteins in the same complexes form the positive set. The interactions between proteins in different complexes form the negative set. Reported accuracies for function prediction are always lower boundaries as there could be true interactions also in the negative set. The protein-protein interactions were scored using socio-affinity and classified in the positive or negative set (fig. S4B). Then we performed a logistic regression on the socio-affinities and used the resulting logistic function to map the socio-affinities to interaction probabilities.

Integration with predicted interactions

Predicted interactions were taken from STRING (COG mode) (S21); predictions from databases and text mining were not taken into account. For every protein pair, we now have an interaction probability from TAP and an interaction probability from STRING (S21). These two probabilities were integrated the same way as in STRING: $P = 1 - (1-p_1)(1-p_2)$. A weak score from TAP and a high score from STRING could indicate physical interactions (S22) without predicting other functional interactions.

For the clustering into complexes only those protein pairs were taken that were seen together in purification at least once. From this we considered a set of 1,058 high confidence interactions for clustering into complexes (table S3).

Clustering

We adopted the clique percolation method from the CFinder method (S23). This is one of the clustering methods where proteins can belong to multiple complexes. We used $k=3$ to find cliques and used overlapping cliques to define the cores of the complexes. We took the high-confidence interactions (table S3) and retrieved protein pairs that have each other as highest scoring interaction partner (stable pairs). If these stable pairs were part of a complex, they were also considered core members. If these stable pairs were not part of a complex, they were added to the complex list as complex of size two. To the core complexes, we added proteins with only one high-confidence interaction and call these attachment proteins. We used different cut-offs for interaction probabilities to define complexes and monitored the number of proteins clustering in complexes as well as the size of the largest complex. We chose a threshold to define complexes such that many proteins were in complexes but still the size of the largest cluster has not exploded yet (fig. S4C). Altogether the study reports 116 heteromultimeric protein complexes (table S4).

Characterization of homomultimeric protein complexes

The TAP-fusions were expressed from an exogenous locus and promoter. As a consequence the TAP-fusions were present together with their untagged wild type allele. It was not rare that in the same purification we could observe both TAP-tagged and untagged versions of the bait, indicating homomultimerization. We manually inspected the 1D SDS-PAGE gel and MS results for instances where the bait was present in more than one band. The bait was considered homomultimeric, if we identified it in two distinct bands corresponding to the MW of the tagged and the untagged versions. Altogether the study reports 62 homomultimeric protein complexes (table S4).

Treatment of the extracts with RNase/DNase

To digest DNA and RNA, some extracts were treated with RNase and DNase prior TAP. We added 3 μ l of RNase (Roche) and 1 spatula tip of DNase (Roche) to a cellular pellet in lysis buffer before mechanical lysis with the douncer. TAP, SDS-PAGE and mass spectrometry were then performed as described above. The association between RpoA-TAP and the ribosome was unaffected by this DNA/RNA treatment indicating the interactions does not imply RNA or DNA (fig. S5, A and B).

Total RNA purification

After growth, surface-attached cells are washed once with PBS at 37°C and immediately lysed in the cultivation flask by adding RLT buffer from the Qiagen™ RNeasy Plus Mini Kit (Cat. Num. 74134). For cell lysis, 1.8 ml of RLT buffer in presence of 18 μ l β -mercaptoethanol was used per cultivation flask. RNA purification was done according to the manufacturer's protocol. Purification of RNA in presence of RNase inhibitors is also shown in fig. S5A.

Evaluating protein complexes by gel filtration chromatography and western blot.

Gel filtration (GF) chromatography was performed at 10°C on a Pharmacia SMART system at a flow rate of 40 µl/minute by using a SuperoseTM 6 PC 3.2/30 column, equilibrated with lysis buffer. The chromatographic profile was monitored at 280 nm by using the µPeak monitor (Pharmacia). Volumes of 50 µl of *M. pneumoniae* lysates were loaded on a column and 60 µl fractions were collected and analyzed by SDS-PAGE and Western blot (Figs. 3 and S7B).

A peroxidase anti-peroxidase antibody (Sigma, P1291) was used to detect the TAP-fusions (alpha subunit of DNA-directed RNA polymerase: RpoA-TAP and segregation and condensation protein A: ScpA-TAP). Polyclonal antibodies produced in rabbits have been used to detect the 30S ribosomal protein S4 (RpsD), the ATP-dependent protease La (Lon) and the protein P115 homolog (P115); final detection was done using secondary antibodies coupled to HRP (Sigma). For the quantification, total band intensity was integrated with Photoshop software (Adobe) and normalized versus the highest detected peak (Fig. 3A).

Multifunctionality index

For each set of two protein complexes and one protein, a multifunctionality index (MI) value was calculated (table S7). This index reflects the tendency for the protein to be part of both complexes and corrects for the possibility that these two complexes should have been one complex, but was separated due to the chosen threshold.

Multifunctionality Index is calculated between each protein and each complex as above where the sums of the interaction scores (P) between the protein under consideration and the members of protein complex 1 (pc1) and protein complex 2 (pc2) are multiplied and divided by the product of the Number of members (N) in complex 1 and complex 2 (equation 10). This is divided by the sum over the interaction scores (P_{c1c2}) between members of complex 1

and complex2 divided by the total number of protein pairs (N_{c1c2}) between complex 1 and complex 2, not taking into account the protein for which the MI is being calculated (equation 10). To benchmark the results from the multifunctionality index a curated list of known multifunctional proteins from the literature was used (table S8).

Electron microscopy and image reconstruction of selected complexes

Selected protein complexes (DNA gyrase, DNA topoisomerase, RNA polymerase) were purified according to the first step of the TAP-protocol (see above). The resulting TEV-eluate was further purified on a glycerol-gradient and fixed by cross-linking following the GRAFIX protocol (S24). Fractions containing the bait were identified by dot-blot using an antibody against calmodulin-binding domain. For electron microscopy, samples were prepared from fractions containing the bait by sandwich negative staining (S25) (fig. S6). For DNA gyrase the first models were determined by random conical tilt reconstruction (S26). The first model of the topoisomerase was determined by projection matching of class averages against the final map of the gyrase. For RNA polymerase a low-pass filtered map of the polymerase from *Thermus aquaticus* (1l9u) was used as first reference (S27). Maps were further refined by iterative projection matching against the current best map using SPIDER (S28).

Cultivation of *M. pneumoniae* strain B176 for electron tomography

The *M. pneumoniae* B176 avirulent, noncytadsorbing strain that lacks the 40- and 90-kDa proteins, the products of the ORF6 gene (Mpn142), was used in this analysis (S29). In comparison to its parental strain, the virulent *M. pneumoniae* M129, B176 does not grow in colonies and hence is more suitable to image individual cells by electron tomography of plunge-frozen samples.

Tilt series acquisition

Single-axis tilt-series were collected covering an angular range from -66° to $+66^{\circ}$ with 1.5 degrees of angular increment in a Tecnai G2 Polara microscope cooled to liquid nitrogen temperature and equipped with a Gatan postcolumn GIF 2002 energy filter. Data acquisition was carried out under low-dose conditions using the UCSF tomography software (S30). Images were recorded on a 2k x 2k pixel CCD camera at a defocus level between -5 and -10 μm . The pixel size at the specimen level was 0.6 nm.

Image processing

The images were interactively aligned with respect to a common origin using 10 nm colloidal gold particles distributed in the sample as fiducial markers. The reconstructions were performed using weighted back-projection algorithm and visualized with the Amira package (S31).

Mapping of complexes in cellular tomograms

We scanned 26 tomograms of whole *M. pneumoniae* cells (with sub-pixel alignment error), with four templates: the ribosome (S32), GroEL (S33), the RNA polymerase, and the structural core of the pyruvate dehydrogenase (PdhC, homomultimer) (S34), by means of locally normalized cross-correlation (S35). We have developed a classification method integrating knowledge on protein complexes networks for the detection of putative complexes inside cellular tomograms that largely overcomes the problem of template bias and significantly increases detection certainty (fig. S8). The principle here is that a tomogram is scanned with a template resembling an incomplete complex. The part of the complex, which is deliberately ignored, does not need to be structurally relevant. Complexes are identified as such, when the ignored part is recovered after the classification.

Here, we illustrate the procedure on the example of the ribosome. The tomogram (fig. S8A) is scanned with a portion of the whole ribosome, in this case the large ribosomal subunit (50S). The resulting probability map (fig. S8B) contains only the probabilities that the large ribosomal subunit is located at particular positions in the electron tomogram. Most of the ribosomes will most probably have the small ribosomal subunit (the 30S) attached to the large one (fig. S8, C and D). Hence, when classifying the hits obtained for the 50S subunit, we expect to identify classes which show the ribosome as a complete entity. We expect that the sub-tomograms composing those classes will have significantly reduced false positive rates, since the additional density resembles the structure of the small ribosomal subunit increases our level of fidelity.

On each cell, the locally normalized cross-correlation (S35) produced a list of putative particles, which were further classified using the missing wedge independent procedure (S33, S36) with a network of 5x5 classification units (fig. S8E). Visual inspection of these classification units reveals the presence of extra-density which was not included in the original template. The similarity of this extra density to the 30S subunit is further asserted by normalized cross correlation and Fourier Shell Correlation. As an effective decision criterion is considered a high cross-correlation value and a Fourier Shell correlation extending into the higher frequencies. The sub-tomograms classifying in classification units with high Fourier Shell Correlation components are accounted as putative particles (fig. S8F). All other hits are rejected.

Quantitative Western blotting

To determine absolute quantities of GroEL and ribosomes in *M. pneumoniae*, levels of RpsD and the GroEL monomer were quantified by Western blotting. *M. pneumoniae* recombinant RpsD and GroEL genes used as standard were cloned into the pET19b vector, and proteins carrying an N-terminal poly-histidine tag were expressed and purified from *E. coli* cells.

Protein concentrations were determined using a commercially available bicinchonic acid- (BCA) based protein assay kit (Pierce #23227) essentially following the instructions in the product manual. Western blotting was carried out with polyclonal antibodies against RpsD and GroEL, respectively. Western blot bands were quantified using the software “ImageJ”.

Relating band intensity values to known protein amounts of the purified standards and to the experimentally determined protein per *M. pneumoniae* cell ratio (10 fg/cell), allowed the calculation of copies of ribosomes and GroEL particles per cell, considering the homooligomeric state of the GroEL particle (S37)(fig. S9, A and B).

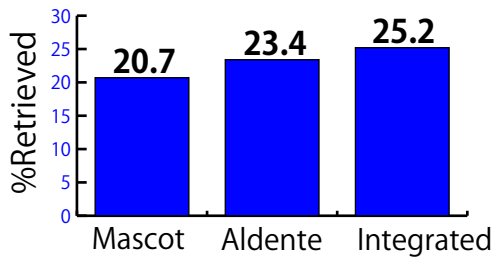
The value for the quantified Western blot band is divided by the slope of the standard curve and multiplied by $1\text{E-}9$ to obtain the amount of RpsD and GroEL in grams per gel lane. Determination of the protein concentration of the *M. pneumoniae* cell lysate (5.07 g/l) and the amount of protein per *M. pneumoniae* cell (10 fg/cell) allows for calculating the number of cells per gel lane, considering dilution factors and the amount of lysate loaded. Dividing the amount of RpsD and GroEL per lane by the number of cells/lane gives a value for RpsD and GroEL in grams per cell. This number is then transformed to moles by dividing it with the molecular weight of the respective proteins in g/mol. The Avogadro constant determines 1 mol to contain $6.022\text{E}+23$ particles. Dividing the amount of protein per cell in moles by the Avogadro constant allows the determination of molecules per cell. Since GroEL is a homotetradecamer, this number has to be divided by 14 to obtain a number for GroEL particles in a single cell. RpsD is believed to be in a 1:1 stoichiometry with ribosomes; hence the number of RpsD molecules directly reflects the number of ribosomes in a single *M. pneumoniae* cell (fig. S9B).

Structural bioinformatics

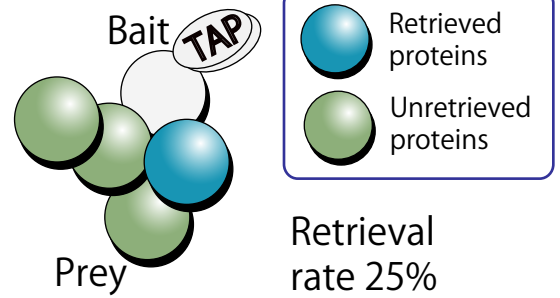
Homologous proteins were collected using PSI-BLAST (S38) with default parameters. Hidden-Markov Models were built and used to find homologues of known structures from PDB (S39) using HHMake and HHSearch modules of the HHPred suite (S40). Matches of at least 50 residues in the query and with an E-value ≤ 0.01 were considered further. Two matches were considered to be in physical contact when they did not overlap on the target sequence and each had at least 10 residues within 5.0 Å of the other. Overlaps between multiple interfaces on a single protein were calculated as the number of such residues that they had in common, and the interfaces clustered in to groups using these overlaps and the Markov Cluster algorithm ‘MCL’ (S41) (table S6).

Figure S1

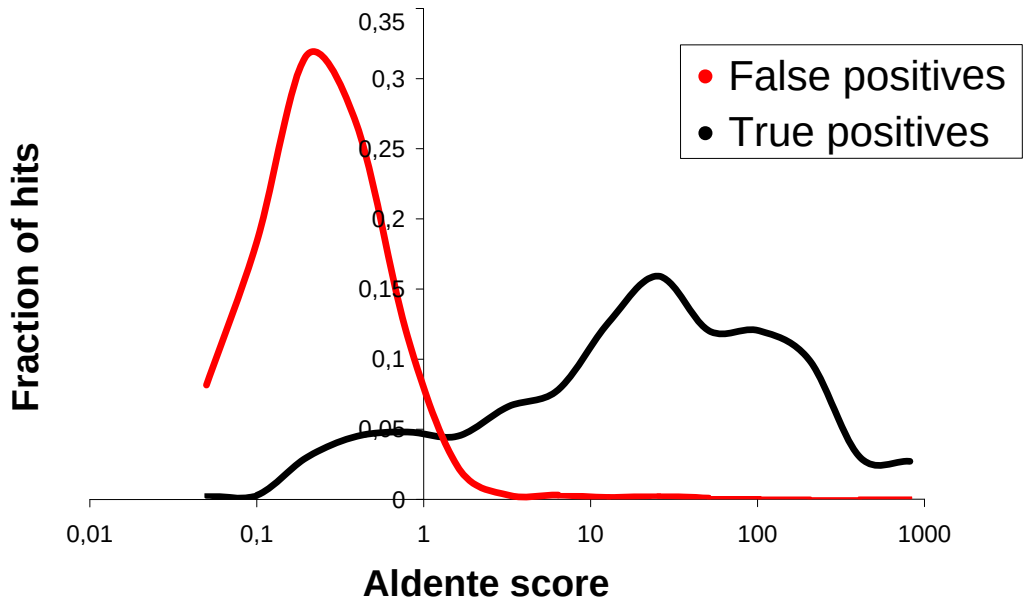
A



B



C



D

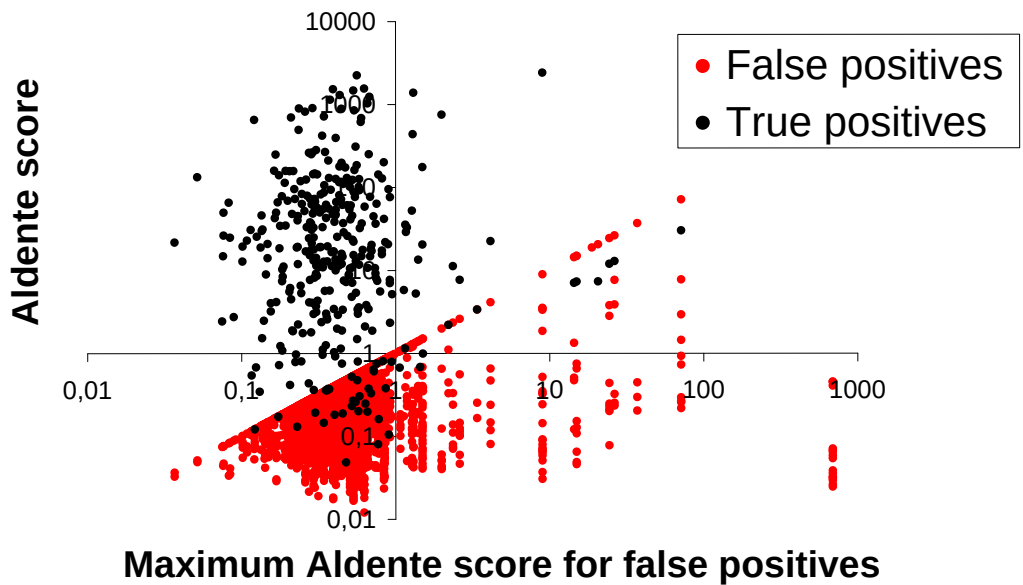
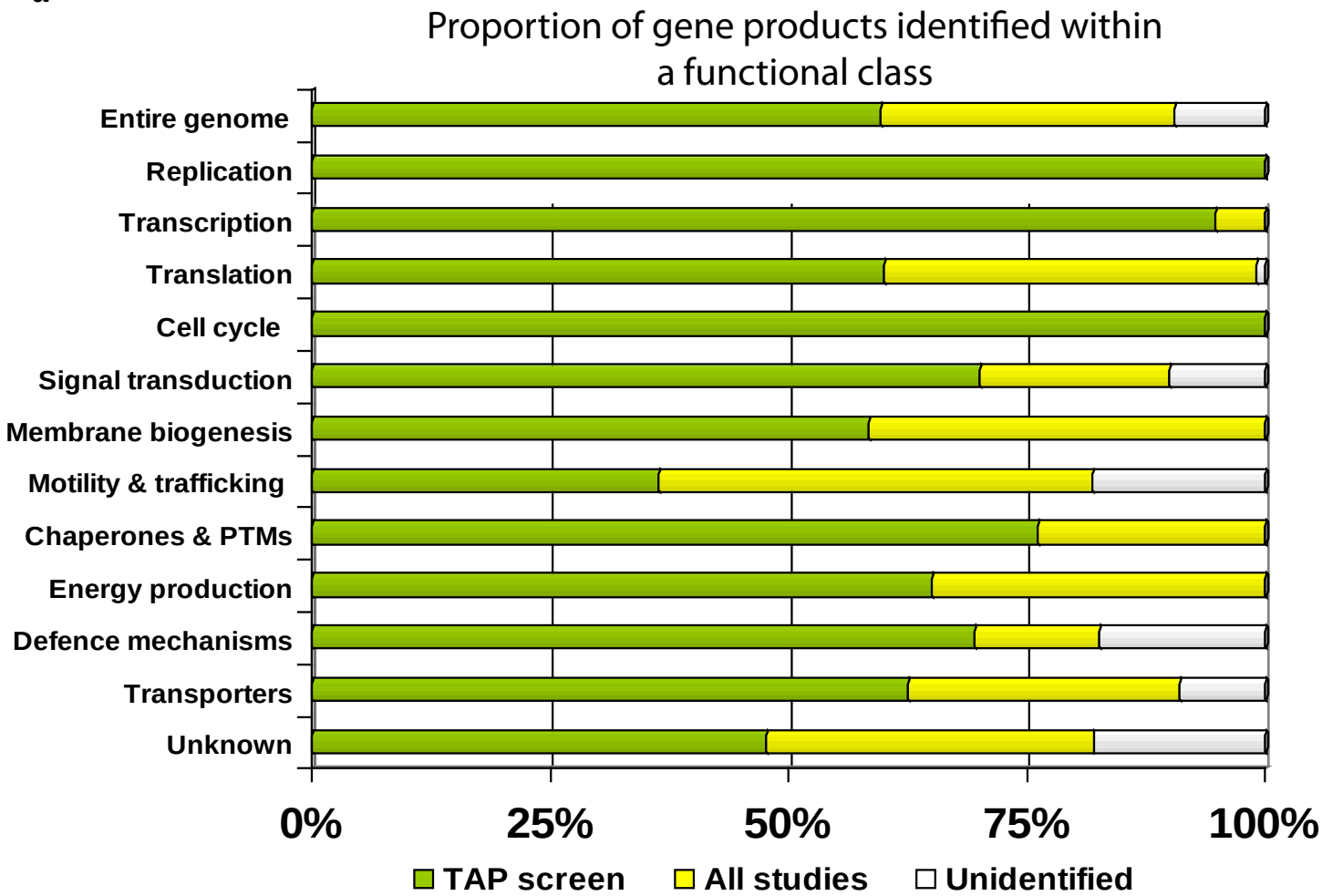


Fig. S1 Protein identification by MS

From all purifications, we processed 10,447 samples for MS. Proteins were identified using a refined approach that integrates both Mascot (*S15*) and Aldente (*S17*) search algorithms. **(A)** This strategy increases the retrieval rate of true positives by 20% compared to only using Mascot. **(B)** The retrieval rate is calculated by comparing to known complexes. Of all the expected subunits that also belong to these known complexes, we retrieved 25%. The retrieval rate is used in the socio-affinity scoring. **(C)** the PeptideAtlas was used to define profilogs; proteins that, in a MS experiment, give rise to overlapping peptide mass profiles. The masses of the sequenced peptides were used to do a Peptide Mass Fingerprinting (PMF), if the actual sequenced protein is retrieved this is a true positive identification. If a different protein is retrieved this is a false positive identification resulting from the fact that the two proteins are profilogs. Black line: distribution of true positive identification scores. Red line: false positive identification scores. **(D)** Same as **(C)** however the Aldente scores are ordered to the maximum false positive score for each protein. In the screen, after using both Aldente and Mascot identifications to find significant proteins, the best-scoring protein is always kept. Consecutive lower scoring hits are removed if a better-scoring protein is a profilog of the lower scoring protein.

Figure S2

a



b

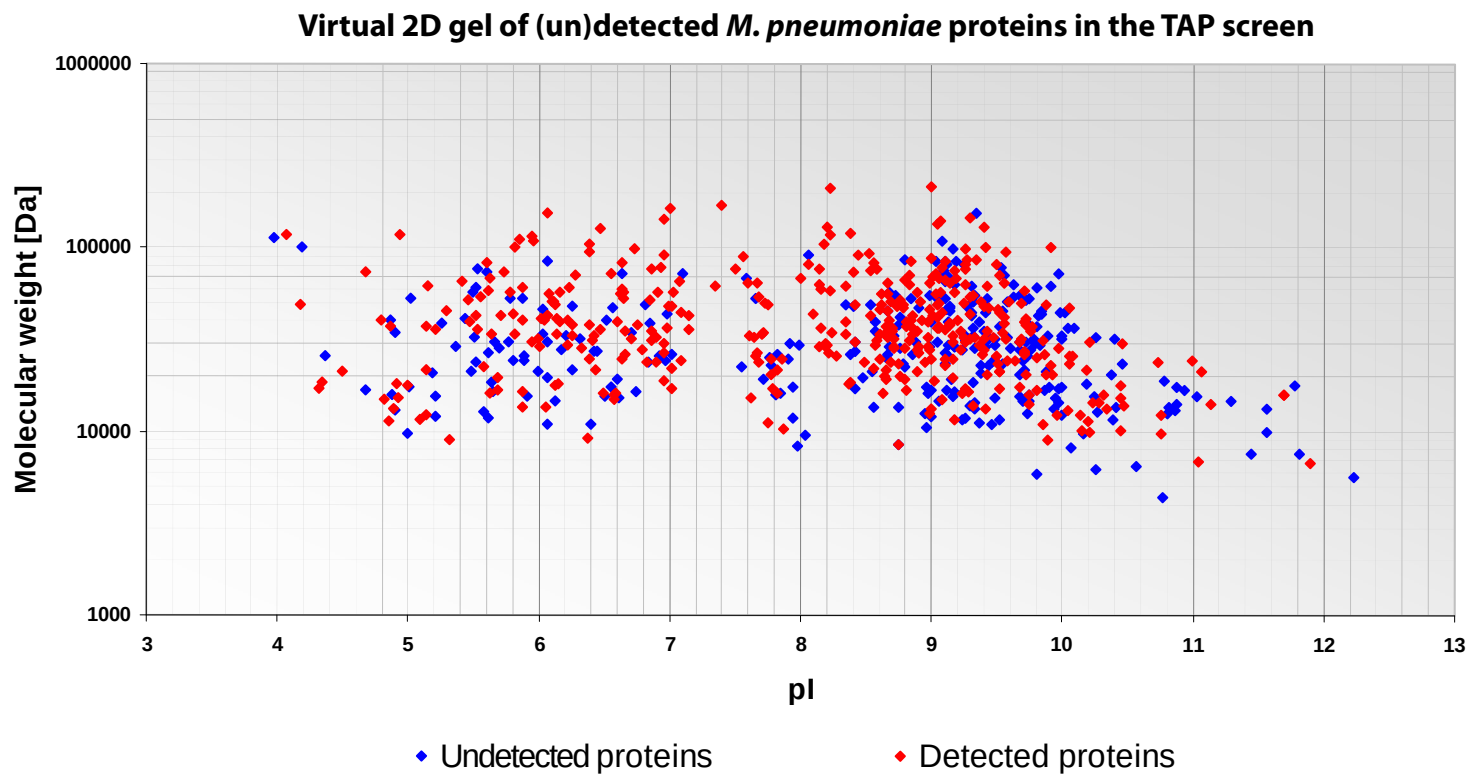
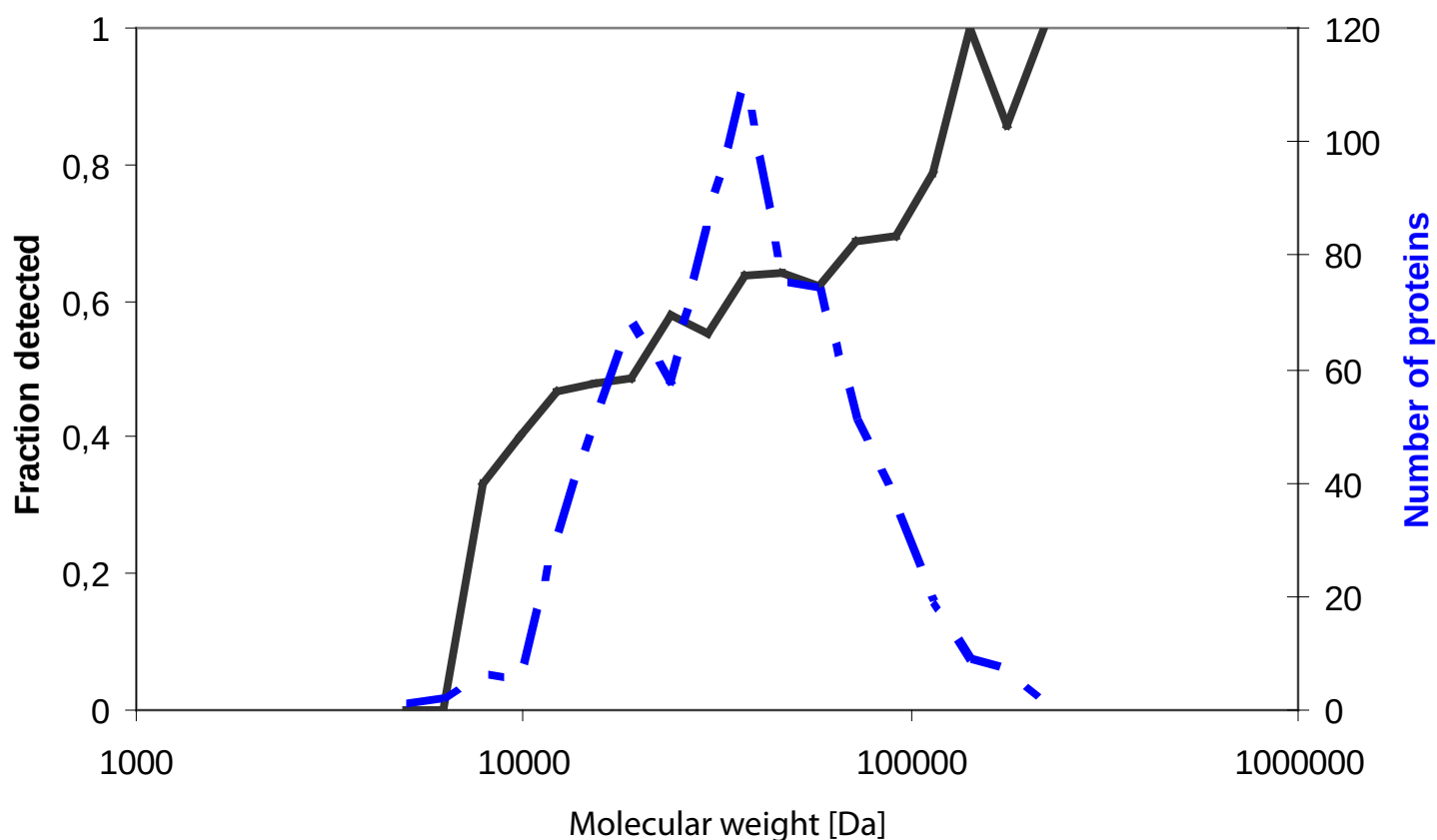


Figure S2

C



D

Domain/motif	Total # proteins	Fraction detected	P=
Signal peptide	127	44%	0.000268
Transmembrane	146	36%	5.56e-9
DUF16	25	32%	0.00436
Lipoprotein_10	10	20%	0.00948
Adhesin_P1	9	11%	0.00279

Figure S2

E

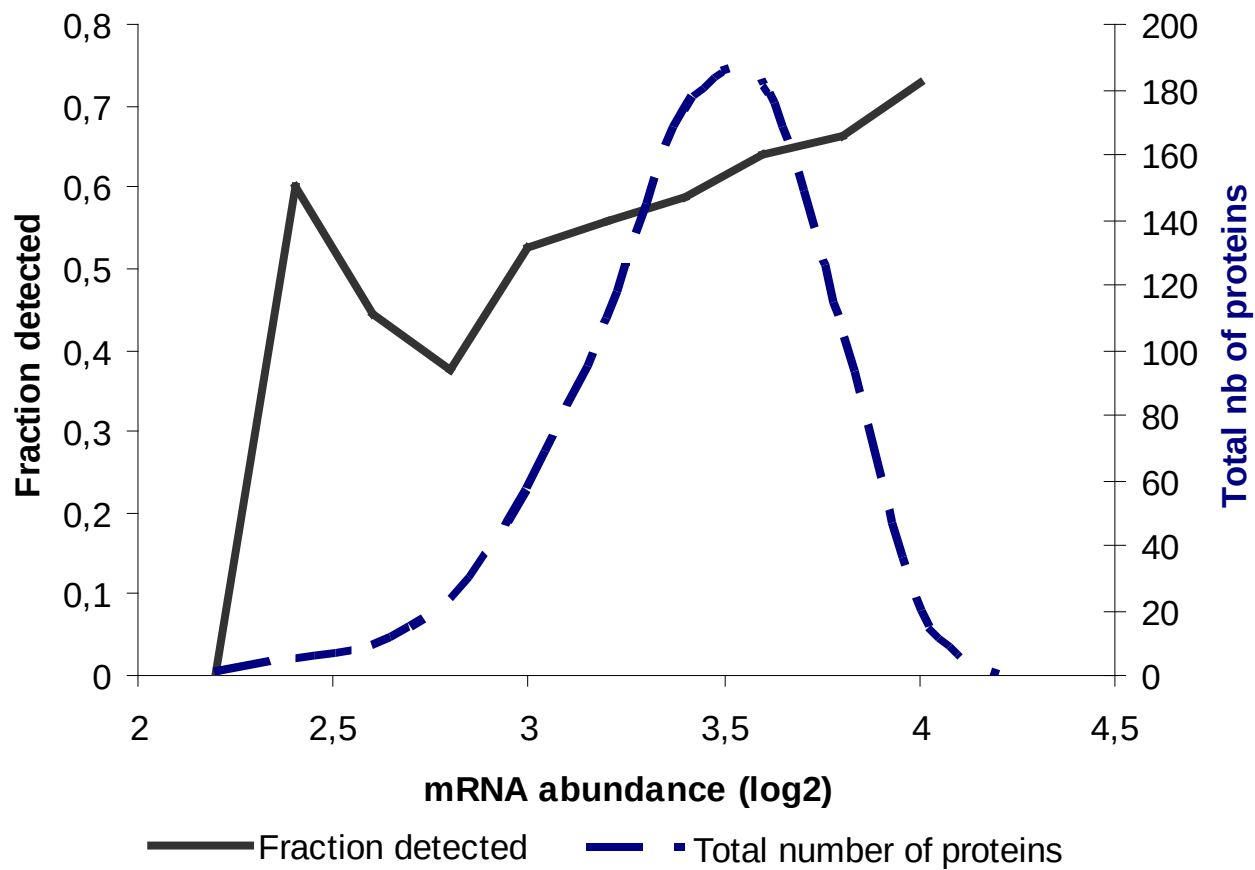


Fig. S2 Proteome coverage (functional & physicochemical)

In **(A)** we represent the fraction of proteins identified per specified COG (Clusters of orthologous groups of genes) class (S42, S43) in our TAP-MS analysis (green), other proteomics analyses (S44-S46) including the PeptideAtlas (yellow), as well as those proteins predicted from the genome, but not yet observed (white). The *M. pneumoniae* genome contains 689 annotated protein coding genes, of which 564, corresponding to 82% of the genome, have been observed by MS-based screens before our study. In the TAP-MS analysis we detected 411 distinct *M. pneumoniae* proteins. Combining the results from the TAP-MS screen, the PeptideAtlas as well as previous screens, a total of 622 proteins (90% of the genome) have been detected. **(B)** Virtual 2D gel where proteins are represented according to their size, y-axis, and isoelectric point, x-axis. The proteins identified in our dataset are colored red the ones not identified are seen in blue. The distribution of identified proteins covers molecular weights from 6 to 215 kDa and the pI range between 4 and 11.8. **(C)** The graph represents the fraction of proteins with a given molecular weight that were identified in the TAP-MS analysis. Molecular weights (MW) were extracted from UniProt. Proteins were grouped into different MW classes by exponential binning and subsequently counted. The plot shows that small proteins, e.g. below 20kDa, are significantly underrepresented ($P = 0.000444$). The R^2 between the molecular weight and the fraction detected is 0.88 (correlation of 0.94). The table in panel **(D)** shows the five protein domains or motifs that were significantly underrepresented in the dataset. Proteins were classified according to their domain or motif composition according to SMART (S47). The percentage of proteins with a given domain or a given motif that were identified in the TAP-MS analysis was calculated. P-values were calculated with the χ^2 -test. **(E)** The graph represents the fraction of proteins with a given mRNA abundance that were identified in the TAP-MS analysis. For abundance, we took the log2 signal of mRNA abundance on a spotted array measured under reference conditions and binned with a bin width of 0.5. Per bin, the number of proteins and the fraction

detected was calculated. Low abundant proteins are underrepresented. There was no relation found between detection probability and hydrophobicity, when hydrophobicity scores were calculated using GRAVY.

Figure S3

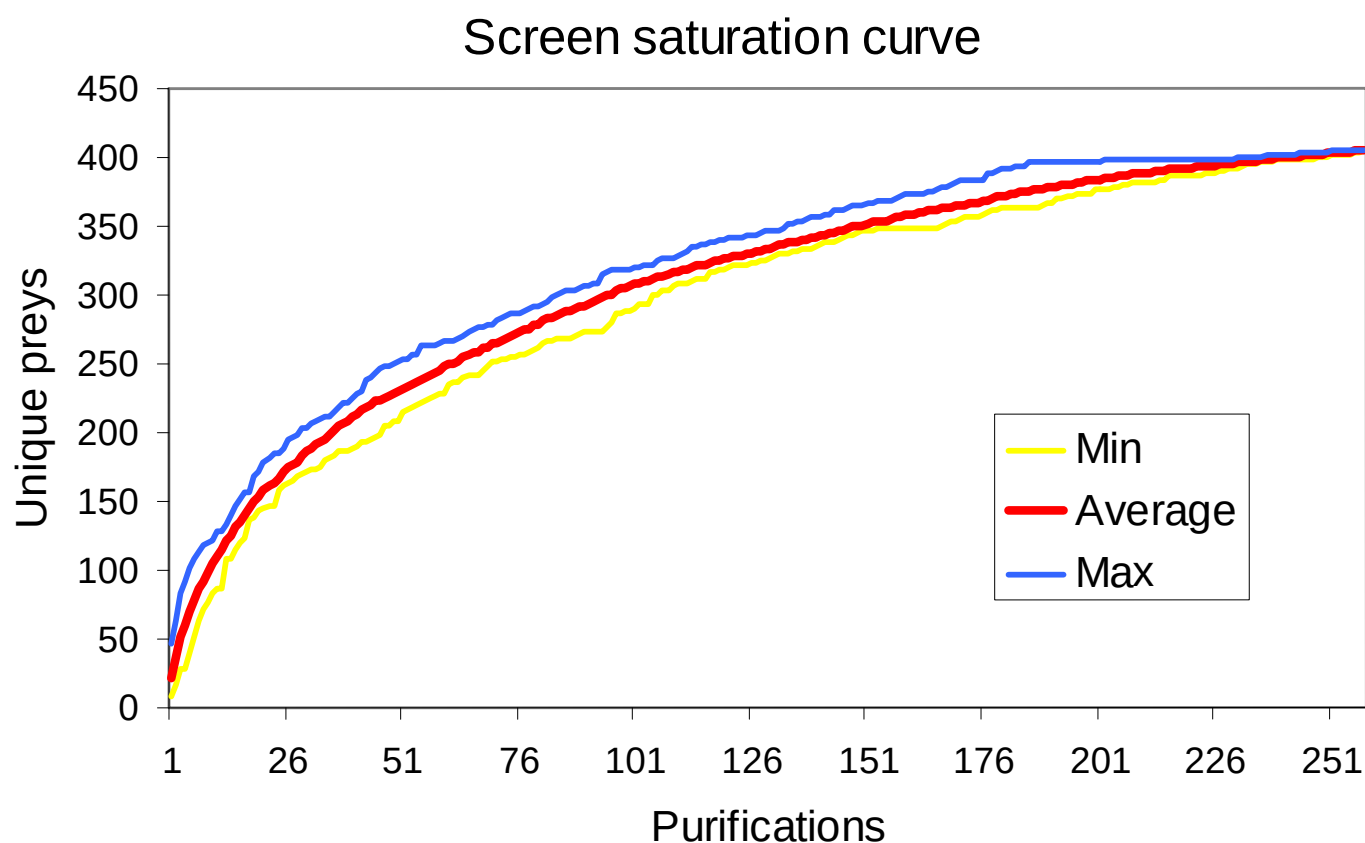


Fig. S3 Screen saturation curve

The graph shows the cumulated number of non-redundant proteins identified per TAP purifications as a function of increasing numbers of purifications performed. The 259 purifications are represented on the x-axis in computationally randomized fashion, while the 411 identified proteins are depicted on the y-axis. The average value, over ten randomization procedures, of proteins identified after a specific amount of purifications is shown in red. The maximal/minimal value, over ten randomization procedures, of proteins identified after a specific amount of purifications is shown in blue and yellow, respectively.

Figure S4

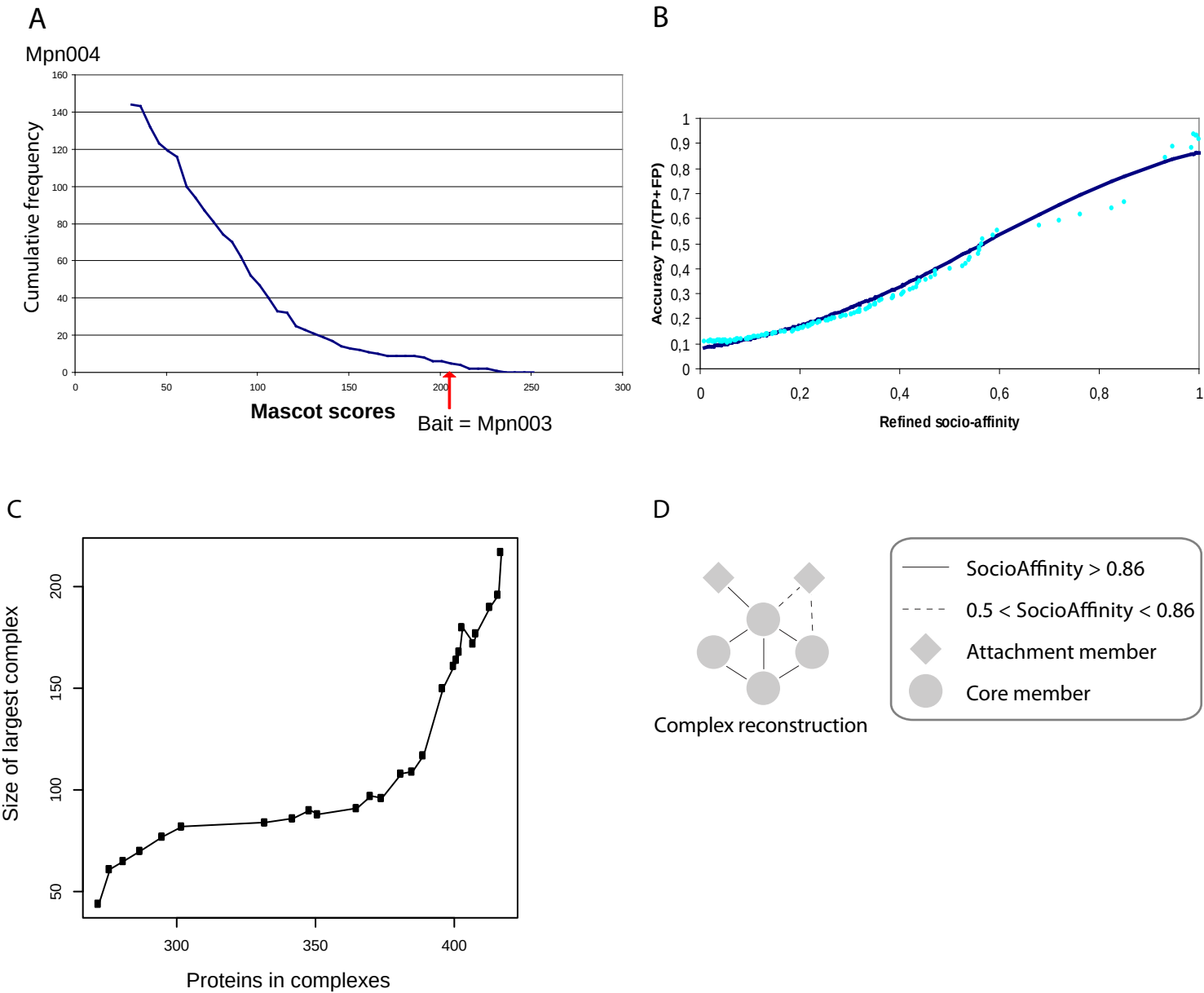


Fig. S4 Refined socio-affinity scoring, benchmarking & clustering

(A) Example of PMF scores for a protein that appears either as contaminant or as cognate interactor in different purifications. On the x-axis is the MASCOT score of all identifications of Mpn004. On the y-axis is the cumulative frequency of identifications of Mpn004. This protein seems to be a contaminant in many purifications (low score). However, if the bait was Mpn003 (a known interactor), Mpn004 (red arrow) is identified with a very high MASCOT score (many identified peptides). This was also the case for other interactors within known complexes. Therefore, in the spoke component of the socio-affinity, we take for fj only the cumulative prey frequency until the PMF score of the prey-identification with bait i. **(B)** The socio-affinity scoring was adjusted from (S19, S20). For every identified prey we ranked the PMF scores. The rank is now used instead of the frequency of the prey in the socio-affinity scoring of bait-prey relations. On the x-axis is the normalized, refined socio-affinity. Accuracy was estimated for each bin of socio-affinity scores. Accuracy was measured by the golden standard set of known complexes (table S5). True positive interactions (TPInt) are interactions between proteins of the same complex, whereas false positives interactions (FPInt) are interactions between proteins of different complexes. Accuracy is $TPInt/(TPInt+FPInt)$, dots. A logistic is done and the logistic curve was used to estimate for each protein-pair, the probability of an interaction. **(C)** During clustering the number of proteins clustering in complexes as well as the size of the largest complex was monitored. We chose a threshold to define complexes such that many proteins are in complexes but still the size of the largest cluster has not exploded yet. This is depicted in **c** where on the x-axis the number of proteins in complexes and on the y-axis the number of proteins in the largest complex is shown. We chose a threshold that includes 348 proteins into the complexes of our analysis. **(D)** Nomenclature and definition: classification of protein complex components into cores (ellipses) and attachments (diamonds).

Figure S5

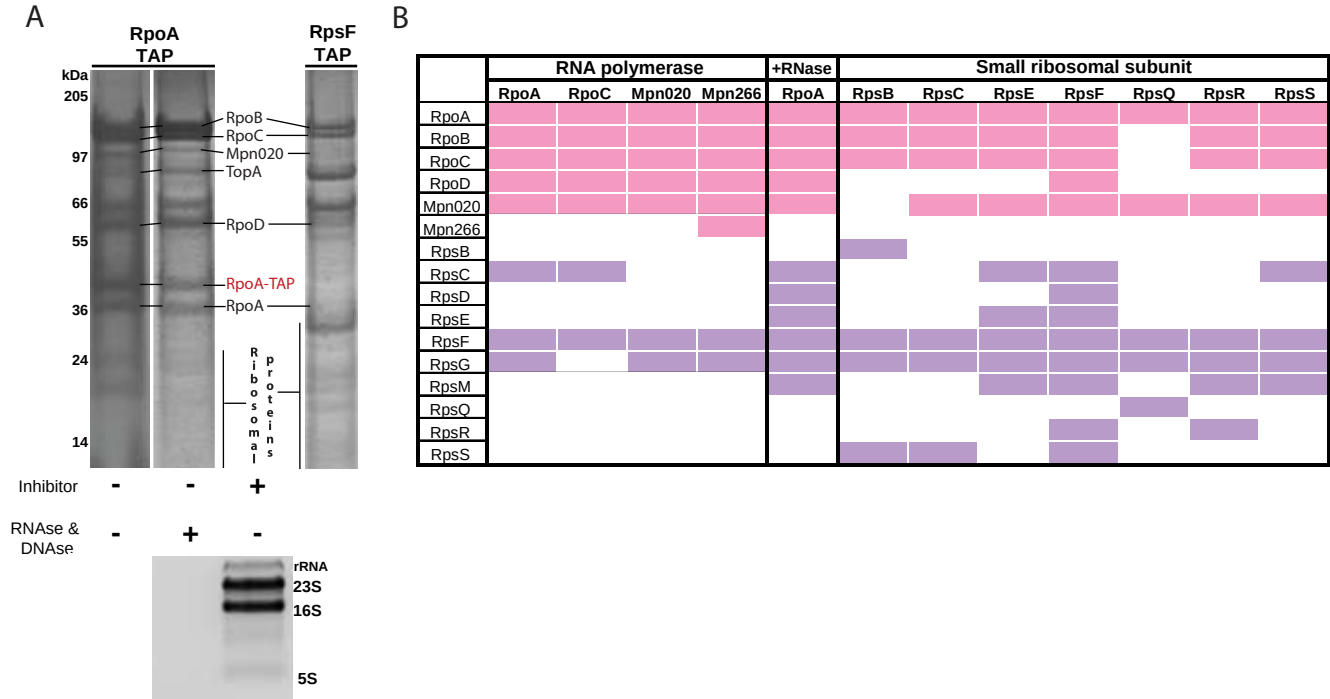


Fig. S5 The association between the RNA polymerase and the ribosome

(A) Ribosomal proteins copurify with RpoA-TAP in extracts treated with RNase and DNase. Upper panels: coomassie-stained SDS-PAGE gels of RpoA-TAP and 30S ribosomal protein RpsF-TAP purifications. Proteins identified by MS are indicated between gel images. Lower panel: Ethidium bromide stained 1% agarose gel of total extracted RNA. Cell extracts were treated either with RNase/RNase inhibitors or RNase/DNase. (B) RNA polymerase and ribosomal proteins copurify in various purifications. The matrix shows the different baits used in the top of each column and the preys identified by MS in each row underneath. This underlines ability to reversely purify the RNA polymerase with ribosomal entry points and vice versa.

Figure S6

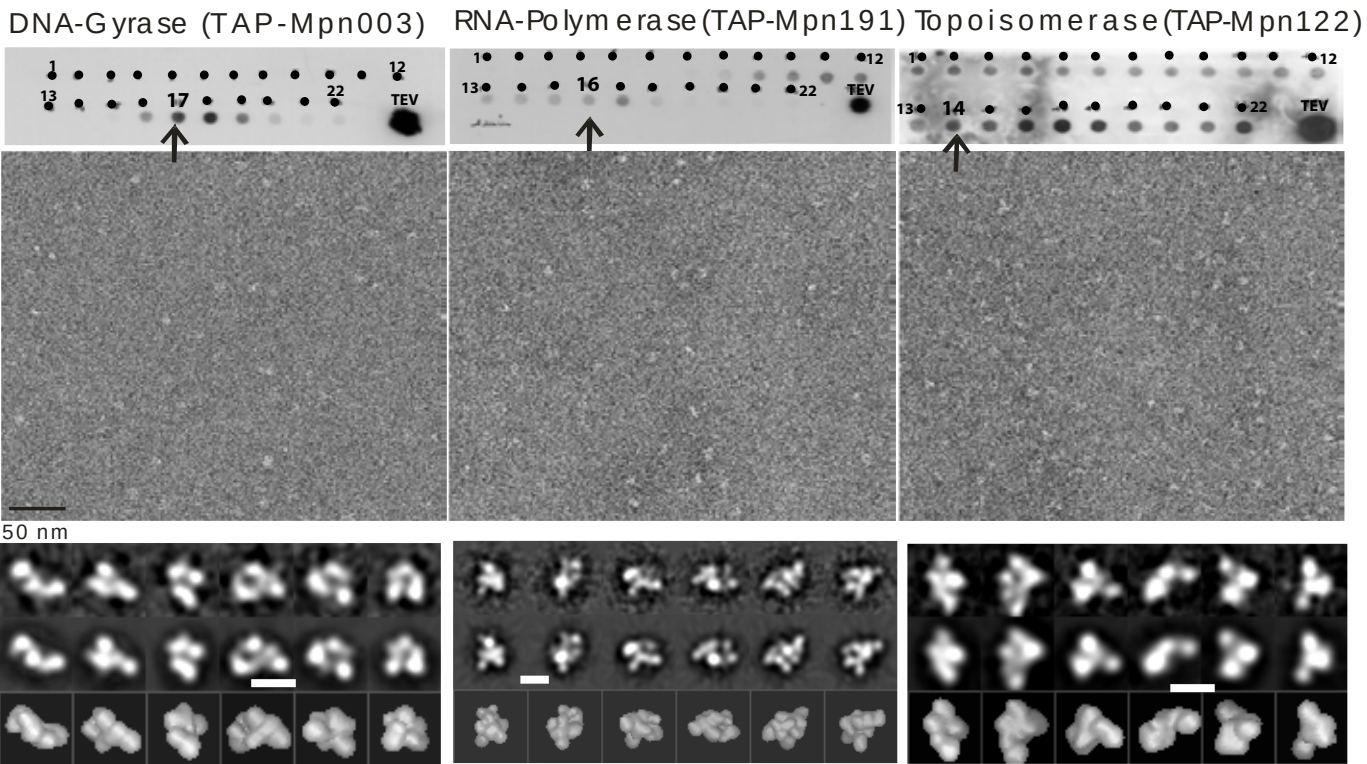


Fig. S6 Electron microscopy particle selection

Electron microscopy and image analysis of negatively stained complexes fixed with glutaraldehyde on a glycerol density gradient. From left to right: DNA-gyrase (Mpn003, Mpn004), RNA-polymerase (Mpn191, Mpn352, Mpn515, Mpn516), DNA-topoisomerase (Mpn122, Mpn123). Upper panel: Fractions of the glycerol gradient were tested by dot blot using an antibody against the calmodulin tag of the bait (fraction 1 is always at the bottom of the gradient). Arrowheads mark the fractions selected for further analysis by electron microscopy and image reconstruction. Representative micrographs of the fractions marked by arrowheads are shown below the dot blots. The samples were prepared by sandwich negative stain using 2% uranyl acetate for staining. The bottom panel shows results from the image analysis. Upper row: Selected class averages calculated by multivariate statistical analysis. The class averages match projections (central row) of the three-dimensional image reconstructions of the respective complexes. Bottom row shows surface representations of the image reconstructions from the same direction as the projections above. The scale bars equal 10 nm.

Figure S7

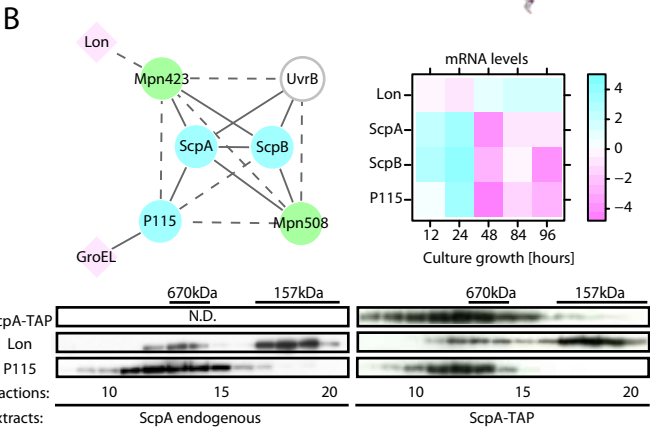
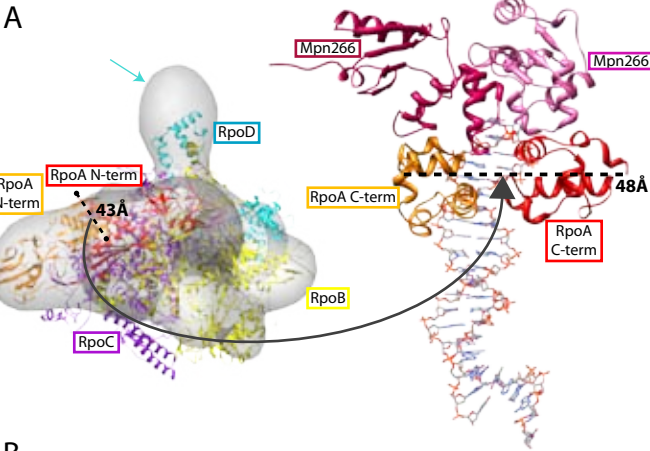
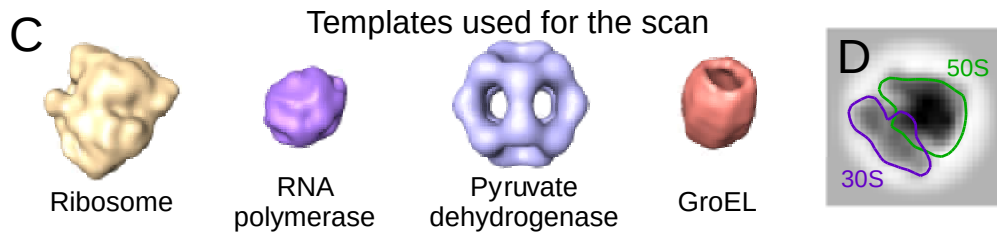
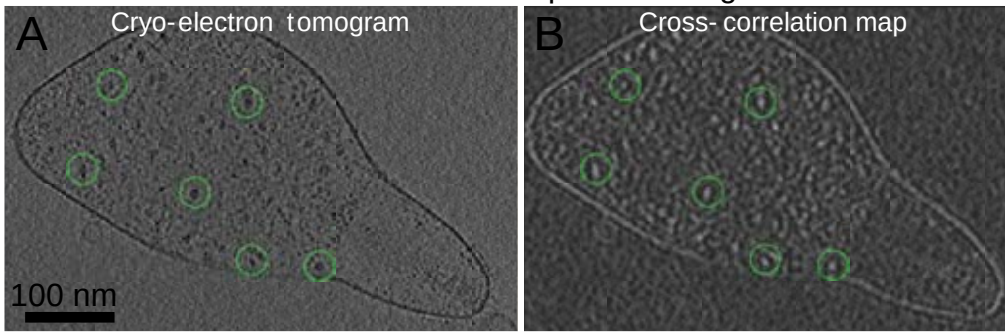


Fig. S7 Novelties in protein complexes

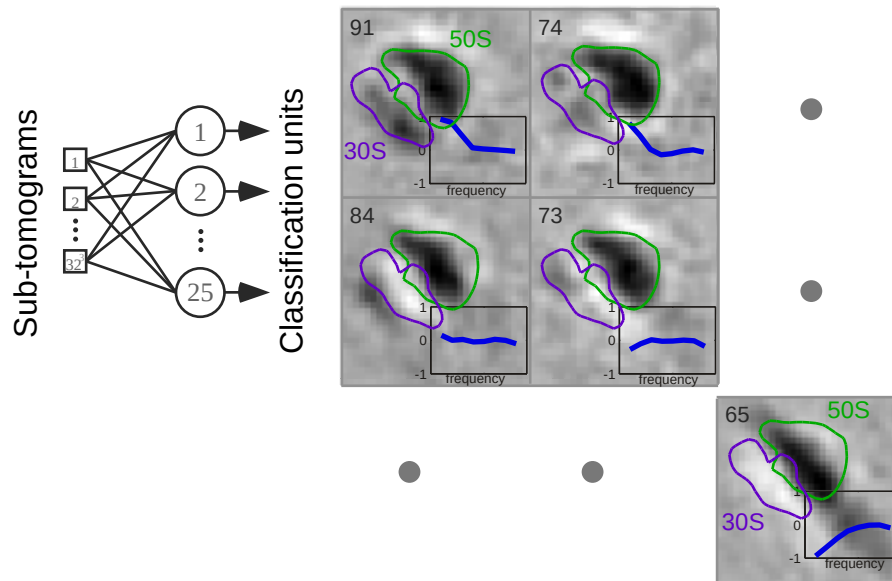
(A) Mpn266 associates with the RNA polymerase as a homodimer. On the left, *M. pneumoniae* RNAPol electron density (~17 nm) with a fitted structure modeled on *T. aquaticus* RNAPol (PDB code: 1l9u). The model fits apart from an extra stalk (cyan arrow), which can be explained by a 200 amino-acids N-terminus insertion in *M. pneumoniae* RpoD. On the right, interaction with an Mpn266 homodimer modeled using a template of *B. subtilis* SPX interacting with RpoA C-terminus (1z3e) and a template of an *E. coli* RpoA C-terminus dimer interacting with DNA (1lb2). The RpoA C-terminus dimer is separated by a distance (48Å) similar to that observed for the RpoA N-terminus from the polymerase fitted on the EM (43Å) **(B)**. The cohesin-like protein complex. The components are color coded according to their belonging to expression clusters; the heat map illustrates some of these expression patterns (accompanying manuscripts by Yus *et al.*). Core components are represented by circles, attachments by diamonds. The line attribute corresponds to socio-affinity indices: dashed lines, 0.5–0.86; plain lines, >0.86. The blots at the bottom show co-elution, during gel filtration chromatography, of ScpA (59kDa), Lon (90kDa) and P115 (111kDa) in extract over-expressing (right) or not (left) ScpA-TAP (79kDa). N.D.: not determined.

Figure S8

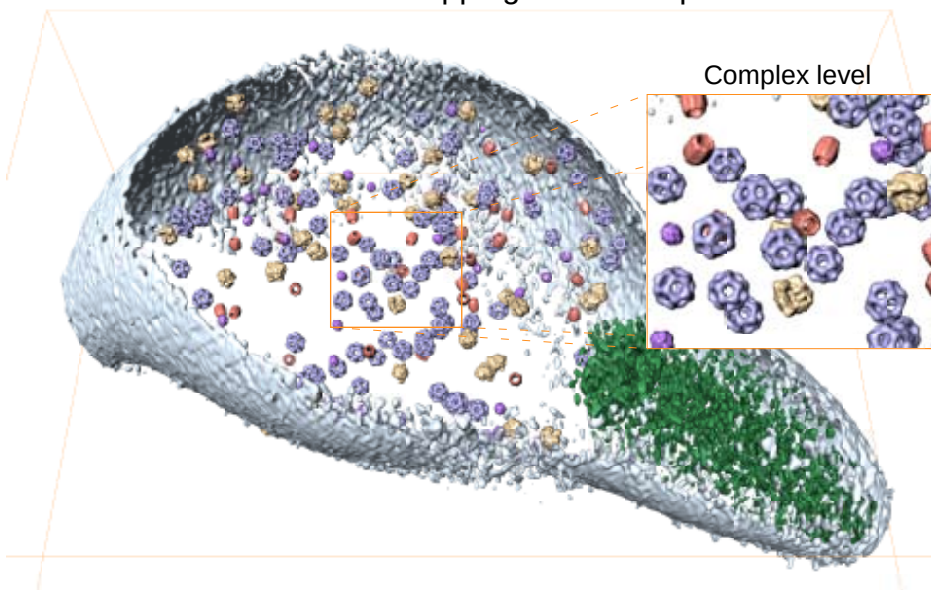
Cross-correlation based pattern recognition



E Classification on the example of the ribosome



F Cell level: Mapping of the complexes



Template matching

Classification

Identification

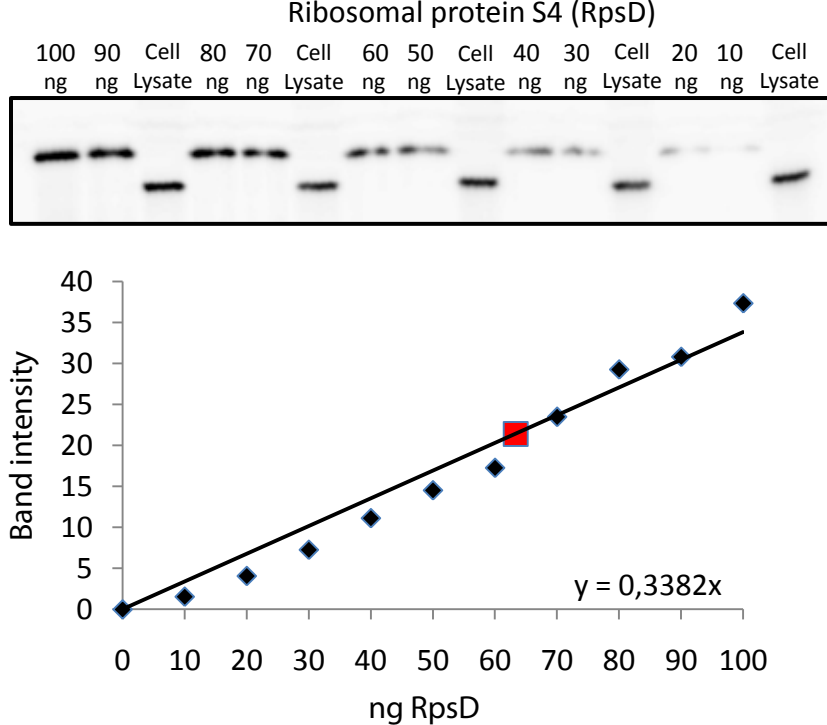
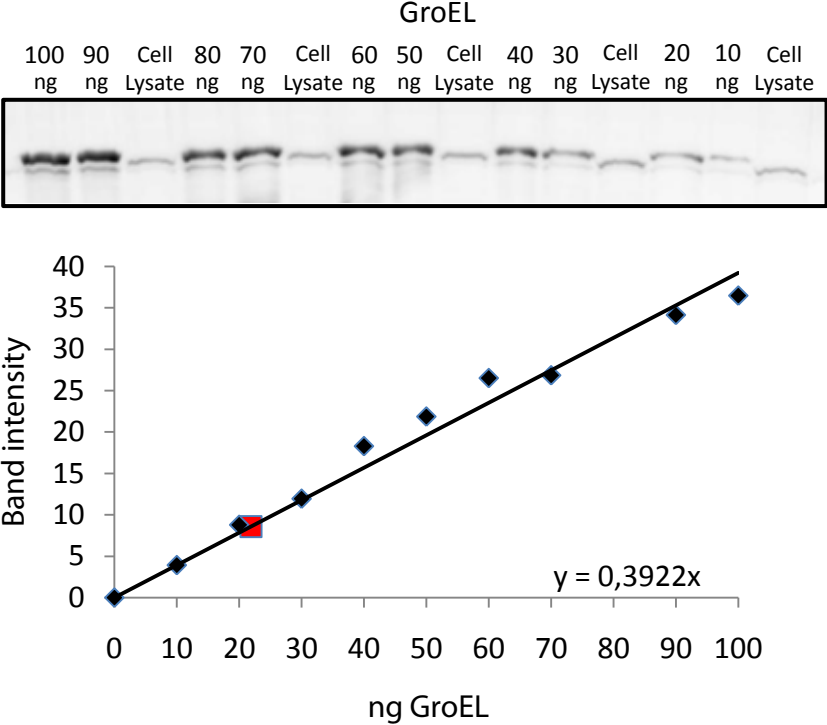
Fig. S8 Complexes detection in *M. pneumoniae* cryoelectron tomograms by template-matching and protein complex-based classification

Mapping the spatial distribution of four selected complexes in one of the cryo-electron tomograms. **(A)** x-y slice of a reconstruction. The protein density is visualized in dark intensity values. Some of the putative complexes after template matching have been highlighted with circles. **(B)** The same slice as in **(A)**, in the normalized cross-correlation map after scanning with the large ribosomal subunit. Higher cross-correlation values are represented in the bright intensity values. High correlation values corresponding to putative complexes are highlighted with circle in **(A)** and **(B)**. **(C)** Isosurface representation of four complexes present in the tomogram. Parts of these complexes only have been used as templates: the large subunit of the ribosome (yellow) only, half a RNA polymerase (purple), half the structural core of the pyruvate dehydrogenase (PdhC, homomultimer) (blue) and 10 out of the 14 apical and equatorial domains of GroEL (red). **(D)** The projection of the whole ribosome is visualized at the same view as the projection images in the classification units. **(E)** In the green overlay contour the 50S is indicated and in the purple the 30S small ribosomal sub-unit. The overlays are also superimposed to the classification units. **(E)** Protein complex-based classification strategy using the ribosome as an example. In our methodology, the 50S subunit is used to search through the whole tomogram (in six degrees of freedom, three rotational and three translational). Sub-Tomograms at all local maxima of the cross-correlation map are extracted and subjected to missing wedge independent classification with KerDenSOM. A cropped region of the total output map of KerDenSOM is shown. A projection image of five out of the 25 classification units is presented. In the first row the Fourier shell correlation, visualized as the blue plot in the bottom right corner of the images, is at the high frequencies indicating high similarity of the density inside the green overlay region to the small ribosomal subunit. The Fourier shell correlation curves of all the other

classification units are either close to zeros (showing no similarity) or even negative similarity (due to the inverted contrast). The visual inspection confirms this result. Hence only the subtomograms classifying within the classes on the upper row are selected in this example to contain true hits for the ribosomes. The number of hits assigned to each class appears in the upper-left corner of the classification unit. **(F)** Isosurface representation of the whole *M. pneumoniae* cell with the putative complexes mapped inside. Membrane (blue) and tip organelle (green) can also be visualized. The color mapping of the complexes is as in **(C)**.

Figure S9

A



B

GroEL		RpsD
8.57525	measured value	21.42391667
0.3922	Slope of standard curve	0.3382
2.18645E-08	g protein/lane	6.33469E-08
380250000	cells/lane	3802500000
5.75003E-17	g/cell	1.66593E-17
58086	MW	23775
9.89916E-22	moles of RpsD & GroEL	7.00705E-22
6.022E+23	molecules in 1 mol	6.022E+23
596	molecules per cell	422
~43	Particles	~420

Fig. S9 Quantitative Western blots of GroEL and ribosomal protein S4 (RpsD) in whole cell extract

(A) Top panels: 0-100 ng affinity purified GroEL and RpsD protein was loaded next to a *M. pneumoniae* cell lysate and blotted with polyclonal antibodies (Antibodies: R. Herrmann). Bottom panels: Standard curves from quantified Western blots. The slope of the trend line was used to determine the amount of GroEL and RpsD protein in a *M. pneumoniae* cell lysate (red square). The error in quantifying the amount of GroEL in the *M. pneumoniae* sample is 2.1%, as judged from five measurements and 6.6% for RpsD, as judged from four measurements. (B) Estimation of the number of molecules of GroEL and RpsD proteins in a *M. pneumonia* cell.

Table S2: List of purifications and proteins identified.

In total 259 successful purifications were performed using 212 distinct TAP-fusion proteins. i) Is the list of TAP purifications. The first column gives the purification ID. The second and third columns provide the annotated systematic and protein names of the TAP-protein used as entry point, respectively. The last column gives the list of prey proteins that co-purify with the bait. ii) Is the list of proteins identified in the PeptideAtlas by FT-MS/MS. The first column provides the annotated systematic name of each identified protein. The second column gives the protein name.

i) List of purifications			
Purification ID	Bait (systematic name)	Bait (protein name)	Preys
EMBL001	Mpn025	Fba	Mpn025 Mpn434 Mpn067 Mpn573 Mpn118 Mpn034 Mpn003
EMBL002	Mpn025	Fba	Mpn025 Mpn390 Mpn516 Mpn430 Mpn315 Mpn665 Mpn261 Mpn365 Mpn034 Mpn141 Mpn669 Mpn210 Mpn434 Mpn573 Mpn207 Mpn515 Mpn426 Mpn546 Mpn608 Mpn191 Mpn020 Mpn252 Mpn452 Mpn621 Mpn004
EMBL003	Mpn567	P200	Mpn210 Mpn434 Mpn095 Mpn567 Mpn390 Mpn295 Mpn003 Mpn393 Mpn004
EMBL004	Mpn230	RpsR	Mpn390 Mpn230 Mpn516 Mpn394 Mpn440 Mpn354 Mpn258 Mpn658 Mpn228 Mpn556 Mpn166 Mpn573 Mpn515 Mpn686 Mpn191 Mpn020 Mpn452 Mpn621 Mpn004 Mpn619 Mpn303 Mpn179 Mpn430 Mpn665 Mpn261 Mpn685 Mpn044 Mpn574 Mpn019 Mpn660 Mpn210 Mpn434 Mpn226 Mpn280 Mpn426 Mpn050 Mpn189
EMBL005	Mpn051	Mpn051	Mpn210 Mpn627 Mpn361 Mpn573 Mpn051 Mpn042 Mpn034
EMBL006	Mpn516	RpoB	Mpn434 Mpn352 Mpn191 Mpn020 Mpn516 Mpn228 Mpn515
EMBL007	Mpn516	RpoB	Mpn328 Mpn352 Mpn516 Mpn665 Mpn171 Mpn330 Mpn571 Mpn228 Mpn210 Mpn226 Mpn024 Mpn515 Mpn673 Mpn239 Mpn191 Mpn020 Mpn004
EMBL008	Mpn379	PolA	Mpn210 Mpn621
EMBL009	Mpn379	PolA	Mpn684 Mpn390 Mpn665 Mpn261 Mpn379 Mpn258 Mpn141 Mpn210 Mpn434 Mpn573 Mpn310 Mpn207 Mpn032 Mpn515 Mpn426 Mpn050 Mpn020 Mpn621 Mpn004
EMBL010	Mpn006	Tmk	Mpn372 Mpn210 Mpn134 Mpn434 Mpn390 Mpn280 Mpn516 Mpn665 Mpn515 Mpn006 Mpn191 Mpn258 Mpn452 Mpn621
EMBL011	Mpn006	Tmk	Mpn210 Mpn434 Mpn204 Mpn390 Mpn118 Mpn665 Mpn426 Mpn006 Mpn034 Mpn040 Mpn452 Mpn213
EMBL012	Mpn006	Tmk	Mpn134 Mpn025 Mpn619 Mpn390 Mpn303 Mpn118 Mpn516 Mpn430 Mpn197 Mpn665 Mpn394 Mpn261 Mpn378 Mpn006 Mpn393 Mpn258 Mpn141 Mpn210 Mpn553 Mpn434 Mpn322 Mpn573 Mpn280 Mpn515 Mpn191 Mpn452 Mpn106 Mpn621 Mpn004
EMBL013	Mpn416	P29	Mpn623 Mpn434 Mpn656 Mpn674 Mpn665 Mpn108 Mpn314 Mpn392 Mpn416 Mpn393 Mpn034 Mpn337
EMBL014	Mpn320	ThyA	Mpn563 Mpn210 Mpn434 Mpn573 Mpn324 Mpn516 Mpn515 Mpn426 Mpn320
EMBL015	Mpn320	ThyA	Mpn284 Mpn434 Mpn320 Mpn573 Mpn555 Mpn076 Mpn032
EMBL016	Mpn493	UlaD	Mpn596 Mpn434 Mpn555 Mpn674 Mpn665 Mpn314 Mpn392 Mpn034
EMBL017	Mpn327	RpmA	Mpn390 Mpn516 Mpn430 Mpn261 Mpn615 Mpn658 Mpn574 Mpn228 Mpn165 Mpn322 Mpn434 Mpn166 Mpn226 Mpn676 Mpn032 Mpn314 Mpn239 Mpn549 Mpn081 Mpn621
EMBL018	Mpn228	RpsF	Mpn352 Mpn230 Mpn516 Mpn103 Mpn518 Mpn169 Mpn115 Mpn228 Mpn553 Mpn616 Mpn515 Mpn178 Mpn020 Mpn191 Mpn621 Mpn004 Mpn446 Mpn480 Mpn179 Mpn295 Mpn520 Mpn665 Mpn171 Mpn182 Mpn465 Mpn685 Mpn660 Mpn035 Mpn210 Mpn434 Mpn226 Mpn622 Mpn280 Mpn239 Mpn189 Mpn252 Mpn561
EMBL019	Mpn389	LplA	Mpn389 Mpn434 Mpn295
EMBL020	Mpn553	ThrS	Mpn210 Mpn553 Mpn204 Mpn573 Mpn595 Mpn034
EMBL021	Mpn322	NrdF	Mpn210 Mpn434 Mpn322 Mpn573 Mpn390 Mpn419 Mpn261 Mpn034 Mpn365 Mpn004
EMBL022	Mpn322	NrdF	Mpn454 Mpn210 Mpn434 Mpn322 Mpn059 Mpn324
EMBL023	Mpn550	Thil	Mpn210 Mpn434 Mpn550 Mpn573 Mpn295 Mpn003 Mpn515
EMBL024	Mpn573	GroL	Mpn062 Mpn392 Mpn573 Mpn264 Mpn194 Mpn430 Mpn008

EMBL025	Mpn495	UlaB	Mpn269 Mpn430 Mpn315 Mpn665 Mpn309 Mpn365 Mpn019 Mpn228 Mpn210 Mpn434 Mpn324 Mpn495 Mpn515 Mpn426 Mpn050 Mpn345 Mpn004 Mpn621
EMBL026	Mpn668	Mpn668	Mpn563 Mpn391 Mpn203 Mpn514 Mpn118 Mpn665 Mpn668
EMBL027	Mpn111	Mpn111	Mpn390 Mpn340 Mpn303 Mpn555 Mpn516 Mpn430 Mpn315 Mpn665 Mpn261 Mpn658 Mpn574 Mpn669 Mpn210 Mpn434 Mpn573 Mpn032 Mpn314 Mpn515 Mpn111 Mpn050 Mpn547 Mpn621 Mpn004
EMBL028	Mpn663	Mpn663	Mpn671 Mpn390 Mpn342 Mpn516 Mpn430 Mpn003 Mpn536 Mpn438 Mpn246 Mpn034 Mpn210 Mpn185 Mpn434 Mpn573 Mpn515 Mpn452 Mpn621 Mpn004
EMBL029	Mpn686	DnaA	Mpn210 Mpn434 Mpn686 Mpn516 Mpn452 Mpn376 Mpn141
EMBL030	Mpn509	Mpn509	Mpn210 Mpn434 Mpn390 Mpn516 Mpn430 Mpn295 Mpn003 Mpn665 Mpn281 Mpn034 Mpn599 Mpn509 Mpn452
EMBL031	Mpn079	FruK	Mpn434 Mpn062 Mpn573 Mpn390 Mpn295 Mpn665 Mpn079
EMBL032	Mpn013	Mpn013	Mpn390 Mpn665 Mpn258 Mpn210 Mpn556 Mpn434 Mpn553 Mpn280 Mpn207 Mpn346 Mpn515 Mpn426 Mpn050 Mpn452 Mpn004 Mpn621
EMBL033	Mpn075	Mpn075	Mpn210 Mpn075 Mpn434 Mpn322 Mpn573 Mpn324 Mpn665 Mpn261 Mpn426 Mpn020 Mpn258 Mpn228 Mpn399
EMBL034	Mpn328	Nfo	Mpn525 Mpn300 Mpn328 Mpn567 Mpn390 Mpn042 Mpn516 Mpn665 Mpn261 Mpn258 Mpn574 Mpn141 Mpn210 Mpn434 Mpn226 Mpn573 Mpn192 Mpn388 Mpn110 Mpn515 Mpn314 Mpn426 Mpn050 Mpn621 Mpn004
EMBL035	Mpn328	Nfo	Mpn210 Mpn434 Mpn328 Mpn573 Mpn118 Mpn516 Mpn515 Mpn034 Mpn072
EMBL036	Mpn156	RbfA	Mpn390 Mpn555 Mpn156 Mpn516 Mpn430 Mpn665 Mpn261 Mpn033 Mpn034 Mpn228 Mpn210 Mpn434 Mpn226 Mpn063 Mpn515 Mpn593 Mpn191 Mpn621 Mpn004
EMBL037	Mpn536	RuvB	Mpn269 Mpn390 Mpn555 Mpn516 Mpn430 Mpn315 Mpn665 Mpn536 Mpn436 Mpn261 Mpn115 Mpn574 Mpn072 Mpn210 Mpn434 Mpn550 Mpn573 Mpn524 Mpn515 Mpn050 Mpn020 Mpn345 Mpn452 Mpn621
EMBL038	Mpn002	Mpn002	Mpn397 Mpn134 Mpn303 Mpn352 Mpn516 Mpn430 Mpn295 Mpn665 Mpn261 Mpn002 Mpn034 Mpn165 Mpn210 Mpn553 Mpn434 Mpn573 Mpn280 Mpn310 Mpn674 Mpn515 Mpn050 Mpn191 Mpn621 Mpn004
EMBL039	Mpn638	Mpn638	Mpn434
EMBL040	Mpn638	Mpn638	Mpn210 Mpn434 Mpn366 Mpn295 Mpn474 Mpn518 Mpn273 Mpn638 Mpn004 Mpn072
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EMBL042	Mpn246	Def	Mpn434 Mpn573 Mpn390 Mpn246 Mpn155 Mpn245
EMBL043	Mpn168	RplB	Mpn210 Mpn434 Mpn390 Mpn315 Mpn295 Mpn665 Mpn261 Mpn207 Mpn060 Mpn252 Mpn621 Mpn004
EMBL044	Mpn275	Mpn275	Mpn596 Mpn568 Mpn390 Mpn118 Mpn520 Mpn516 Mpn430 Mpn315 Mpn665 Mpn394 Mpn261 Mpn275 Mpn556 Mpn210 Mpn434 Mpn676 Mpn036 Mpn515 Mpn392 Mpn050 Mpn621 Mpn004
EMBL045	Mpn172	RplP	Mpn642 Mpn390 Mpn430 Mpn211 Mpn103 Mpn665 Mpn261 Mpn658 Mpn228 Mpn210 Mpn418 Mpn434 Mpn166 Mpn063 Mpn032 Mpn426 Mpn050 Mpn020 Mpn252 Mpn452 Mpn621
EMBL046	Mpn268	Mpn268	Mpn062 Mpn671 Mpn390 Mpn516 Mpn665 Mpn261 Mpn379 Mpn259 Mpn258 Mpn210 Mpn322 Mpn434 Mpn268 Mpn207 Mpn252 Mpn452
EMBL047	Mpn268	Mpn268	Mpn210 Mpn434 Mpn268 Mpn390 Mpn555 Mpn295 Mpn003 Mpn665 Mpn261 Mpn314 Mpn426 Mpn489 Mpn034 Mpn040 Mpn452
EMBL048	Mpn192	RplQ	Mpn253 Mpn434 Mpn034 Mpn665 Mpn108
EMBL049	Mpn559	Mpn559	Mpn210 Mpn434 Mpn516 Mpn430 Mpn665 Mpn436 Mpn640 Mpn559 Mpn426 Mpn514 Mpn452 Mpn072
EMBL050	Mpn472	Mpn472	Mpn210 Mpn434 Mpn429 Mpn390 Mpn516 Mpn430 Mpn665 Mpn261 Mpn472 Mpn317 Mpn426 Mpn034 Mpn452
EMBL051	Mpn479	AzoR	Mpn390 Mpn479 Mpn040

EMBL052	Mpn461	Mpn461	Mpn434 Mpn390 Mpn034 Mpn295 Mpn476
EMBL053	Mpn208	RpsB	Mpn390 Mpn555 Mpn516 Mpn430 Mpn315 Mpn665 Mpn261 Mpn169 Mpn115 Mpn034 Mpn658 Mpn228 Mpn210 Mpn434 Mpn226 Mpn622 Mpn674 Mpn515 Mpn314 Mpn426 Mpn239 Mpn050 Mpn191 Mpn208 Mpn621 Mpn004
EMBL054	Mpn558	GidB	Mpn558 Mpn434 Mpn573 Mpn390 Mpn280 Mpn562 Mpn059 Mpn452 Mpn621
EMBL055	Mpn117	RplT	Mpn210 Mpn434 Mpn269 Mpn303 Mpn180 Mpn430 Mpn665 Mpn050 Mpn222 Mpn034 Mpn658 Mpn574 Mpn452 Mpn650 Mpn621 Mpn004
EMBL056	Mpn117	RplT	Mpn165 Mpn423 Mpn390 Mpn436 Mpn515 Mpn034 Mpn191 Mpn658 Mpn189 Mpn228 Mpn332 Mpn167
EMBL057	Mpn390	PdhD	Mpn210 Mpn390
EMBL058	Mpn245	Gmk	Mpn434
EMBL059	Mpn419	AlaS	Mpn434 Mpn011 Mpn573 Mpn118 Mpn555 Mpn516 Mpn419 Mpn003 Mpn582
EMBL060	Mpn100	Mpn100	Mpn210 Mpn434 Mpn390 Mpn118 Mpn295 Mpn100 Mpn547 Mpn034 Mpn509 Mpn650 Mpn004
EMBL061	Mpn108	Mpn108	Mpn390 Mpn516 Mpn430 Mpn665 Mpn261 Mpn108 Mpn529 Mpn210 Mpn434 Mpn124 Mpn524 Mpn346 Mpn515 Mpn426 Mpn452 Mpn646 Mpn004 Mpn621
EMBL062	Mpn108	Mpn108	Mpn210 Mpn434 Mpn619 Mpn155 Mpn430 Mpn665 Mpn640 Mpn032 Mpn034 Mpn004 Mpn072
EMBL063	Mpn471	RpmG1	Mpn390 Mpn555 Mpn516 Mpn430 Mpn261 Mpn115 Mpn228 Mpn210 Mpn462 Mpn434 Mpn207 Mpn294 Mpn515 Mpn020 Mpn252 Mpn004 Mpn621
EMBL064	Mpn631	Tsf	Mpn631 Mpn434 Mpn295 Mpn665
EMBL065	Mpn521	Mpn521	Mpn573
EMBL066	Mpn023	MetG	Mpn025 Mpn619 Mpn390 Mpn303 Mpn118 Mpn516 Mpn430 Mpn401 Mpn315 Mpn665 Mpn261 Mpn354 Mpn638 Mpn393 Mpn023 Mpn141 Mpn061 Mpn210 Mpn434 Mpn322 Mpn007 Mpn280 Mpn324 Mpn515 Mpn032 Mpn426 Mpn347 Mpn191 Mpn106 Mpn621 Mpn004
EMBL067	Mpn001	DnaN	Mpn434 Mpn001
EMBL068	Mpn001	DnaN	Mpn434 Mpn001 Mpn516 Mpn621
EMBL069	Mpn064	DeoA	Mpn210 Mpn033 Mpn434 Mpn064
EMBL070	Mpn029	Efp	Mpn555 Mpn516 Mpn430 Mpn295 Mpn665 Mpn261 Mpn321 Mpn393 Mpn258 Mpn029 Mpn586 Mpn210 Mpn322 Mpn434 Mpn573 Mpn280 Mpn207 Mpn314 Mpn515 Mpn621 Mpn004
EMBL071	Mpn122	ParE	Mpn402 Mpn390 Mpn516 Mpn665 Mpn436 Mpn006 Mpn060 Mpn351 Mpn122 Mpn141 Mpn210 Mpn434 Mpn424 Mpn674 Mpn426 Mpn123 Mpn392 Mpn050 Mpn266 Mpn191 Mpn452 Mpn252 Mpn387 Mpn621 Mpn282
EMBL072	Mpn122	ParE	Mpn062 Mpn516 Mpn430 Mpn665 Mpn262 Mpn044 Mpn034 Mpn658 Mpn122 Mpn210 Mpn434 Mpn573 Mpn515 Mpn050 Mpn549 Mpn191 Mpn020 Mpn345 Mpn621 Mpn004
EMBL073	Mpn563	Mpn563	Mpn563 Mpn210 Mpn434 Mpn486 Mpn573 Mpn390 Mpn251 Mpn665 Mpn006 Mpn125 Mpn034 Mpn658 Mpn650 Mpn399
EMBL074	Mpn165	RplC	Mpn390 Mpn430 Mpn261 Mpn228 Mpn141 Mpn165 Mpn210 Mpn475 Mpn434 Mpn166 Mpn226 Mpn487 Mpn207 Mpn515 Mpn426 Mpn050 Mpn191 Mpn020 Mpn252 Mpn413 Mpn452 Mpn621 Mpn004
EMBL075	Mpn251	Rpe	Mpn434 Mpn390 Mpn555 Mpn315 Mpn251 Mpn674 Mpn665 Mpn261 Mpn640 Mpn034 Mpn040 Mpn660
EMBL076	Mpn223	HprK	Mpn341 Mpn434 Mpn573 Mpn044 Mpn555 Mpn658 Mpn223
EMBL077	Mpn067	Mpn067	Mpn210 Mpn434 Mpn067 Mpn550 Mpn504 Mpn516 Mpn248 Mpn261
EMBL078	Mpn067	Mpn067	Mpn210 Mpn434 Mpn595 Mpn545 Mpn516 Mpn003 Mpn261 Mpn008 Mpn515 Mpn426 Mpn067 Mpn452 Mpn678

EMBL079	Mpn067	Mpn067	Mpn430 Mpn315 Mpn665 Mpn261 Mpn258 Mpn210 Mpn322 Mpn434 Mpn573 Mpn324 Mpn426 Mpn472 Mpn067 Mpn050 Mpn191 Mpn020
EMBL080	Mpn392	PdhB	Mpn434 Mpn392 Mpn306 Mpn393
EMBL081	Mpn266	Mpn266	Mpn352 Mpn516 Mpn665 Mpn261 Mpn247 Mpn228 Mpn210 Mpn669 Mpn434 Mpn280 Mpn515 Mpn673 Mpn266 Mpn191 Mpn020 Mpn004 Mpn621
EMBL082	Mpn266	Mpn266	Mpn434 Mpn226 Mpn352 Mpn516 Mpn665 Mpn515 Mpn266 Mpn020 Mpn191 Mpn658 Mpn509 Mpn621 Mpn004
EMBL083	Mpn427	Mpn427	Mpn210 Mpn434 Mpn573 Mpn516 Mpn419 Mpn003 Mpn034 Mpn427 Mpn004
EMBL084	Mpn020	Mpn020	Mpn025 Mpn062 Mpn352 Mpn516 Mpn394 Mpn354 Mpn386 Mpn258 Mpn141 Mpn669 Mpn553 Mpn322 Mpn573 Mpn598 Mpn515 Mpn392 Mpn020 Mpn191 Mpn106 Mpn153 Mpn621 Mpn004 Mpn014 Mpn303 Mpn315 Mpn430 Mpn520 Mpn665 Mpn261 Mpn393 Mpn034 Mpn061 Mpn210 Mpn434 Mpn280 Mpn324 Mpn688
EMBL085	Mpn020	Mpn020	Mpn390 Mpn555 Mpn352 Mpn264 Mpn516 Mpn003 Mpn665 Mpn394 Mpn261 Mpn571 Mpn228 Mpn476 Mpn072 Mpn210 Mpn429 Mpn434 Mpn226 Mpn592 Mpn674 Mpn515 Mpn032 Mpn314 Mpn673 Mpn191 Mpn020 Mpn509 Mpn004
EMBL086	Mpn191	RpoA	Mpn437 Mpn352 Mpn191 Mpn516 Mpn515
EMBL087	Mpn191	RpoA	Mpn118 Mpn352 Mpn516 Mpn443 Mpn171 Mpn228 Mpn210 Mpn265 Mpn434 Mpn226 Mpn640 Mpn362 Mpn515 Mpn673 Mpn239 Mpn191 Mpn020 Mpn621 Mpn004
EMBL088	Mpn106	PheT	Mpn210 Mpn434 Mpn322 Mpn516 Mpn295 Mpn665 Mpn261 Mpn034 Mpn106 Mpn141
EMBL089	Mpn106	PheT	Mpn269 Mpn390 Mpn470 Mpn516 Mpn430 Mpn295 Mpn665 Mpn394 Mpn261 Mpn141 Mpn165 Mpn434 Mpn573 Mpn324 Mpn515 Mpn608 Mpn191 Mpn466 Mpn106 Mpn621
EMBL090	Mpn225	RpsL	Mpn390 Mpn516 Mpn430 Mpn100 Mpn665 Mpn034 Mpn228 Mpn277 Mpn210 Mpn434 Mpn280 Mpn515 Mpn173 Mpn050 Mpn191 Mpn040 Mpn345 Mpn452 Mpn621 Mpn004
EMBL091	Mpn397	SpoT	Mpn397 Mpn062 Mpn390 Mpn142 Mpn555 Mpn430 Mpn665 Mpn171 Mpn297 Mpn354 Mpn441 Mpn021 Mpn434 Mpn573 Mpn559 Mpn538 Mpn546 Mpn547 Mpn079
EMBL092	Mpn300	ScpA	Mpn300 Mpn210 Mpn434 Mpn423 Mpn573 Mpn390 Mpn211 Mpn665 Mpn551 Mpn032 Mpn301 Mpn426 Mpn508 Mpn040
EMBL093	Mpn231	RplI	Mpn231 Mpn390 Mpn303 Mpn118 Mpn516 Mpn430 Mpn154 Mpn665 Mpn261 Mpn365 Mpn658 Mpn574 Mpn228 Mpn210 Mpn434 Mpn063 Mpn515 Mpn426 Mpn173 Mpn050 Mpn020 Mpn452 Mpn621
EMBL094	Mpn231	RplI	Mpn231 Mpn352 Mpn516 Mpn665 Mpn365 Mpn034 Mpn658 Mpn574 Mpn210 Mpn434 Mpn166 Mpn515 Mpn640 Mpn020 Mpn252 Mpn452 Mpn004
EMBL095	Mpn303	Pyk	Mpn303 Mpn191 Mpn430
EMBL096	Mpn118	Mpn118	Mpn563 Mpn298 Mpn269 Mpn390 Mpn155 Mpn124 Mpn430 Mpn346 Mpn041 Mpn050 Mpn331 Mpn400
EMBL097	Mpn470	PepP	Mpn210 Mpn434 Mpn470 Mpn004
EMBL098	Mpn381	Mpn381	Mpn434 Mpn573 Mpn656 Mpn324 Mpn034 Mpn381 Mpn665
EMBL099	Mpn295	Mpn295	Mpn434
EMBL100	Mpn120	GrpE	Mpn210 Mpn434 Mpn390 Mpn430 Mpn295 Mpn120 Mpn032 Mpn452 Mpn621 Mpn004
EMBL101	Mpn171	RpsC	Mpn558 Mpn390 Mpn555 Mpn516 Mpn430 Mpn315 Mpn665 Mpn261 Mpn309 Mpn297 Mpn169 Mpn034 Mpn574 Mpn228 Mpn165 Mpn210 Mpn265 Mpn434 Mpn166 Mpn226 Mpn573 Mpn280 Mpn380 Mpn008 Mpn515 Mpn426 Mpn191 Mpn020 Mpn621 Mpn004
EMBL102	Mpn330	Mpn330	Mpn303 Mpn352 Mpn516 Mpn315 Mpn665 Mpn330 Mpn115 Mpn258 Mpn658 Mpn574 Mpn228 Mpn141 Mpn660 Mpn210 Mpn434 Mpn226 Mpn063 Mpn622 Mpn280 Mpn380 Mpn515 Mpn032 Mpn294 Mpn426 Mpn445 Mpn050 Mpn191 Mpn020 Mpn252 Mpn621 Mpn004
EMBL103	Mpn677	Mpn677	Mpn434 Mpn677 Mpn573 Mpn034 Mpn658 Mpn223 Mpn665
EMBL104	Mpn351	Mpn351	Mpn351 Mpn034
EMBL105	Mpn009	Mpn009	Mpn619 Mpn390 Mpn516 Mpn430 Mpn665 Mpn261 Mpn141 Mpn210 Mpn434 Mpn573 Mpn310 Mpn380 Mpn515 Mpn426 Mpn191 Mpn452 Mpn004

EMBL106	Mpn009	Mpn009	Mpn619 Mpn390 Mpn516 Mpn430 Mpn665 Mpn261 Mpn009 Mpn141 Mpn210 Mpn434 Mpn573 Mpn515 Mpn426 Mpn050 Mpn191 Mpn452 Mpn004
EMBL107	Mpn365	Mpn365	Mpn210 Mpn434 Mpn615 Mpn343 Mpn365 Mpn040 Mpn034 Mpn665
EMBL108	Mpn278	GlF	Mpn210 Mpn434 Mpn573 Mpn555 Mpn324 Mpn430 Mpn665 Mpn314 Mpn050 Mpn278 Mpn542 Mpn228 Mpn621 Mpn004
EMBL109	Mpn061	Ffh	Mpn390 Mpn516 Mpn665 Mpn261 Mpn600 Mpn258 Mpn228 Mpn210 Mpn434 Mpn226 Mpn280 Mpn032 Mpn515 Mpn191 Mpn004 Mpn621
EMBL110	Mpn618	DnaX	Mpn684 Mpn516 Mpn430 Mpn665 Mpn465 Mpn169 Mpn685 Mpn258 Mpn658 Mpn141 Mpn618 Mpn322 Mpn434 Mpn573 Mpn280 Mpn515 Mpn032 Mpn317 Mpn426 Mpn514 Mpn191 Mpn020 Mpn252 Mpn452 Mpn153
EMBL111	Mpn429	Pgk	Mpn210 Mpn434 Mpn429
EMBL112	Mpn185	Adk	Mpn210 Mpn434 Mpn185 Mpn034
EMBL113	Mpn595	RpiB	Mpn269 Mpn595 Mpn295
EMBL114	Mpn324	NrdE	Mpn434 Mpn322 Mpn573 Mpn390 Mpn555 Mpn324 Mpn516 Mpn430 Mpn003 Mpn665 Mpn515 Mpn050 Mpn222 Mpn621
EMBL115	Mpn007	Mpn007	Mpn618 Mpn434 Mpn573 Mpn007 Mpn665 Mpn032 Mpn351 Mpn034 Mpn040 Mpn629
EMBL116	Mpn007	Mpn007	Mpn618 Mpn210 Mpn434 Mpn280 Mpn007 Mpn516 Mpn430 Mpn665 Mpn515 Mpn258 Mpn621
EMBL117	Mpn007	Mpn007	Mpn303 Mpn516 Mpn430 Mpn665 Mpn171 Mpn440 Mpn246 Mpn258 Mpn034 Mpn148 Mpn618 Mpn556 Mpn210 Mpn434 Mpn011 Mpn573 Mpn007 Mpn140 Mpn207 Mpn598 Mpn515 Mpn392 Mpn311 Mpn466 Mpn020 Mpn450 Mpn452 Mpn153 Mpn621 Mpn004
EMBL118	Mpn545	Rnc	Mpn434
EMBL119	Mpn257	GalE	Mpn210 Mpn497 Mpn434 Mpn089 Mpn269 Mpn390 Mpn118 Mpn656 Mpn516 Mpn257 Mpn207 Mpn034 Mpn040 Mpn621 Mpn004
EMBL120	Mpn329	Mpn329	Mpn099 Mpn390 Mpn303 Mpn516 Mpn430 Mpn665 Mpn261 Mpn440 Mpn609 Mpn258 Mpn658 Mpn228 Mpn210 Mpn434 Mpn573 Mpn324 Mpn207 Mpn329 Mpn515 Mpn314 Mpn050 Mpn191 Mpn020 Mpn452 Mpn252 Mpn621 Mpn004
EMBL121	Mpn317	FtsZ	Mpn062 Mpn390 Mpn516 Mpn430 Mpn315 Mpn665 Mpn261 Mpn258 Mpn141 Mpn228 Mpn210 Mpn434 Mpn322 Mpn573 Mpn066 Mpn515 Mpn032 Mpn317 Mpn426 Mpn546 Mpn191 Mpn020 Mpn621 Mpn004
EMBL122	Mpn317	FtsZ	Mpn062 Mpn118 Mpn555 Mpn516 Mpn430 Mpn003 Mpn665 Mpn361 Mpn685 Mpn258 Mpn210 Mpn434 Mpn071 Mpn573 Mpn008 Mpn032 Mpn515 Mpn317 Mpn426 Mpn273 Mpn020
EMBL123	Mpn483	Mpn483	Mpn210 Mpn434 Mpn573 Mpn430 Mpn665 Mpn515 Mpn483 Mpn609 Mpn050 Mpn258 Mpn141 Mpn004
EMBL124	Mpn483	Mpn483	Mpn210 Mpn497 Mpn434 Mpn306 Mpn390 Mpn430 Mpn665 Mpn515 Mpn032 Mpn027 Mpn483 Mpn452
EMBL125	Mpn158	RibF	Mpn158 Mpn390 Mpn295 Mpn665
EMBL126	Mpn292	Mpn292	Mpn210 Mpn434 Mpn573 Mpn390 Mpn516 Mpn003 Mpn665 Mpn515 Mpn426 Mpn292 Mpn094 Mpn547 Mpn034 Mpn106 Mpn221 Mpn305 Mpn678
EMBL127	Mpn273	Mpn273	Mpn210 Mpn434 Mpn390 Mpn261 Mpn207 Mpn515 Mpn273 Mpn050 Mpn040 Mpn658 Mpn452 Mpn153 Mpn621 Mpn004
EMBL128	Mpn526	Mpn526	Mpn340 Mpn516 Mpn003 Mpn141 Mpn228 Mpn210 Mpn434 Mpn592 Mpn573 Mpn116 Mpn324 Mpn257 Mpn674 Mpn346 Mpn515 Mpn347 Mpn254 Mpn526 Mpn232 Mpn004
EMBL129	Mpn387	Mpn387	Mpn434 Mpn390 Mpn387 Mpn293
EMBL130	Mpn221	Pth	Mpn434 Mpn226 Mpn340 Mpn430 Mpn665 Mpn649 Mpn034 Mpn658 Mpn221 Mpn228 Mpn004 Mpn621
EMBL131	Mpn541	RpsT	Mpn390 Mpn516 Mpn430 Mpn574 Mpn228 Mpn434 Mpn226 Mpn063 Mpn280 Mpn207 Mpn032 Mpn515 Mpn239 Mpn191 Mpn020 Mpn452 Mpn621 Mpn004
EMBL132	Mpn089	Mpn089	Mpn210 Mpn497 Mpn434 Mpn089 Mpn573 Mpn003 Mpn674 Mpn665

EMBL133	Mpn269	Mpn269	Mpn210 Mpn434 Mpn684 Mpn269 Mpn573 Mpn207 Mpn114 Mpn341 Mpn262 Mpn050 Mpn547 Mpn258 Mpn241 Mpn004 Mpn277
EMBL134	Mpn568	Era	Mpn210 Mpn434 Mpn573 Mpn568 Mpn390 Mpn665 Mpn640 Mpn663 Mpn034
EMBL135	Mpn307	Mpn307	Mpn210 Mpn434 Mpn063 Mpn516 Mpn430 Mpn207 Mpn515 Mpn346 Mpn472 Mpn040 Mpn621 Mpn004
EMBL136	Mpn307	Mpn307	Mpn027 Mpn434 Mpn034 Mpn665
EMBL137	Mpn628	GpmI	Mpn218 Mpn210 Mpn390 Mpn516 Mpn430 Mpn628 Mpn027 Mpn350 Mpn191 Mpn599 Mpn252
EMBL138	Mpn378	DnaE	Mpn434 Mpn353 Mpn573 Mpn118 Mpn003 Mpn665 Mpn551 Mpn108 Mpn378 Mpn034 Mpn141
EMBL139	Mpn533	AckA	Mpn210 Mpn533 Mpn434 Mpn516
EMBL140	Mpn321	FoIA	Mpn210 Mpn434 Mpn573 Mpn390 Mpn555 Mpn516 Mpn430 Mpn457 Mpn207 Mpn321 Mpn050 Mpn034 Mpn252 Mpn621 Mpn004
EMBL141	Mpn361	PrfA	Mpn210 Mpn434 Mpn361 Mpn034 Mpn419 Mpn003
EMBL142	Mpn499	Mpn499	Mpn210 Mpn645 Mpn434 Mpn674 Mpn665 Mpn314 Mpn200 Mpn392 Mpn514 Mpn499 Mpn034 Mpn367
EMBL143	Mpn187	InfA	Mpn099 Mpn352 Mpn516 Mpn665 Mpn536 Mpn171 Mpn416 Mpn685 Mpn034 Mpn187 Mpn660 Mpn078 Mpn210 Mpn429 Mpn434 Mpn226 Mpn515 Mpn559 Mpn191 Mpn189 Mpn252 Mpn621 Mpn004
EMBL144	Mpn155	InfB	Mpn179 Mpn042 Mpn516 Mpn171 Mpn685 Mpn571 Mpn115 Mpn034 Mpn658 Mpn228 Mpn226 Mpn155 Mpn515 Mpn616 Mpn191 Mpn189 Mpn621 Mpn167 Mpn004
EMBL145	Mpn124	HrcA	Mpn210 Mpn434 Mpn573 Mpn390 Mpn124 Mpn516 Mpn515 Mpn452 Mpn621 Mpn004
EMBL146	Mpn229	Ssb	Mpn034 Mpn229 Mpn665
EMBL147	Mpn487	Csd	Mpn210 Mpn434 Mpn390 Mpn034 Mpn436 Mpn399
EMBL148	Mpn349	Mpn349	Mpn210 Mpn434 Mpn573 Mpn063 Mpn656 Mpn555 Mpn665 Mpn349 Mpn044 Mpn034
EMBL149	Mpn220	RplA	Mpn390 Mpn516 Mpn430 Mpn315 Mpn103 Mpn665 Mpn261 Mpn115 Mpn258 Mpn034 Mpn658 Mpn574 Mpn228 Mpn210 Mpn434 Mpn226 Mpn063 Mpn357 Mpn280 Mpn310 Mpn515 Mpn294 Mpn426 Mpn050 Mpn191 Mpn020 Mpn452 Mpn621 Mpn004
EMBL150	Mpn081	Mpn081	Mpn434 Mpn573 Mpn295 Mpn081 Mpn665 Mpn515 Mpn505
EMBL151	Mpn446	RpsD	Mpn390 Mpn118 Mpn295 Mpn261 Mpn456 Mpn258 Mpn228 Mpn210 Mpn434 Mpn524 Mpn487 Mpn294 Mpn426 Mpn222 Mpn050 Mpn020 Mpn252 Mpn621 Mpn004
EMBL152	Mpn353	DnaG	Mpn353 Mpn142 Mpn118 Mpn555 Mpn516 Mpn003 Mpn665 Mpn551 Mpn378 Mpn034 Mpn258 Mpn141 Mpn434 Mpn083 Mpn686 Mpn040 Mpn582
EMBL153	Mpn340	Mpn340	Mpn005 Mpn521 Mpn340 Mpn034 Mpn674 Mpn108
EMBL154	Mpn042	Mpn042	Mpn390 Mpn042 Mpn516 Mpn430 Mpn315 Mpn665 Mpn261 Mpn354 Mpn115 Mpn258 Mpn574 Mpn046 Mpn228 Mpn210 Mpn434 Mpn063 Mpn207 Mpn515 Mpn294 Mpn426 Mpn445 Mpn050 Mpn040 Mpn252 Mpn621 Mpn004
EMBL155	Mpn126	Mpn126	Mpn390 Mpn303 Mpn126 Mpn516 Mpn430 Mpn665 Mpn394 Mpn365 Mpn034 Mpn658 Mpn574 Mpn019 Mpn210 Mpn434 Mpn573 Mpn398 Mpn515 Mpn050 Mpn040 Mpn452 Mpn621 Mpn004
EMBL156	Mpn126	Mpn126	Mpn210 Mpn434 Mpn269 Mpn573 Mpn390 Mpn126 Mpn516 Mpn665 Mpn580 Mpn346 Mpn034
EMBL157	Mpn430	GapA	Mpn390 Mpn516 Mpn430 Mpn665 Mpn379 Mpn258 Mpn019 Mpn141 Mpn210 Mpn434 Mpn280 Mpn207 Mpn515 Mpn426 Mpn546 Mpn050 Mpn020 Mpn252 Mpn621 Mpn004
EMBL158	Mpn430	GapA	Mpn210 Mpn434 Mpn619 Mpn390 Mpn555 Mpn516 Mpn430 Mpn683 Mpn426 Mpn361 Mpn542 Mpn034 Mpn452 Mpn072
EMBL159	Mpn401	GreA	Mpn210 Mpn434 Mpn063 Mpn352 Mpn516 Mpn401 Mpn665 Mpn515 Mpn673 Mpn321 Mpn350 Mpn191 Mpn621 Mpn004

EMBL160	Mpn003	GyrB	Mpn210 Mpn434 Mpn486 Mpn516 Mpn003 Mpn020 Mpn004 Mpn621
EMBL161	Mpn003	GyrB	Mpn210 Mpn434 Mpn486 Mpn573 Mpn390 Mpn280 Mpn003 Mpn665 Mpn426 Mpn677 Mpn020 Mpn258 Mpn004
EMBL162	Mpn665	Tuf	Mpn434 Mpn137 Mpn658 Mpn665
EMBL163	Mpn177	RplE	Mpn623 Mpn434 Mpn573 Mpn665
EMBL164	Mpn033	Upp	Mpn210 Mpn133 Mpn434 Mpn155 Mpn295 Mpn665 Mpn346 Mpn635 Mpn016 Mpn072
EMBL165	Mpn033	Upp	Mpn390 Mpn516 Mpn430 Mpn315 Mpn295 Mpn665 Mpn261 Mpn456 Mpn034 Mpn133 Mpn210 Mpn434 Mpn155 Mpn515 Mpn346 Mpn426 Mpn635 Mpn050 Mpn191 Mpn020 Mpn016 Mpn621 Mpn004
EMBL166	Mpn297	Mpn297	Mpn297 Mpn210 Mpn434 Mpn042 Mpn034
EMBL167	Mpn606	Eno	Mpn210 Mpn612 Mpn434 Mpn390 Mpn118 Mpn516 Mpn003 Mpn606 Mpn629 Mpn316 Mpn452 Mpn569
EMBL168	Mpn235	Ung	Mpn210 Mpn434 Mpn390 Mpn014 Mpn674 Mpn515 Mpn222 Mpn235 Mpn034 Mpn678
EMBL169	Mpn494	UlaC	Mpn390 Mpn555 Mpn352 Mpn516 Mpn430 Mpn665 Mpn261 Mpn494 Mpn210 Mpn434 Mpn314 Mpn515 Mpn426 Mpn050 Mpn191 Mpn452 Mpn621 Mpn004
EMBL170	Mpn574	GroS	Mpn434 Mpn226 Mpn535 Mpn352 Mpn280 Mpn516 Mpn024 Mpn665 Mpn515 Mpn191 Mpn658 Mpn228 Mpn621
EMBL171	Mpn653	MtlF	Mpn210 Mpn434 Mpn042 Mpn516 Mpn295 Mpn665 Mpn551 Mpn653
EMBL172	Mpn544	Mpn544	Mpn210 Mpn544 Mpn434 Mpn361 Mpn390 Mpn295 Mpn003 Mpn665
EMBL173	Mpn539	RplL	Mpn480 Mpn516 Mpn430 Mpn003 Mpn665 Mpn551 Mpn261 Mpn065 Mpn658 Mpn210 Mpn434 Mpn643 Mpn207 Mpn515 Mpn426 Mpn073 Mpn191 Mpn020 Mpn621 Mpn004
EMBL174	Mpn066	ManB	Mpn434 Mpn118 Mpn066 Mpn042 Mpn295 Mpn674 Mpn564 Mpn394 Mpn048 Mpn365
EMBL175	Mpn280	Mpn280	Mpn430 Mpn103 Mpn665 Mpn261 Mpn562 Mpn278 Mpn228 Mpn434 Mpn573 Mpn280 Mpn239 Mpn549 Mpn191 Mpn686 Mpn345 Mpn621
EMBL176	Mpn492	UlaE	Mpn210 Mpn434 Mpn573 Mpn390 Mpn283 Mpn665 Mpn436 Mpn492 Mpn034 Mpn277
EMBL177	Mpn602	AtpF	Mpn665 Mpn261 Mpn600 Mpn258 Mpn072 Mpn210 Mpn434 Mpn207 Mpn515 Mpn598 Mpn426 Mpn020 Mpn452 Mpn153 Mpn004 Mpn621
EMBL178	Mpn073	Prs	Mpn210 Mpn434 Mpn573 Mpn404 Mpn674 Mpn515 Mpn073 Mpn034 Mpn004
EMBL179	Mpn050	GlpK	Mpn210 Mpn434 Mpn322 Mpn324 Mpn516 Mpn419 Mpn295 Mpn665 Mpn261 Mpn050 Mpn266
EMBL180	Mpn050	GlpK	Mpn062 Mpn269 Mpn390 Mpn555 Mpn516 Mpn430 Mpn665 Mpn141 Mpn210 Mpn434 Mpn032 Mpn515 Mpn426 Mpn483 Mpn050 Mpn020 Mpn452 Mpn079
EMBL181	Mpn107	Mpn107	Mpn390 Mpn555 Mpn516 Mpn665 Mpn261 Mpn379 Mpn258 Mpn556 Mpn210 Mpn434 Mpn573 Mpn207 Mpn032 Mpn515 Mpn426 Mpn444 Mpn107 Mpn050 Mpn608 Mpn020 Mpn452 Mpn621 Mpn004
EMBL182	Mpn107	Mpn107	Mpn434 Mpn573 Mpn118 Mpn555 Mpn003 Mpn665 Mpn032 Mpn426 Mpn107 Mpn141
EMBL183	Mpn222	TilS	Mpn623 Mpn434 Mpn222 Mpn516 Mpn430 Mpn665 Mpn621 Mpn515 Mpn004
EMBL184	Mpn323	NrdI	Mpn407 Mpn390 Mpn118 Mpn516 Mpn430 Mpn261 Mpn034 Mpn072 Mpn210 Mpn322 Mpn434 Mpn573 Mpn515 Mpn323 Mpn050 Mpn452 Mpn621 Mpn004
EMBL185	Mpn395	Apt	Mpn434 Mpn514 Mpn388 Mpn042 Mpn001 Mpn395 Mpn367 Mpn295 Mpn195
EMBL186	Mpn652	MtlD	Mpn210 Mpn434 Mpn573 Mpn516 Mpn515 Mpn032 Mpn426 Mpn191 Mpn652 Mpn228 Mpn004

EMBL187	Mpn383	Mpn383	Mpn210 Mpn434 Mpn656 Mpn674 Mpn665 Mpn314 Mpn533 Mpn034 Mpn383
EMBL188	Mpn167	RplW	Mpn390 Mpn516 Mpn665 Mpn261 Mpn428 Mpn115 Mpn034 Mpn658 Mpn574 Mpn228 Mpn399 Mpn165 Mpn210 Mpn434 Mpn226 Mpn063 Mpn515 Mpn426 Mpn191 Mpn020 Mpn621 Mpn167 Mpn004
EMBL189	Mpn105	PheS	Mpn210 Mpn105 Mpn434 Mpn322 Mpn062 Mpn573 Mpn568 Mpn516 Mpn665 Mpn515 Mpn472 Mpn426 Mpn247 Mpn020 Mpn106 Mpn141
EMBL190	Mpn105	PheS	Mpn105 Mpn434 Mpn573 Mpn555 Mpn516 Mpn106 Mpn329 Mpn141 Mpn515
EMBL191	Mpn504	Mpn504	Mpn434 Mpn555 Mpn331 Mpn295 Mpn674 Mpn665
EMBL192	Mpn180	RplF	Mpn434 Mpn180 Mpn658
EMBL193	Mpn555	Mpn555	Mpn390 Mpn555 Mpn516 Mpn665 Mpn261 Mpn141 Mpn210 Mpn434 Mpn324 Mpn515 Mpn426 Mpn020 Mpn452 Mpn153 Mpn652 Mpn004
EMBL194	Mpn555	Mpn555	Mpn210 Mpn434 Mpn555 Mpn324 Mpn665 Mpn261 Mpn072
EMBL195	Mpn531	ClpB	Mpn434 Mpn686 Mpn531 Mpn280 Mpn258 Mpn516 Mpn621
EMBL196	Mpn531	ClpB	Mpn531 Mpn352 Mpn516 Mpn665 Mpn182 Mpn171 Mpn571 Mpn228 Mpn434 Mpn226 Mpn515 Mpn616 Mpn640 Mpn673 Mpn239 Mpn191 Mpn020 Mpn189
EMBL197	Mpn283	Mpn283	Mpn470 Mpn352 Mpn283 Mpn516 Mpn430 Mpn315 Mpn665 Mpn394 Mpn261 Mpn354 Mpn258 Mpn556 Mpn210 Mpn434 Mpn007 Mpn380 Mpn515 Mpn050 Mpn191 Mpn020 Mpn621 Mpn004
EMBL198	Mpn283	Mpn283	Mpn210 Mpn434 Mpn118 Mpn643 Mpn283 Mpn216 Mpn003 Mpn665
EMBL199	Mpn517	Mpn517	Mpn434 Mpn226 Mpn517
EMBL200	Mpn551	Mpn551	Mpn210 Mpn434 Mpn573 Mpn295 Mpn003 Mpn551 Mpn683 Mpn059 Mpn034 Mpn452
EMBL201	Mpn562	NadE	Mpn426 Mpn210 Mpn562 Mpn434 Mpn573 Mpn516 Mpn515 Mpn097
EMBL202	Mpn247	Mpn247	Mpn210 Mpn434 Mpn429 Mpn619 Mpn390 Mpn665 Mpn426 Mpn247 Mpn094 Mpn461 Mpn004
EMBL203	Mpn169	RpsS	Mpn269 Mpn390 Mpn516 Mpn430 Mpn665 Mpn261 Mpn171 Mpn363 Mpn341 Mpn571 Mpn115 Mpn228 Mpn210 Mpn434 Mpn226 Mpn280 Mpn515 Mpn426 Mpn191 Mpn020 Mpn452 Mpn621 Mpn004
EMBL204	Mpn658	RplS	Mpn573 Mpn571 Mpn034 Mpn665 Mpn536
EMBL205	Mpn658	RplS	Mpn568 Mpn516 Mpn034 Mpn658 Mpn574 Mpn228 Mpn078 Mpn165 Mpn210 Mpn434 Mpn226 Mpn515 Mpn292 Mpn222 Mpn020 Mpn547 Mpn252 Mpn189 Mpn004
EMBL206	Mpn022	Pip	Mpn062 Mpn619 Mpn303 Mpn516 Mpn430 Mpn665 Mpn261 Mpn321 Mpn393 Mpn023 Mpn365 Mpn034 Mpn022 Mpn141 Mpn210 Mpn322 Mpn573 Mpn388 Mpn324 Mpn310 Mpn598 Mpn515 Mpn472 Mpn426 Mpn050 Mpn191 Mpn452 Mpn106 Mpn621 Mpn004
EMBL207	Mpn556	ArgS	Mpn556 Mpn361 Mpn073 Mpn665
EMBL208	Mpn265	TrpS	Mpn300 Mpn390 Mpn555 Mpn516 Mpn430 Mpn665 Mpn320 Mpn210 Mpn265 Mpn322 Mpn434 Mpn573 Mpn207 Mpn644 Mpn515 Mpn191 Mpn020 Mpn452 Mpn621 Mpn004
EMBL209	Mpn265	TrpS	Mpn265 Mpn434 Mpn573 Mpn118 Mpn003 Mpn665
EMBL210	Mpn063	DeoC	Mpn210 Mpn434 Mpn063 Mpn047 Mpn555 Mpn516 Mpn430 Mpn665 Mpn515 Mpn426 Mpn392 Mpn191 Mpn658 Mpn621 Mpn004
EMBL211	Mpn656	Mpn656	Mpn516 Mpn665 Mpn536 Mpn034 Mpn658 Mpn210 Mpn434 Mpn573 Mpn674 Mpn515 Mpn426 Mpn392 Mpn050 Mpn663 Mpn452 Mpn243 Mpn004 Mpn621
EMBL212	Mpn557	GidA	Mpn210 Mpn434 Mpn573 Mpn390 Mpn155 Mpn118 Mpn557 Mpn042 Mpn516 Mpn295 Mpn003 Mpn665 Mpn515 Mpn621
EMBL213	Mpn219	RplK	Mpn555 Mpn516 Mpn430 Mpn665 Mpn394 Mpn261 Mpn606 Mpn533 Mpn574 Mpn141 Mpn228 Mpn078 Mpn556 Mpn210 Mpn389 Mpn289 Mpn434 Mpn573 Mpn324 Mpn674 Mpn668 Mpn314 Mpn426 Mpn020 Mpn479 Mpn621 Mpn004

EMBL214	Mpn598	AtpD	Mpn602 Mpn516 Mpn598
EMBL215	Mpn032	Mpn032	Mpn556 Mpn434 Mpn573 Mpn555 Mpn516 Mpn665 Mpn515 Mpn032 Mpn191 Mpn574 Mpn621
EMBL216	Mpn314	MraZ	Mpn210 Mpn434 Mpn390 Mpn516 Mpn430 Mpn339 Mpn314 Mpn032 Mpn521 Mpn034 Mpn040 Mpn633 Mpn621 Mpn004
EMBL217	Mpn109	Mpn109	Mpn596 Mpn390 Mpn516 Mpn430 Mpn665 Mpn261 Mpn309 Mpn169 Mpn115 Mpn658 Mpn574 Mpn210 Mpn434 Mpn550 Mpn578 Mpn032 Mpn314 Mpn515 Mpn426 Mpn444 Mpn635 Mpn109 Mpn073 Mpn050 Mpn191 Mpn020 Mpn621 Mpn004
EMBL218	Mpn109	Mpn109	Mpn210 Mpn434 Mpn118 Mpn109 Mpn034 Mpn516
EMBL219	Mpn360	RpmE	Mpn390 Mpn531 Mpn516 Mpn430 Mpn401 Mpn307 Mpn261 Mpn177 Mpn297 Mpn365 Mpn658 Mpn228 Mpn669 Mpn210 Mpn434 Mpn590 Mpn226 Mpn063 Mpn280 Mpn207 Mpn329 Mpn515 Mpn426 Mpn050 Mpn020 Mpn452 Mpn252 Mpn621
EMBL220	Mpn332	Lon	Mpn269 Mpn390 Mpn555 Mpn516 Mpn443 Mpn430 Mpn665 Mpn606 Mpn210 Mpn434 Mpn207 Mpn515 Mpn040 Mpn452 Mpn332 Mpn004 Mpn621
EMBL221	Mpn332	Lon	Mpn210 Mpn434 Mpn390 Mpn034 Mpn665
EMBL222	Mpn560	Mpn560	Mpn560 Mpn434 Mpn573 Mpn555
EMBL223	Mpn480	ValS	Mpn480 Mpn118 Mpn516 Mpn295 Mpn003 Mpn665 Mpn551 Mpn034 Mpn228 Mpn210 Mpn434 Mpn226 Mpn573 Mpn032 Mpn515 Mpn454 Mpn004
EMBL224	Mpn342	Mpn342	Mpn210 Mpn089 Mpn507 Mpn390 Mpn342 Mpn295 Mpn003 Mpn551 Mpn361 Mpn034
EMBL225	Mpn182	RpsE	Mpn516 Mpn295 Mpn182 Mpn171 Mpn518 Mpn571 Mpn441 Mpn228 Mpn210 Mpn434 Mpn226 Mpn573 Mpn032 Mpn515 Mpn547 Mpn191 Mpn020 Mpn189 Mpn252 Mpn621 Mpn004
EMBL226	Mpn662	MsrB	Mpn434 Mpn555 Mpn537 Mpn516 Mpn665 Mpn662
EMBL227	Mpn540	RpmF	Mpn390 Mpn470 Mpn051 Mpn516 Mpn430 Mpn315 Mpn665 Mpn261 Mpn571 Mpn115 Mpn658 Mpn228 Mpn210 Mpn434 Mpn226 Mpn535 Mpn515 Mpn032 Mpn294 Mpn426 Mpn191 Mpn020 Mpn621 Mpn004
EMBL228	Mpn391	PdhC	Mpn391
EMBL229	Mpn060	MetK	Mpn048 Mpn434 Mpn060 Mpn034
EMBL230	Mpn508	Mpn508	Mpn426 Mpn210 Mpn434 Mpn034 Mpn516 Mpn452 Mpn430 Mpn665
EMBL231	Mpn034	PolC	Mpn336 Mpn210 Mpn434 Mpn390 Mpn324 Mpn034 Mpn516 Mpn261 Mpn207 Mpn004
EMBL232	Mpn034	PolC	Mpn210 Mpn612 Mpn434 Mpn573 Mpn280 Mpn516 Mpn008 Mpn145 Mpn620 Mpn677 Mpn514 Mpn526 Mpn034 Mpn088 Mpn621
EMBL233	Mpn034	PolC	Mpn218 Mpn269 Mpn352 Mpn516 Mpn609 Mpn258 Mpn228 Mpn141 Mpn553 Mpn598 Mpn515 Mpn358 Mpn238 Mpn250 Mpn392 Mpn611 Mpn452 Mpn572 Mpn153 Mpn004 Mpn671 Mpn303 Mpn470 Mpn197 Mpn295 Mpn315 Mpn430 Mpn261 Mpn665 Mpn393 Mpn034 Mpn210 Mpn372 Mpn434 Mpn207
EMBL234	Mpn034	PolC	Mpn645 Mpn390 Mpn516 Mpn430 Mpn665 Mpn261 Mpn034 Mpn258 Mpn210 Mpn434 Mpn573 Mpn280 Mpn207 Mpn515 Mpn426 Mpn546 Mpn050 Mpn490 Mpn191 Mpn020 Mpn621 Mpn004
EMBL235	Mpn331	Tig	Mpn210 Mpn165 Mpn434 Mpn209 Mpn516 Mpn003 Mpn665 Mpn158 Mpn331 Mpn452 Mpn228 Mpn004
EMBL236	Mpn302	PfkA	Mpn210 Mpn434 Mpn034 Mpn302 Mpn665 Mpn644 Mpn097
EMBL237	Mpn263	TrxA	Mpn321 Mpn034 Mpn452 Mpn263 Mpn314
EMBL238	Mpn188	RpmJ	Mpn434 Mpn390 Mpn665
EMBL239	Mpn072	Mpn072	Mpn210 Mpn434 Mpn042 Mpn516 Mpn295 Mpn177 Mpn515 Mpn059 Mpn191 Mpn599 Mpn004 Mpn072
EMBL240	Mpn226	RpsG	Mpn390 Mpn555 Mpn516 Mpn430 Mpn003 Mpn665 Mpn428 Mpn034 Mpn658 Mpn228 Mpn210 Mpn434 Mpn226 Mpn324 Mpn294 Mpn314 Mpn452 Mpn621 Mpn004

EMBL241	Mpn110	Mpn110	Mpn062 Mpn390 Mpn555 Mpn516 Mpn430 Mpn419 Mpn315 Mpn665 Mpn261 Mpn297 Mpn169 Mpn115 Mpn034 Mpn574 Mpn660 Mpn210 Mpn434 Mpn226 Mpn524 Mpn110 Mpn280 Mpn515 Mpn426 Mpn239 Mpn050 Mpn191 Mpn020 Mpn452 Mpn252 Mpn621 Mpn004
EMBL242	Mpn267	PpnK	Mpn210 Mpn434 Mpn573 Mpn280 Mpn264 Mpn516 Mpn267 Mpn515 Mpn050 Mpn621
EMBL243	Mpn674	Ldh	Mpn210 Mpn434 Mpn516 Mpn674 Mpn621
EMBL244	Mpn380	MutM	Mpn434 Mpn555 Mpn295 Mpn665 Mpn380
EMBL245	Mpn008	TrmE	Mpn390 Mpn516 Mpn430 Mpn665 Mpn261 Mpn258 Mpn228 Mpn210 Mpn322 Mpn434 Mpn553 Mpn573 Mpn008 Mpn515 Mpn317 Mpn050 Mpn191 Mpn547
EMBL246	Mpn008	TrmE	Mpn210 Mpn434 Mpn573 Mpn516 Mpn665 Mpn261 Mpn008 Mpn050
EMBL247	Mpn174	RpsQ	Mpn434 Mpn226 Mpn390 Mpn430 Mpn295 Mpn665 Mpn174 Mpn386 Mpn020 Mpn191 Mpn228
EMBL248	Mpn362	Mpn362	Mpn434 Mpn573 Mpn555 Mpn516 Mpn665 Mpn108 Mpn362
EMBL249	Mpn121	Mpn121	Mpn210 Mpn434 Mpn366 Mpn390 Mpn135 Mpn430 Mpn346 Mpn121 Mpn050 Mpn034
EMBL250	Mpn377	Mpn377	Mpn377 Mpn434 Mpn014 Mpn034
EMBL251	Mpn490	RecA	Mpn210 Mpn434 Mpn402 Mpn490 Mpn034 Mpn252 Mpn079 Mpn621
EMBL252	Mpn490	RecA	Mpn284 Mpn210 Mpn480 Mpn434 Mpn573 Mpn003 Mpn490 Mpn034 Mpn452
EMBL253	Mpn420	Mpn420	Mpn210 Mpn434 Mpn573 Mpn555 Mpn516 Mpn419 Mpn394 Mpn668 Mpn515 Mpn314 Mpn420 Mpn034
EMBL254	Mpn232	DnaB	Mpn210 Mpn434 Mpn390 Mpn665 Mpn415 Mpn207 Mpn232 Mpn258 Mpn141 Mpn241
EMBL255	Mpn232	DnaB	Mpn298 Mpn210 Mpn390 Mpn118 Mpn295 Mpn108 Mpn521 Mpn526 Mpn232 Mpn034 Mpn452
EMBL256	Mpn311	Mpn311	Mpn210 Mpn434 Mpn204 Mpn095 Mpn390 Mpn324 Mpn280 Mpn516 Mpn430 Mpn665 Mpn261 Mpn621 Mpn004
EMBL257	Mpn176	RplX	Mpn210 Mpn434 Mpn665 Mpn615 Mpn220 Mpn685 Mpn034 Mpn658 Mpn189 Mpn091 Mpn228 Mpn004 Mpn097
EMBL258	Mpn189	RpsM	Mpn434 Mpn629 Mpn034 Mpn228
EMBL259	Mpn316	Mpn316	Mpn390 Mpn555 Mpn516 Mpn430 Mpn665 Mpn261 Mpn440 Mpn258 Mpn141 Mpn556 Mpn210 Mpn553 Mpn434 Mpn573 Mpn192 Mpn636 Mpn329 Mpn515 Mpn546 Mpn549 Mpn050 Mpn191 Mpn252 Mpn316 Mpn452 Mpn621

ii) Proteins identified in the PeptideAtlas by FT-MS/MS	
Systematic name	Protein name
Mpn001	DnaN
Mpn002	Mpn002
Mpn003	GyrB
Mpn004	GyrA
Mpn005	SerS
Mpn006	Tmk
Mpn007	Mpn007
Mpn008	TrmE
Mpn009	Mpn009
Mpn011	Mpn011
Mpn012	Mpn012
Mpn013	Mpn013
Mpn015	Mpn015
Mpn016	Mpn016
Mpn017	FoID
Mpn018	Mpn018
Mpn019	Mpn019
Mpn020	Mpn020
Mpn021	DnaJ
Mpn022	Pip
Mpn023	MetG
Mpn024	RpoE
Mpn025	Fba
Mpn026	Mpn026
Mpn027	Mpn027
Mpn029	Efp
Mpn030	Mpn030
Mpn031	Mpn031
Mpn033	Upp
Mpn034	PolC
Mpn035	Mpn035
Mpn036	Mpn036
Mpn043	GlpF
Mpn044	Tdk
Mpn045	HisS
Mpn046	AspS
Mpn047	Mpn047
Mpn050	GlpK
Mpn051	Mpn051
Mpn052	Mpn052
Mpn053	PtsH
Mpn055	PotA
Mpn058	Mpn058
Mpn059	Gcp
Mpn060	MetK
Mpn061	Ffh
Mpn062	DeoD
Mpn063	DeoC
Mpn064	DeoA
Mpn065	Cdd
Mpn066	ManB
Mpn067	Mpn067
Mpn068	Mpn068
Mpn070	Mpn070
Mpn071	Mpn071
Mpn072	Mpn072
Mpn073	Prs
Mpn075	Mpn075
Mpn076	Mpn076
Mpn077	Mpn077
Mpn078	FruA
Mpn079	FruK
Mpn080	Mpn080
Mpn081	Mpn081
Mpn082	Tkt
Mpn083	Mpn083
Mpn084	Mpn084
Mpn090	Mpn090
Mpn099	Mpn099
Mpn105	PheS
Mpn106	PheT
Mpn109	Mpn109
Mpn115	InfC
Mpn116	RpmI
Mpn117	RplT
Mpn118	Mpn118
Mpn119	Mpn119
Mpn120	GrpE
Mpn121	Mpn121
Mpn122	ParE
Mpn123	ParC
Mpn124	HrcA
Mpn125	UvrC
Mpn126	Mpn126
Mpn131	Mpn131
Mpn132	Mpn132
Mpn133	Mpn133
Mpn134	Mpn134

Mpn135	Mpn135
Mpn136	Mpn136
Mpn140	Mpn140
Mpn141	MgpA
Mpn142	Mpn142
Mpn144	Mpn144
Mpn149	Mpn149
Mpn152	Mpn152
Mpn153	Mpn153
Mpn154	NusA
Mpn155	InfB
Mpn156	RbfA
Mpn157	Mpn157
Mpn158	RibF
Mpn159	Mpn159
Mpn160	Mpn160
Mpn161	Mpn161
Mpn162	Mpn162
Mpn163	Mpn163
Mpn164	RpsJ
Mpn165	RplC
Mpn166	RplD
Mpn167	RplW
Mpn168	RplB
Mpn169	RpsS
Mpn170	RplV
Mpn171	RpsC
Mpn172	RplP
Mpn173	RpmC
Mpn174	RpsQ
Mpn175	RplN
Mpn176	RplX
Mpn177	RplE
Mpn178	RpsZ
Mpn179	RpsH
Mpn180	RplF
Mpn181	RplR
Mpn182	RpsE
Mpn183	RplO
Mpn184	SecY
Mpn185	Adk
Mpn186	Map
Mpn187	InfA
Mpn188	RpmJ
Mpn189	RpsM
Mpn190	RpsK
Mpn191	RpoA
Mpn192	RplQ
Mpn193	CbiO1
Mpn194	CbiO2
Mpn195	Mpn195
Mpn196	TruA
Mpn197	PepF
Mpn199	Mpn199
Mpn200	Mpn200
Mpn202	Mpn202
Mpn205	Mpn205
Mpn207	PtsG
Mpn208	RpsB
Mpn209	PacL
Mpn210	SecA
Mpn211	UvrB
Mpn213	Mpn213
Mpn214	Mpn214
Mpn215	OppB
Mpn216	OppC
Mpn217	OppD
Mpn218	OppF
Mpn219	RplK
Mpn220	RplA
Mpn221	Pth
Mpn222	TilS
Mpn223	HprK
Mpn224	Lgt
Mpn225	RpsL
Mpn226	RpsG
Mpn227	FusA
Mpn228	RpsF
Mpn229	Ssb
Mpn230	RpsR
Mpn231	RplI
Mpn232	DnaB
Mpn233	Mpn233
Mpn235	Ung
Mpn236	Mpn236
Mpn237	GatA
Mpn238	GatB
Mpn239	Mpn239
Mpn240	TrxB
Mpn241	Mpn241

Mpn243	RnR
Mpn244	Mpn244
Mpn245	Gmk
Mpn246	Def
Mpn247	Mpn247
Mpn248	Mpn248
Mpn250	Pgi
Mpn251	Rpe
Mpn252	AsnS
Mpn255	Mpn255
Mpn256	Mpn256
Mpn257	GalE
Mpn258	Mpn258
Mpn259	Mpn259
Mpn260	Mpn260
Mpn261	TopA
Mpn262	Mpn262
Mpn263	TrxA
Mpn264	Mpn264
Mpn265	TrpS
Mpn266	Mpn266
Mpn267	PpnK
Mpn268	Mpn268
Mpn269	Mpn269
Mpn271	Mpn271
Mpn273	Mpn273
Mpn275	Mpn275
Mpn276	Mpn276
Mpn277	LysS
Mpn278	Glf
Mpn279	LepA
Mpn280	Mpn280
Mpn284	Mpn284
Mpn286	Mpn286
Mpn288	Mpn288
Mpn291	Mpn291
Mpn292	Mpn292
Mpn293	LspA
Mpn294	Mpn294
Mpn295	Mpn295
Mpn296	RpsU
Mpn297	Mpn297
Mpn298	AcpS
Mpn299	PlsC
Mpn300	ScpA
Mpn301	ScpB
Mpn302	PfkA
Mpn303	Pyk
Mpn307	Mpn307
Mpn308	Mpn308
Mpn309	P65
Mpn310	Hmw2
Mpn311	Mpn311
Mpn312	Mpn312
Mpn314	MraZ
Mpn315	MraW
Mpn316	Mpn316
Mpn317	FtsZ
Mpn318	Mpn318
Mpn319	Mpn319
Mpn320	ThyA
Mpn321	FolA
Mpn322	NrdF
Mpn323	NrdI
Mpn324	NrdE
Mpn325	RplU
Mpn326	Mpn326
Mpn327	RpmA
Mpn328	Nfo
Mpn329	Mpn329
Mpn330	Mpn330
Mpn331	Tig
Mpn332	Lon
Mpn336	Mpn336
Mpn337	Mpn337
Mpn338	Mpn338
Mpn339	Mpn339
Mpn340	Mpn340
Mpn342	Mpn342
Mpn343	Mpn343
Mpn346	Mpn346
Mpn349	Mpn349
Mpn352	RpoD
Mpn353	DnaG
Mpn354	GlyQS
Mpn355	Mpn355
Mpn356	CysS
Mpn357	LigA
Mpn358	Mpn358
Mpn359	Mpn359

Mpn360	RpmE
Mpn361	PrfA
Mpn362	Mpn362
Mpn366	Mpn366
Mpn370	Mpn370
Mpn372	Mpn372
Mpn376	Mpn376
Mpn378	DnaE
Mpn379	PolA
Mpn380	MutM
Mpn381	Mpn381
Mpn383	Mpn383
Mpn384	LeuS
Mpn386	Mpn386
Mpn387	Mpn387
Mpn389	LplA
Mpn390	PdhD
Mpn391	PdhC
Mpn392	PdhB
Mpn393	PdhA
Mpn394	Nox
Mpn395	Apt
Mpn396	Mpn396
Mpn397	SpoT
Mpn398	Mpn398
Mpn399	Mpn399
Mpn400	Mpn400
Mpn401	GreA
Mpn402	ProS
Mpn406	Mpn406
Mpn407	Mpn407
Mpn408	Mpn408
Mpn409	Mpn409
Mpn415	P37
Mpn419	AlaS
Mpn420	Mpn420
Mpn421	Mpn421
Mpn422	TrnU
Mpn423	Mpn423
Mpn425	FtsY
Mpn426	P115
Mpn427	Mpn427
Mpn428	Pta
Mpn429	Pgk
Mpn430	GapA
Mpn432	Mpn432
Mpn434	DnaK
Mpn436	Mpn436
Mpn438	Mpn438
Mpn443	Mpn443
Mpn444	Mpn444
Mpn445	Mpn445
Mpn446	RpsD
Mpn447	Hmw1
Mpn449	Mpn449
Mpn452	Hmw3
Mpn453	P30
Mpn454	Mpn454
Mpn456	Mpn456
Mpn459	Mpn459
Mpn460	Mpn460
Mpn461	Mpn461
Mpn469	Mpn469
Mpn470	PepP
Mpn471	RpmG1
Mpn472	Mpn472
Mpn473	Mpn473
Mpn474	Mpn474
Mpn475	EngA
Mpn476	Cmk
Mpn477	Mpn477
Mpn478	Mpn478
Mpn479	AzoR
Mpn480	ValS
Mpn481	EngB
Mpn483	Mpn483
Mpn487	Csd
Mpn488	Mpn488
Mpn489	Mpn489
Mpn490	RecA
Mpn491	Mpn491
Mpn492	UlaE
Mpn493	UlaD
Mpn494	UlaC
Mpn496	UlaA
Mpn497	Mpn497
Mpn499	Mpn499
Mpn500	Mpn500
Mpn505	Mpn505
Mpn506	Mpn506

Mpn509	Mpn509
Mpn512	Mpn512
Mpn514	Mpn514
Mpn515	RpoC
Mpn516	RpoB
Mpn517	Mpn517
Mpn518	Mpn518
Mpn519	Mpn519
Mpn520	IleS
Mpn521	Mpn521
Mpn522	TrmB
Mpn523	Mpn523
Mpn524	Mpn524
Mpn525	Mpn525
Mpn526	Mpn526
Mpn528	Ppa
Mpn529	Mpn529
Mpn530	Mpn530
Mpn531	ClpB
Mpn532	Mpn532
Mpn533	AckA
Mpn538	RplJ
Mpn539	RplL
Mpn540	RpmF
Mpn541	RpsT
Mpn542	Mpn542
Mpn543	Fmt
Mpn544	Mpn544
Mpn545	Rnc
Mpn546	PlsX
Mpn547	Mpn547
Mpn548	Mpn548
Mpn549	Mpn549
Mpn550	ThiI
Mpn551	Mpn551
Mpn552	Mpn552
Mpn553	ThrS
Mpn554	Mpn554
Mpn555	Mpn555
Mpn556	ArgS
Mpn557	GidA
Mpn558	GidB
Mpn560	Mpn560
Mpn561	Udk
Mpn562	NadE
Mpn563	Mpn563
Mpn564	Adh
Mpn566	Mpn566
Mpn567	P200
Mpn568	Era
Mpn572	PepA
Mpn573	GroL
Mpn574	GroS
Mpn576	GlyA
Mpn578	Mpn578
Mpn591	Mpn591
Mpn592	Mpn592
Mpn593	Mpn593
Mpn595	RpiB
Mpn597	AtpC
Mpn598	AtpD
Mpn599	AtpG
Mpn600	AtpA
Mpn601	AtpH
Mpn602	AtpF
Mpn603	AtpE
Mpn604	AtpB
Mpn606	Eno
Mpn607	MsrA
Mpn608	Mpn608
Mpn609	PstB
Mpn610	PstA
Mpn611	Mpn611
Mpn612	Mpn612
Mpn616	RpsI
Mpn617	RplM
Mpn618	DnaX
Mpn619	UvrA
Mpn620	Mpn620
Mpn621	Mpn621
Mpn622	RpsO
Mpn623	Mpn623
Mpn624	RpmB
Mpn625	Mpn625
Mpn627	PtsI
Mpn628	GpmI
Mpn629	TpiA
Mpn630	Mpn630
Mpn631	Tsf
Mpn632	PyrH

Mpn636	Frr
Mpn638	Mpn638
Mpn639	Mpn639
Mpn641	Mpn641
Mpn642	Mpn642
Mpn643	Mpn643
Mpn652	MtlD
Mpn653	MtlF
Mpn655	Mpn655
Mpn658	RplS
Mpn659	TrnD
Mpn660	RpsP
Mpn661	Mpn661
Mpn662	MsrB
Mpn663	Mpn663
Mpn664	Mpn664
Mpn665	Tuf
Mpn666	Mpn666
Mpn667	GalU
Mpn668	Mpn668
Mpn669	TyrS
Mpn670	Mpn670
Mpn671	FtsH
Mpn672	Hpt
Mpn673	Mpn673
Mpn674	Ldh
Mpn677	Mpn677
Mpn678	GltX
Mpn679	KsgA
Mpn680	OxaA
Mpn682	RpmH
Mpn683	Mpn683
Mpn684	Mpn684
Mpn685	Mpn685
Mpn686	DnaA
Mpn687	Mpn687
Mpn688	Mpn688

Table S3: List of high confidence interactions and corresponding socio-affinity scores.

In total 2,116 redundant high confidence interactions, i.e. 1,058 non-redundant interactions, are reported in this list. The first two columns give the proteins involved in these interactions. The third column provides the socio-affinity score attributed to this interaction.

Protein 1	Protein 2	Socio-affinity score x1000
Mpn230	Mpn178	999
Mpn230	Mpn228	999
Mpn230	Mpn660	999
Mpn230	Mpn179	999
Mpn230	Mpn616	999
Mpn516	Mpn515	999
Mpn516	Mpn191	999
Mpn322	Mpn324	999
Mpn573	Mpn574	999
Mpn178	Mpn230	999
Mpn178	Mpn171	999
Mpn178	Mpn446	999
Mpn178	Mpn182	999
Mpn178	Mpn179	999
Mpn178	Mpn169	999
Mpn023	Mpn106	999
Mpn122	Mpn004	999
Mpn122	Mpn123	999
Mpn392	Mpn393	999
Mpn106	Mpn023	999
Mpn106	Mpn105	999
Mpn171	Mpn178	999
Mpn171	Mpn182	999
Mpn171	Mpn189	999
Mpn171	Mpn169	999
Mpn171	Mpn226	999
Mpn622	Mpn446	999
Mpn622	Mpn182	999
Mpn622	Mpn228	999
Mpn622	Mpn660	999
Mpn622	Mpn208	999
Mpn622	Mpn179	999
Mpn622	Mpn169	999
Mpn622	Mpn616	999
Mpn622	Mpn226	999
Mpn207	Mpn268	999
Mpn089	Mpn342	999
Mpn378	Mpn353	999
Mpn515	Mpn516	999
Mpn515	Mpn191	999
Mpn611	Mpn609	999
Mpn446	Mpn178	999
Mpn446	Mpn622	999
Mpn446	Mpn182	999
Mpn446	Mpn189	999
Mpn446	Mpn228	999
Mpn446	Mpn179	999
Mpn446	Mpn169	999
Mpn446	Mpn616	999
Mpn446	Mpn226	999
Mpn003	Mpn004	999
Mpn602	Mpn598	999
Mpn602	Mpn600	999
Mpn167	Mpn165	999
Mpn167	Mpn179	999
Mpn166	Mpn172	999
Mpn166	Mpn165	999
Mpn598	Mpn602	999
Mpn598	Mpn600	999
Mpn182	Mpn178	999
Mpn182	Mpn171	999
Mpn182	Mpn622	999
Mpn182	Mpn446	999
Mpn182	Mpn189	999
Mpn182	Mpn228	999
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Mpn549	Mpn556	850
Mpn621	Mpn622	850
Mpn621	Mpn619	850
Mpn621	Mpn618	850
Mpn621	Mpn616	850
Mpn621	Mpn620	850
Mpn002	Mpn004	850
Mpn246	Mpn258	850
Mpn294	Mpn303	850
Mpn294	Mpn295	850
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Mpn303	Mpn295	850
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Mpn185	Mpn663	850
Mpn622	Mpn621	850
Mpn350	Mpn352	850
Mpn221	Mpn226	850

Mpn262	Mpn258	850
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Mpn033	Mpn034	850
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Mpn222	Mpn226	850
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Mpn555	Mpn553	850
Mpn555	Mpn550	850
Mpn555	Mpn556	850
Mpn551	Mpn555	850
Mpn258	Mpn246	850
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Mpn656	Mpn665	850
Mpn656	Mpn658	850
Mpn250	Mpn258	850
Mpn250	Mpn261	850
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Mpn684	Mpn685	850
Mpn197	Mpn191	850
Mpn261	Mpn251	850
Mpn261	Mpn258	850
Mpn261	Mpn250	850
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Mpn394	Mpn392	850
Mpn394	Mpn393	850
Mpn247	Mpn261	850
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Mpn556	Mpn553	850
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Mpn252	Mpn259	850
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Mpn266	Mpn424	849
Mpn434	Mpn674	849
Mpn434	Mpn191	849
Mpn674	Mpn434	849
Mpn191	Mpn434	849
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Mpn033	Mpn155	848
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Mpn602	Mpn207	847
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Mpn317	Mpn426	846
Mpn354	Mpn007	845
Mpn487	Mpn222	845
Mpn515	Mpn169	845
Mpn222	Mpn487	845
Mpn007	Mpn354	845
Mpn169	Mpn515	845
Mpn168	Mpn060	844
Mpn294	Mpn540	844
Mpn399	Mpn487	844
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Mpn517	Mpn226	844
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Mpn540	Mpn294	844
Mpn493	Mpn596	844
Mpn223	Mpn044	844
Mpn618	Mpn684	844
Mpn618	Mpn252	844
Mpn060	Mpn168	844
Mpn226	Mpn517	844
Mpn252	Mpn618	844
Mpn638	Mpn474	843
Mpn365	Mpn066	843
Mpn297	Mpn397	843
Mpn078	Mpn389	843

Mpn066	Mpn365	843
Mpn504	Mpn331	843
Mpn232	Mpn415	843
Mpn389	Mpn078	843
Mpn474	Mpn638	843
Mpn415	Mpn232	843
Mpn397	Mpn297	843
Mpn331	Mpn504	843
Mpn416	Mpn656	842
Mpn573	Mpn663	842
Mpn238	Mpn671	842
Mpn238	Mpn609	842
Mpn178	Mpn518	842
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Mpn358	Mpn609	842
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Mpn572	Mpn671	842
Mpn213	Mpn204	842
Mpn035	Mpn622	842
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Mpn688	Mpn520	842
Mpn656	Mpn416	842
Mpn250	Mpn671	842
Mpn250	Mpn609	842
Mpn678	Mpn305	842
Mpn623	Mpn337	842
Mpn596	Mpn578	842
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Mpn684	Mpn114	842
Mpn438	Mpn671	842
Mpn388	Mpn525	842
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Mpn363	Mpn341	842
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Mpn081	Mpn327	839
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Mpn445	Mpn042	838
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Mpn636	Mpn316	838
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Mpn042	Mpn046	838
Mpn042	Mpn445	838
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Mpn118	Mpn025	836
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Mpn155	Mpn189	836
Mpn383	Mpn674	836
Mpn063	Mpn307	836
Mpn480	Mpn520	836
Mpn674	Mpn383	836
Mpn189	Mpn155	836
Mpn025	Mpn118	836
Mpn004	Mpn072	836
Mpn081	Mpn615	836
Mpn520	Mpn480	836
Mpn072	Mpn004	836
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Mpn023	Mpn324	835
Mpn106	Mpn573	835
Mpn683	Mpn430	835
Mpn324	Mpn023	835
Mpn430	Mpn683	835
Mpn516	Mpn177	834
Mpn487	Mpn446	834
Mpn446	Mpn487	834
Mpn177	Mpn516	834
Mpn674	Mpn526	834
Mpn526	Mpn674	834
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Mpn615	Mpn176	831
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Mpn394	Mpn020	830
Mpn520	Mpn106	830
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Mpn350	Mpn218	829
Mpn480	Mpn178	829
Mpn153	Mpn618	829
Mpn618	Mpn153	829
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Mpn166	Mpn516	828
Mpn571	Mpn171	828
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Mpn021	Mpn354	827
Mpn306	Mpn483	826
Mpn020	Mpn430	826

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Mpn140	Mpn618	826
Mpn619	Mpn378	826
Mpn618	Mpn140	826
Mpn483	Mpn306	826
Mpn430	Mpn020	826
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Mpn668	Mpn219	825
Mpn671	Mpn342	825
Mpn580	Mpn126	825
Mpn329	Mpn636	825
Mpn492	Mpn283	825
Mpn636	Mpn329	825
Mpn173	Mpn516	825
Mpn126	Mpn580	825
Mpn283	Mpn492	825
Mpn556	Mpn520	825
Mpn289	Mpn219	825
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Mpn239	Mpn191	824
Mpn073	Mpn065	824
Mpn394	Mpn420	824
Mpn658	Mpn117	824
Mpn520	Mpn230	824
Mpn573	Mpn050	823
Mpn573	Mpn105	823
Mpn340	Mpn116	823
Mpn490	Mpn430	823
Mpn430	Mpn490	823
Mpn050	Mpn573	823
Mpn105	Mpn573	823
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Mpn155	Mpn635	822
Mpn656	Mpn623	822
Mpn346	Mpn124	822
Mpn623	Mpn656	822
Mpn197	Mpn393	822
Mpn182	Mpn239	822
Mpn393	Mpn197	822
Mpn165	Mpn516	822
Mpn239	Mpn182	822
Mpn309	Mpn169	822
Mpn124	Mpn346	822
Mpn635	Mpn155	822
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Mpn515	Mpn167	821
Mpn340	Mpn221	821
Mpn434	Mpn283	821
Mpn167	Mpn515	821
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Mpn426	Mpn508	821
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Mpn283	Mpn434	821
Mpn658	Mpn571	821
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Mpn480	Mpn551	820
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Mpn340	Mpn254	819
Mpn653	Mpn551	819
Mpn544	Mpn361	819
Mpn551	Mpn653	819
Mpn590	Mpn360	819
Mpn254	Mpn340	819

Mpn686	Mpn280	819
Mpn361	Mpn544	819
Mpn497	Mpn257	819
Mpn280	Mpn686	819
Mpn360	Mpn307	819
Mpn360	Mpn590	819
Mpn243	Mpn515	817
Mpn278	Mpn280	817
Mpn515	Mpn243	817
Mpn636	Mpn546	817
Mpn546	Mpn636	817
Mpn280	Mpn278	817
Mpn165	Mpn331	816
Mpn331	Mpn165	816
Mpn006	Mpn123	815
Mpn294	Mpn115	815
Mpn171	Mpn515	815
Mpn515	Mpn171	815
Mpn123	Mpn006	815
Mpn115	Mpn294	815
Mpn315	Mpn357	814
Mpn023	Mpn061	814
Mpn061	Mpn023	814
Mpn357	Mpn315	814
Mpn393	Mpn662	814
Mpn662	Mpn393	814
Mpn222	Mpn324	813
Mpn324	Mpn222	813
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Mpn397	Mpn142	812
Mpn220	Mpn357	812
Mpn061	Mpn032	811
Mpn515	Mpn179	811
Mpn167	Mpn228	811
Mpn228	Mpn167	811
Mpn179	Mpn515	811
Mpn032	Mpn061	811
Mpn567	Mpn192	810
Mpn567	Mpn300	810
Mpn416	Mpn559	810
Mpn668	Mpn389	810
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Mpn387	Mpn351	810
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Mpn606	Mpn389	810
Mpn367	Mpn001	810
Mpn423	Mpn332	810
Mpn311	Mpn246	810
Mpn232	Mpn347	810
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Mpn490	Mpn645	810
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Mpn389	Mpn668	810
Mpn389	Mpn606	810
Mpn593	Mpn033	810
Mpn300	Mpn567	810
Mpn300	Mpn110	810
Mpn545	Mpn595	810
Mpn024	Mpn328	810
Mpn644	Mpn320	810
Mpn645	Mpn490	810

Mpn562	Mpn278	810
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Mpn342	Mpn246	810
Mpn060	Mpn387	810
Mpn110	Mpn192	810
Mpn110	Mpn300	810
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Mpn383	Mpn656	809
Mpn656	Mpn383	809
Mpn678	Mpn292	809
Mpn352	Mpn401	809
Mpn292	Mpn678	809
Mpn629	Mpn434	809
Mpn595	Mpn269	808
Mpn269	Mpn595	808
Mpn515	Mpn166	808
Mpn606	Mpn316	808
Mpn166	Mpn515	808
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Mpn061	Mpn393	804
Mpn393	Mpn061	804
Mpn268	Mpn259	804
Mpn259	Mpn268	804
Mpn638	Mpn061	803
Mpn061	Mpn638	803
Mpn257	Mpn089	803
Mpn089	Mpn257	803
Mpn166	Mpn228	803
Mpn228	Mpn166	803
Mpn207	Mpn321	802
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Mpn243	Mpn426	801
Mpn020	Mpn003	801
Mpn003	Mpn020	801
Mpn426	Mpn243	801
Mpn324	Mpn050	801
Mpn050	Mpn324	801

Table S4: List of protein complexes.

This table is a repository of all protein complexes that have been observed in this study. It is divided into i) a set of 116 heteromultimeric complexes and ii) a set of 62 homomultimeric complexes. In part i) the first column gives the assembly ID that specifies the higher order molecular assembly in which a heteromultimeric protein complex clusters. The second column specifies the unique complex ID for each protein complex. Columns 3 and 4 provide complex name and complex function, respectively. In columns 5-8 the systematic name, protein name, UniProt protein name and subunit type of each protein component of a complex are indicated. In part ii) the first column on the left shows the complex ID of the homomultimeric protein complexes. The second column indicates the complex name. Columns 3 and 4 give the annotated systematic and protein name of each homomultimer.

Assembly ID	Complex ID	Complex name	Complex function	Systematic name	Protein name	UniProt protein name	Subunit type
j) Heteromultimeric complexes							
XXVIII	40	Cohesin like complex	Cell division and chromosome partitioning	Mpn300	ScpA	Segregation and condensation protein A	core
				Mpn301	ScpB	Segregation and condensation protein B	core
				Mpn426	P115	Protein P115 homolog	core
				Mpn423	Mpn423	Uncharacterized protein MG296 homolog	core
				Mpn211	UvrB	UvrABC system protein B	core
				Mpn508	Mpn508	Uncharacterized protein MPN508	core
				Mpn332	Lon	ATP-dependent protease La	attachment
				Mpn573	GroL	60 kDa chaperonin	attachment
XXX	42	Complex 42	Cell division and chromosome partitioning	Mpn314	MraZ	Protein MraZ	core
				Mpn339	Mpn339	Uncharacterized protein MG243 homolog	core
				Mpn633	Mpn633	Uncharacterized protein Mpn633	core
				Mpn315	MraW	S-adenosyl-L-methionine-dependent methyltransferase	attachment
XXX	43	Complex 43	Cell division and chromosome partitioning	Mpn314	MraZ	Protein MraZ	core
				Mpn315	MraW	S-adenosyl-L-methionine-dependent methyltransferase	core
III	3	DNA Polymerase III γ complex	DNA replication, recombination and repair	Mpn007	Mpn007	Uncharacterized protein MG007 homolog	core
				Mpn618	DnaX	DNA polymerase III subunit γ T	core
				Mpn450	Mpn450	Uncharacterized protein MG315 homolog	core
XX	29	DNA Polymerase III core complex	DNA replication, recombination and repair	Mpn353	DnaG	DNA primase	core
				Mpn378	DnaE	DNA polymerase III subunit α	core
				Mpn551	Mpn551	Uncharacterized protein MG373 homolog	core
				Mpn118	Mpn118	Uncharacterized protein MG199 homolog	core
				Mpn563	Mpn563	Uncharacterized GTP-binding protein MG384 homolog	attachment
				Mpn557	GidA	tRNA uridine 5-carboxymethylaminomethyl modification enzyme	attachment
				Mpn141	MgpA	Adhesin P1 precursor	attachment
				Mpn059	Gcp	Probable O-sialoglycoprotein endopeptidase	attachment
				Mpn653	Mlf	Mannitol-specific phosphotransferase enzyme IIA component	attachment
				Mpn683	Mpn683	Putative ABC transporter ATP-binding protein MG467 homolog	attachment
	59	DnaA complex	DNA replication, recombination and repair	Mpn376	Mpn376	Uncharacterized protein Mpn376	core
				Mpn686	DnaA	Chromosomal replication initiator protein dnaA	core
	82	DNA Gyrase complex	DNA replication, recombination and repair	Mpn003	GyrB	DNA gyrase subunit B	core
				Mpn004	GyrA	DNA gyrase subunit A	core
	102	ParE-GyrA complex	DNA replication, recombination and repair	Mpn122	ParE	DNA topoisomerase 4 subunit B	core
				Mpn004	GyrA	DNA gyrase subunit A	core
XI	17	DNA Topoisomerase IV complex	DNA replication, recombination and repair	Mpn123	ParC	DNA topoisomerase 4 subunit A	core
				Mpn122	ParE	DNA topoisomerase 4 subunit B	core
	86	DNA Recombination complex	DNA replication, recombination and repair	Mpn034	PolC	DNA polymerase III PolC-type	core
				Mpn490	RecA	Protein RecA	core
	101	GyrB-Mpn486 complex	DNA replication, recombination and repair	Mpn003	GyrB	DNA gyrase subunit B	core
				Mpn486	Mpn486	Uncharacterized protein Mpn486	core
	84	DNA Primase complex	DNA replication, recombination and repair	Mpn014	Mpn014	Uncharacterized protein MG010 homolog	core
				Mpn377	Mpn377	Uncharacterized protein Mpn377	core
	103	DNA Primase complex	DNA replication, recombination and repair	Mpn014	Mpn014	Uncharacterized protein MG010 homolog	core
				Mpn235	Ung	Uracil-DNA glycosylase	core
	90	Complex 90	DNA replication, recombination and repair	Mpn125	UvrC	UvrABC system protein C	core
				Mpn563	Mpn563	Uncharacterized GTP-binding protein MG384 homolog	core
	105	Complex 105	DNA replication, recombination and repair	Mpn023	MetG	Methionyl-tRNA synthetase	core
				Mpn619	UvrA	UvrABC system protein A	core
	100	Complex 100	DNA replication, recombination and repair	Mpn536	RuvB	Holliday junction ATP-dependent DNA helicase RuvB	core
				Mpn663	Mpn663	Uncharacterized protein MG449 homolog	core
	104	Helicase complex	DNA replication, recombination and repair	Mpn020	Mpn020	Uncharacterized ATP-dependent helicase MPN020	core
				Mpn341	UvrD	Probable DNA helicase II homolog	core
XI	15	DNA Recombination complex	DNA replication, recombination and repair	Mpn123	ParC	DNA topoisomerase 4 subunit A	core
				Mpn282	Mpn282	Uncharacterized protein Mpn282	core
				Mpn424	Mpn424	UPF0122 protein Mpn424	core
				Mpn060	MetK	S-adenosylmethionine synthetase	core
				Mpn402	ProS	Prolyl-tRNA synthetase	core
				Mpn351	Mpn351	Uncharacterized protein MG248 homolog	core
				Mpn387	Mpn387	Uncharacterized protein MG269 homolog	core
				Mpn293	LspA	Lipoprotein signal peptidase	attachment
				Mpn490	RecA	Protein RecA	attachment
				Mpn122	ParE	DNA topoisomerase 4 subunit B	attachment
				Mpn352	RpoD	RNA polymerase σ factor RpoD	core
				Mpn434	DnaK	Chaperone protein DnaK	core

XXXV	49	RNA polymerase complex	Transcription	Mpn020	Mpn020	Uncharacterized ATP-dependent helicase Mpn020	core
				Mpn261	TopA	DNA topoisomerase 1	core
				Mpn352	RpoD	RNA polymerase σ factor RpoD σ -A)	core
				Mpn515	RpoC	DNA-directed RNA polymerase β' chain	core
				Mpn516	RpoB	DNA-directed RNA polymerase β chain	core
				Mpn191	RpoA	DNA-directed RNA polymerase α chain	core
				Mpn178	RpsZ	30S ribosomal protein S14 type Z	core
				Mpn154	NusA	Transcription elongation protein NusA	core
				Mpn002	Mpn002	DnaJ-like protein MG002 homolog	core
				Mpn539	RplL	50S ribosomal protein L7/L12	core
				Mpn014	Mpn014	Uncharacterized protein MG010 homolog	attachment
				Mpn673	Mpn673	Uncharacterized protein MG459 homolog	attachment
				Mpn437	Mpn437	Uncharacterized protein Mpn437	attachment
				Mpn067	Mpn067	Protein MG054 homolog	attachment
				Mpn265	TrpS	Tryptophanyl-tRNA synthetase	attachment
				Mpn678	GltX	Glutamyl-tRNA synthetase	attachment
				Mpn379	PolA	Probable 5'-3' exonuclease	attachment
				Mpn153	Mpn153	Uncharacterized ATP-dependent helicase MG140 homolog	attachment
				Mpn490	RecA	Protein RecA	attachment
				Mpn266	Mpn266	Uncharacterized protein MG127 homolog	attachment
				Mpn009	Mpn009	Uncharacterized deoxyribonuclease MG009 homolog	attachment
				Mpn443	Mpn443	Probable ATP-dependent RNA helicase MG308 homolog	attachment
				Mpn155	InfB	Translation initiation factor IF-2	attachment
				Mpn187	InfA	Translation initiation factor IF-1	attachment
				Mpn252	AsnS	Asparaginyt-tRNA synthetase	attachment
				Mpn354	GlyQS	Glycyl-tRNA synthetase	attachment
				Mpn219	RplK	50S ribosomal protein L11	attachment
				Mpn173	RpmC	50S ribosomal protein L29	attachment
V	7	Complex 7	Translation, ribosomal structure and biogenesis	Mpn016	Mpn016	Uncharacterized protein MG012 homolog	core
				Mpn033	Upp	Uracil phosphoribosyltransferase	core
				Mpn133	Mpn133	Uncharacterized lipoprotein MG186 homolog precursor	core
				Mpn635	Mpn635	Uncharacterized protein Mpn635	core
				Mpn155	InfB	Translation initiation factor IF-2	core
				Mpn346	Mpn346	Uncharacterized protein Mpn346	core
				Mpn064	DeoA	Thymidine phosphorylase	attachment
				Mpn156	RbfA	Ribosome-binding factor A	attachment
VII	10	Aminoacyl-tRNA synthetase complex	Translation, ribosomal structure and biogenesis	Mpn025	Fba	Fructose-bisphosphate aldolase	core
				Mpn105	PheS	Phenylalanyl-tRNA synthetase α chain	core
				Mpn669	TyrS	Tyrosyl-tRNA synthetase	core
				Mpn061	Ffh	Signal recognition particle protein	core
				Mpn023	MetG	Methionyl-tRNA synthetase	core
				Mpn678	GltX	Glutamyl-tRNA synthetase	attachment
				Mpn106	PheT	Phenylalanyl-tRNA synthetase β chain	attachment
				Mpn322	NrdF	Ribonucleoside-diphosphate reductase subunit β	attachment
				Mpn141	MgpA	Adhesin P1 precursor	attachment
				Mpn553	ThrS	Threonyl-tRNA synthetase	attachment
VII	11	Phenylalanine-tRNA synthetase complex	Translation, ribosomal structure and biogenesis	Mpn106	PheT	Phenylalanyl-tRNA synthetase β chain	core
				Mpn105	PheS	Phenylalanyl-tRNA synthetase α chain	core
XXXV	51	Complex 51	Translation, ribosomal structure and biogenesis	Mpn035	Mpn035	Uncharacterized protein Mpn035	core
				Mpn561	Udk	Uridine kinase	core
				Mpn178	RpsZ	30S ribosomal protein S14 type Z	core
				Mpn553	ThrS	Threonyl-tRNA synthetase	attachment
XXXV	52	Translation elongation factor complex	Translation, ribosomal structure and biogenesis	Mpn631	Tsf	Elongation factor Ts	core
				Mpn665	Tuf	Elongation factor Tu	core
				Mpn006	Tmk	Thymidylate kinase	attachment
				Mpn137	Mpn137	UPF0134 protein Mpn137	attachment
				Mpn219	RplK	50S ribosomal protein L11	attachment
				Mpn168	RplB	50S ribosomal protein L2	attachment
				Mpn173	RpmC	50S ribosomal protein L29	attachment
				Mpn225	RpsL	30S ribosomal protein S12	attachment
XXXV	53	L29-S12 complex	Translation, ribosomal structure and biogenesis	Mpn173	RpmC	50S ribosomal protein L29	core
				Mpn225	RpsL	30S ribosomal protein S12	core
	96	Complex 96	Translation, ribosomal structure and biogenesis	Mpn361	PrfA	Peptide chain release factor 1	core
				Mpn556	ArgS	Arginyl-tRNA synthetase	core
XII	18	Cytidine deamination-ribosome complex	Translation, ribosomal structure and biogenesis	Mpn065	Cdd	Cytidine deaminase	core
				Mpn643	Mpn643	Uncharacterized lipoprotein MG440 homolog 2 precursor	core
				Mpn539	RplL	50S ribosomal protein L7/L12	core
				Mpn216	OppC	Oligopeptide transport system permease protein OppC	attachment

XXXV	50	Ribosome complex	Translation, ribosomal structure and biogenesis	Mpn616	RpsI	30S ribosomal protein S9	core
				Mpn622	RpsO	30S ribosomal protein S15	core
				Mpn115	InfC	Translation initiation factor IF-3	core
				Mpn165	RplC	50S ribosomal protein L3	core
				Mpn166	RplD	50S ribosomal protein L4	core
				Mpn167	RplW	50S ribosomal protein L23	core
				Mpn169	RpsS	30S ribosomal protein S19	core
				Mpn171	RpsC	30S ribosomal protein S3	core
				Mpn172	RplP	50S ribosomal protein L16	core
				Mpn174	RpsQ	30S ribosomal protein S17	core
				Mpn176	RplX	50S ribosomal protein L24	core
				Mpn177	RplE	50S ribosomal protein L5	core
				Mpn179	RpsH	30S ribosomal protein S8	core
				Mpn182	RpsE	30S ribosomal protein S5	core
				Mpn189	RpsM	30S ribosomal protein S13	core
				Mpn192	RplQ	50S ribosomal protein L17	core
				Mpn208	RpsB	30S ribosomal protein S2	core
				Mpn220	RplA	50S ribosomal protein L1	core
				Mpn226	RpsG	30S ribosomal protein S7	core
				Mpn228	RpsF	30S ribosomal protein S6	core
				Mpn230	RpsR	30S ribosomal protein S18	core
				Mpn327	RpmA	50S ribosomal protein L27	core
				Mpn446	RpsD	30S ribosomal protein S4	core
				Mpn658	RplS	50S ribosomal protein L19	core
				Mpn221	Pth	Peptidyl-tRNA hydrolase	core
				Mpn660	RpsP	30S ribosomal protein S16	core
				Mpn685	Mpn685	Putative ABC transporter ATP-binding protein MG468.1 homolog	core
				Mpn518	Mpn518	Uncharacterized protein MG343 homolog	core
				Mpn571	Mpn571	Putative ABC transporter ATP-binding MG390 homolog	core
				Mpn665	Tuf	Elongation factor Tu	core
				Mpn178	RpsZ	30S ribosomal protein S14 type Z	core
				Mpn155	InfB	Translation initiation factor IF-2	core
				Mpn520	IleS	Isoleucyl-tRNA synthetase	core
				Mpn091	Mpn091	Uncharacterized protein Mpn091	core
				Mpn428	Pta	Phosphate acetyltransferase	core
				Mpn465	Mpn465	Uncharacterized protein Mpn465	core
				Mpn541	RpsT	30S ribosomal protein S20	attachment
				Mpn154	NusA	Transcription elongation protein NusA	attachment
				Mpn517	Mpn517	Uncharacterized protein MG342 homolog	attachment
				Mpn538	RplJ	50S ribosomal protein L10	attachment
				Mpn231	RplI	50S ribosomal protein L9	attachment
				Mpn180	RplF	50S ribosomal protein L6	attachment
				Mpn117	RplT	50S ribosomal protein L20	attachment
				Mpn475	EngA	GTP-binding protein EngA	attachment
				Mpn636	Frr	Ribosome recycling factor	attachment
				Mpn676	Mpn676	Uncharacterized protein Mpn676	attachment
				Mpn103	Mpn103	Uncharacterized protein Mpn103	attachment
				Mpn386	Mpn386	Uncharacterized protein MG268 homolog	attachment
				Mpn536	RuvB	Holliday junction ATP-dependent DNA helicase RuvB	attachment
				Mpn623	Mpn623	Probable ATP-dependent RNA helicase MG425 homolog	attachment
				Mpn253	PgsA	CDP-diacylglycerol-glycerol-3-phosphate 3-phosphatidyltransferase	attachment
				Mpn524	Mpn524	UPF0134 protein Mpn524	attachment
				Mpn539	RplL	50S ribosomal protein L7/L12	attachment
				Mpn553	ThrS	Threonyl-tRNA synthetase	attachment
				Mpn072	Mpn072	Uncharacterized protein MG057 homolog	attachment
				Mpn219	RplK	50S ribosomal protein L11	attachment
				Mpn006	Tmk	Thymidylate kinase	attachment
				Mpn137	Mpn137	UPF0134 protein Mpn137	attachment
				Mpn188	RpmJ	50S ribosomal protein L36	attachment
				Mpn168	RplB	50S ribosomal protein L2	attachment
				Mpn317	FtsZ	Cell division protein FtsZ	attachment
				Mpn225	RpsL	30S ribosomal protein S12	attachment
XXV & XXXI	37	GroEL-GroES complex	Posttranslational modification, protein turnover, chaperones	Mpn573	GroL	60 kDa chaperonin	core
				Mpn574	GroS	10 kDa chaperonin	core
VI & XXXII	9	Dnak-GrpE complex	Posttranslational modification, protein turnover, chaperones	Mpn120	GrpE	Protein GrpE	core
				Mpn434	DnaK	Chaperone protein DnaK	core
VI	8	Protein chaperone complex	Posttranslational modification, protein turnover, chaperones	Mpn538	RplJ	50S ribosomal protein L10	core
				Mpn434	DnaK	Chaperone protein DnaK	core
				Mpn021	DnaJ	Chaperone protein DnaJ	core
				Mpn142	Mpn142	Mgp-operon protein 3 precursor	core
				Mpn397	SpoT	Probable guanosine-3'-5'-bis(diphosphate) 3'-pyrophosphohydrolase	core
				Mpn441	Mpn441	Uncharacterized protein Mpn441	core
				Mpn297	Mpn297	Uncharacterized protein MG211 homolog	attachment
				Mpn002	Mpn002	DnaJ-like protein MG002 homolog	attachment
				Mpn531	ClpB	Chaperone ClpB	attachment
				Mpn124	HrcA	Heat-inducible transcription repressor HrcA	attachment
				Mpn247	Mpn247	Putative protein phosphatase	attachment
				Mpn354	GlyQS	Glycyl-tRNA synthetase	attachment
				Mpn394	Nox	Probable NADH oxidase	attachment
				Mpn429	Pgk	Phosphoglycerate kinase	attachment
				Mpn120	GrpE	Protein GrpE	attachment
				Mpn547	Mpn547	Uncharacterized protein MG369 homolog	attachment
VIII	12	Glycolytic enzyme complex 1	Carbohydrate transport and metabolism	Mpn025	Fba	Fructose-bisphosphate aldolase	core
				Mpn430	GapA	Glyceraldehyde-3-phosphate dehydrogenase	core
				Mpn303	Pyk	Pyruvate kinase	core
XXXIII	46	Complex 46	Carbohydrate transport and metabolism	Mpn606	Eno	Enolase	core
				Mpn629	TpiA	Triosephosphate isomerase	core
				Mpn569	Mpn569	Putative metalloprotease Mpn569	core
				Mpn316	Mpn316	Uncharacterized protein MG223 homolog	core
	92	Complex 92	Carbohydrate transport and metabolism	Mpn258	Mpn258	Putative carbohydrate transport ATP-binding protein Mpn258	core
				Mpn259	Mpn259	Uncharacterized protein MG120 homolog	core

XXXII	45	Glycolytic enzyme complex 2	Carbohydrate transport and metabolism	Mpn430	GapA	Glyceraldehyde-3-phosphate dehydrogenase	core
				Mpn429	Pgk	Phosphoglycerate kinase	core
				Mpn606	Eno	Enolase	core
				Mpn434	DnaK	Chaperone protein DnaK	core
				Mpn120	GrpE	Protein GrpE	attachment
				Mpn002	Mpn002	DnaJ-like protein MG002 homolog	attachment
				Mpn531	ClpB	Chaperone ClpB	attachment
				Mpn124	HrcA	Heat-inducible transcription repressor	attachment
				Mpn247	Mpn247	Putative protein phosphatase	attachment
				Mpn354	GlyQS	Glycyl-tRNA synthetase	attachment
XXVI	38	Peptidase complex	Amino acid transport and metabolism	Mpn394	Nox	Probable NADH oxidase	attachment
				Mpn238	GatB	Aspartyl/glutamyl-tRNA(Asn/Gln) amidotransferase subunit B	core
				Mpn250	Pgi	Glucose-6-phosphate isomerase	core
				Mpn358	Mpn358	Uncharacterized protein MG255 homolog	core
				Mpn572	PepA	Probable cytosol aminopeptidase	core
				Mpn609	PstB	Phosphate import ATP-binding protein PstB	core
				Mpn611	Mpn611	Uncharacterized lipoprotein MG412 homolog precursor	core
				Mpn197	PepF	Oligoendopeptidase F homolog	core
				Mpn218	OppF	Oligopeptide transport ATP-binding protein OppF	core
				Mpn372	Mpn372	ADP-ribosylating toxin CARDS	core
XXVII	39	Complex 39	Metabolism	Mpn671	FtsH	Cell division protease FtsH homolog	core
				Mpn628	GpmI	2,3-bisphosphoglycerate-independent phosphoglycerate mutase	attachment
				Mpn203	Mpn203	Putative adhesin P1-like protein Mpn203	core
				Mpn668	Mpn668	Organic hydroperoxide resistance protein-like	core
	94	Ribonucleoside-diphosphate reductase complex	Metabolism	Mpn391	PdhC	Dihydrolipoyllysine-residue acetyltransferase	core
				Mpn420	Mpn420	Uncharacterized protein MG293 homolog	attachment
XXXI	44	Pyruvate dehydrogenase complex	Metabolism	Mpn322	NrdF	Ribonucleoside-diphosphate reductase subunit β	core
				Mpn324	NrdE	Ribonucleoside-diphosphate reductase α subunit	core
				Mpn392	PdhB	Pyruvate dehydrogenase E1 component subunit β	core
				Mpn393	PdhA	Pyruvate dehydrogenase E1 component subunit α	core
				Mpn573	GroL	60 kDa chaperonin	core
				Mpn390	PdhD	Dihydrolipoyl dehydrogenase	core
				Mpn574	GroS	10 kDa chaperonin	attachment
				Mpn006	Tmk	Thymidylate kinase	attachment
				Mpn306	ArcB	Ornithine carbamoyltransferase, catabolic	attachment
				Mpn023	MetG	Methionyl-tRNA synthetase	attachment
				Mpn008	TrmE	Probable tRNA modification GTPase TrmE	attachment
				Mpn009	Mpn009	Uncharacterized deoxyribonuclease MG009 homolog	attachment
				Mpn321	FolA	Dihydrofolate reductase	attachment
				Mpn426	P115	Protein P115 homolog	attachment
	54	Phosphotransfer system complex	Metabolism	Mpn553	ThrS	Threonyl-tRNA synthetase	attachment
				Mpn207	PtsG	PTS system glucose-specific EIICBA component	core
	93	ScpA-ThyA complex	Metabolism	Mpn268	Mpn268	Putative phosphotransferase enzyme IIB component	core
				Mpn300	ScpA	Segregation and condensation protein A	core
XXXIV	47	ATP synthase complex	Energy production and conversion	Mpn320	ThyA	Thymidylate synthase	core
				Mpn598	AtpD	ATP synthase subunit β	core
				Mpn600	AtpA	ATP synthase subunit α	core
				Mpn602	AtpF	ATP synthase B chain precursor	core
				Mpn153	Mpn153	Uncharacterized ATP-dependent helicase MG140 homolog	attachment
				Mpn688	Mpn688	ParA family protein Mpn688	attachment
XIX	28	Restriction enzyme complex	Defense mechanisms	Mpn347	Mpn347	Putative type I restriction enzyme MpnORFDP R protein part 1	core
				Mpn638	Mpn638	Putative type I restriction enzyme specificity protein Mpn638 (S protein)	core
XXIV	34	Restriction enzyme complex	Defense mechanisms	Mpn185	Adk	Adenylate kinase	core
				Mpn438	Mpn438	Uncharacterized protein Mpn438	core
				Mpn342	Mpn342	Putative type I restriction enzyme MpnORFDP M protein	core
				Mpn663	Mpn663	Uncharacterized protein MG449 homolog	core
				Mpn671	FtsH	Cell division protease FtsH homolog	core
XXIV	35	Restriction enzyme complex	Defense mechanisms	Mpn342	Mpn342	Putative type I restriction enzyme MpnORFDP M protein	core
				Mpn507	Mpn507	Putative type I restriction enzyme specificity protein Mpn507 (S protein)	core
				Mpn089	Mpn089	Putative type I restriction enzyme specificity protein Mpn089 (S protein)	core
	95	Restriction enzyme complex	Defense mechanisms	Mpn345	Mpn345	Putative type-1 restriction enzyme MpnORFDP R protein part 2	core
				Mpn365	Mpn365	Putative type I restriction enzyme specificity protein Mpn365 (S protein)	core
	109	Restriction enzyme complex	Defense mechanisms	Mpn089	Mpn089	Putative type I restriction enzyme specificity protein Mpn089 (S protein)	core
				Mpn497	Mpn497	Uncharacterized protein Mpn497	core
	113	Restriction enzyme complex	Defense mechanisms	Mpn365	Mpn365	Putative type I restriction enzyme specificity protein Mpn365 (S protein)	core
				Mpn343	Mpn343	Putative type I restriction enzyme specificity protein Mpn343 (S protein)	core
	115	Restriction enzyme complex	Defense mechanisms	Mpn365	Mpn365	Putative type I restriction enzyme specificity protein Mpn365 (S protein)	core
				Mpn615	Mpn615	Putative type I restriction enzyme specificity protein Mpn615 (S protein)	core
XIX	27	Restriction enzyme complex	Defense mechanisms	Mpn116	RpmI	50S ribosomal protein L35	core
				Mpn254	Mpn254	Protein MG115 homolog	core
				Mpn232	DnaB	Replicative DNA helicase	core
				Mpn526	Mpn526	Uncharacterized protein MG350 homolog	core
				Mpn347	Mpn347	Putative type I restriction enzyme MpnORFDP R protein part 1	core
				Mpn257	GalE	UDP-glucose 4-epimerase	core
				Mpn592	Mpn592	Uncharacterized lipoprotein Mpn592 precursor	core
				Mpn638	Mpn638	Putative type I restriction enzyme specificity protein Mpn638 (S protein)	attachment
				Mpn656	Mpn656	Uncharacterized protein MG442 homolog	attachment

II	2	Complex 2	Cell envelope biogenesis, outer membrane	Mpn367	Mpn367	Putative mgpC-like protein Mpn367	core
				Mpn514	Mpn514	Uncharacterized protein Mpn514	core
				Mpn200	Mpn200	Uncharacterized lipoprotein Mpn200 precursor	core
				Mpn499	Mpn499	Uncharacterized protein Mpn499	core
				Mpn645	Mpn645	Uncharacterized lipoprotein MG439 homolog 2 precursor	core
	81	Complex 81	Cell envelope biogenesis, outer membrane	Mpn100	Mpn100	UPF0134 protein Mpn100	core
				Mpn650	Mpn650	Uncharacterized lipoprotein Mpn650 precursor	core
IV	4	Complex 4	Function unknown	Mpn140	Mpn140	Mgp-operon protein 1	core
				Mpn148	Mpn148	Uncharacterized protein Mpn148	core
				Mpn450	Mpn450	Uncharacterized protein MG315 homolog	core
				Mpn311	Mpn311	Uncharacterized protein MG218.1 homolog	core
				Mpn246	Def	Peptide deformylase	core
				Mpn011	Mpn011	Uncharacterized lipoprotein Mpn011 precursor	core
				Mpn466	Mpn466	Uncharacterized protein Mpn466	core
				Mpn204	Mpn204	UPF0134 protein Mpn204	attachment
				Mpn245	Gmk	Guanylate kinase	attachment
				Mpn419	AlaS	Alanyl-tRNA synthetase	attachment
				Mpn095	Mpn095	Uncharacterized protein Mpn095	attachment
				Mpn246	Def	Peptide deformylase	core
IV	5	Complex 5	Function unknown	Mpn245	Gmk	Guanylate kinase	core
IV	6	Complex 6	Function unknown	Mpn095	Mpn095	Uncharacterized protein Mpn095	core
				Mpn311	Mpn311	Uncharacterized protein MG218.1 homolog	core
IX	13	Complex 13	Function unknown	Mpn041	Mpn041	Uncharacterized protein Mpn041	core
				Mpn400	Mpn400	Uncharacterized protein MG281 homolog	core
				Mpn118	Mpn118	Uncharacterized protein MG199 homolog	core
				Mpn298	AcpS	Holo-[acyl carrier protein] synthase	core
				Mpn563	Mpn563	Uncharacterized GTP-binding protein MG384 homolog	attachment
X	14	Complex 14	Function unknown	Mpn557	GidA	tRNA uridine 5-carboxymethylaminomethyl modification enzyme	attachment
				Mpn066	ManB	Phosphomannomutase	core
				Mpn564	Adh	Probable NADP-dependent alcohol dehydrogenase	core
				Mpn048	Mpn048	Uncharacterized protein Mpn048	core
				Mpn320	ThyA	Thymidylate synthase	core
XIII	19	Complex 19	Function unknown	Mpn076	Mpn076	Uncharacterized protein MG061 homolog 1	core
				Mpn284	Mpn284	Uncharacterized lipoprotein Mpn284 precursor	core
XIV	20	Complex 20	Function unknown	Mpn353	DnaG	DNA primase	core
				Mpn083	Mpn083	Uncharacterized lipoprotein Mpn083 precursor	core
				Mpn142	Mpn142	Mgp-operon protein 3 precursor	core
				Mpn582	Mpn582	Uncharacterized lipoprotein Mpn582 precursor	attachment
XV	21	Complex 21	Function unknown	Mpn088	Mpn088	Uncharacterized protein Mpn088	core
				Mpn145	Mpn145	UPF0134 protein Mpn145	core
				Mpn620	Mpn620	Uncharacterized protein MG422 homolog	core
				Mpn612	Mpn612	Uncharacterized protein MG414 homolog	core
XVI	22	Complex 22	Function unknown	Mpn221	Pth	Peptidyl-tRNA hydrolase	core
				Mpn292	Mpn292	Uncharacterized RNA pseudouridine synthase MG209 homolog	core
				Mpn094	Mpn094	UPF0134 protein Mpn094	core
				Mpn305	ArcA	Putative arginine deiminase	core
				Mpn547	Mpn547	Uncharacterized protein MG369 homolog	attachment
				Mpn247	Mpn247	Putative protein phosphatase	attachment
XVI	23	Complex 23	Function unknown	Mpn292	Mpn292	Uncharacterized RNA pseudouridine synthase MG209 homolog	core
				Mpn547	Mpn547	Uncharacterized protein MG369 homolog	core
XVII	24	Complex 24	Function unknown	Mpn525	Mpn525	Uncharacterized protein MG349 homolog	core
				Mpn567	P200	Protein P200	core
				Mpn110	Mpn110	Uncharacterized protein Mpn110	core
				Mpn295	Mpn295	Uncharacterized protein MG210.1 homolog	attachment
				Mpn095	Mpn095	Uncharacterized protein Mpn095	attachment
XVIII	25	Complex 25	Function unknown	Mpn258	Mpn258	Putative carbohydrate transport ATP-binding protein	core
				Mpn341	UvrD	Probable DNA helicase II homolog	core
				Mpn114	Mpn114	Putative acetyltransferase	core
				Mpn241	Mpn241	Uncharacterized protein MG103 homolog	core
				Mpn262	Mpn262	Uncharacterized protein MG123 homolog	core
				Mpn269	Mpn269	UPF0144 protein MG130 homolog	core
				Mpn223	HprK	HPr kinase/phosphorylase	attachment
				Mpn415	P37	High affinity transport system protein p37 precursor	attachment
XVIII	26	Complex 26	Function unknown	Mpn341	UvrD	Probable DNA helicase II homolog	core
				Mpn223	HprK	HPr kinase/phosphorylase	core
XXI	31	Complex 31	Function unknown	Mpn121	Mpn121	Uncharacterized protein MG202 homolog	core
				Mpn135	Mpn135	Probable ABC transporter permease protein MG188 homolog	core
				Mpn366	Mpn366	Putative mgpC-like protein Mpn366	core
				Mpn474	Mpn474	Uncharacterized protein MG328 homolog	attachment
XXII	32	Complex 32	Function unknown	Mpn158	RibF	Putative riboflavin biosynthesis protein	core
				Mpn209	Pacl	Probable cation-transporting P-type ATPase	core
				Mpn331	Tig	Trigger factor	core
XXIII	33	Complex 33	Function unknown	Mpn172	RplP	50S ribosomal protein L16	core
				Mpn418	Mpn418	Putative Holliday junction resolvase	core
				Mpn642	Mpn642	Uncharacterized lipoprotein MG439 homolog 4 precursor	core
				Mpn211	UvrB	UvrABC system protein B	core
XXV	36	Complex 36	Function unknown	Mpn573	GroL	60 kDa chaperonin	core
				Mpn194	CbiO2	Cobalt import ATP-binding protein CbiO 2	core
				Mpn264	Mpn264	Uncharacterized protein MG125 homolog	core
				Mpn574	GroS	10 kDa chaperonin	attachment
				Mpn267	PpnK	Probable inorganic polyphosphate/ATP-NAD kinase	attachment
				Mpn008	TrmE	Probable tRNA modification GTPase TrmE	attachment
				Mpn009	Mpn009	Uncharacterized deoxyribonuclease MG009 homolog	attachment
				Mpn321	FolA	Dihydrofolate reductase	attachment
				Mpn426	P115	Protein P115 homolog	attachment
				Mpn553	ThrS	Threonyl-tRNA synthetase	attachment
XI	16	Complex 16	Function unknown	Mpn387	Mpn387	Uncharacterized protein MG269 homolog	core
				Mpn293	LspA	Lipoprotein signal peptidase	core
XXIX	41	Complex 41	Function unknown	Mpn289	Mpn289	Putative type I restriction enzyme specificity protein Mpn289 (S protein)	core
				Mpn479	AzoR	FMN-dependent NADH-azoreductase	core
				Mpn389	LplA	Probable lipote-protein ligase A	core
				Mpn040	Mpn040	Uncharacterized protein Mpn040	attachment
XXXIV	48	Complex 48	Function unknown	Mpn602	AtpF	ATP synthase B chain precursor	core
				Mpn153	Mpn153	Uncharacterized ATP-dependent helicase MG140 homolog	core

I	1	Complex 1	Function unknown	Mpn195	Mpn195	Uncharacterized protein MG181 homolog	core
				Mpn395	Apt	Adenine phosphoribosyltransferase	core
				Mpn367	Mpn367	Putative mqpC-like protein Mpn367	core
				Mpn001	DnaN	DNA polymerase III subunit β	core
				Mpn388	Mpn388	Uncharacterized protein MG269.1 homolog	core
XX	30	Complex 30	Function unknown	Mpn551	Mpn551	Uncharacterized protein MG373 homolog	core
				Mpn653	Mif	Mannitol-specific phosphotransferase enzyme IIA component	core
	55	Complex 55	Function unknown	Mpn444	Mpn444	Putative esterase/lipase 1	core
				Mpn578	Mpn578	Uncharacterized protein Mpn578	core
	56	Complex 56	Function unknown	Mpn071	Mpn071	UPF0011 protein MG056 homolog	core
				Mpn273	Mpn273	Uncharacterized 16.1 kDa HIT-like protein	core
	57	Complex 57	Function unknown	Mpn462	Mpn462	Uncharacterized protein Mpn462	core
				Mpn471	RpmG1	50S ribosomal protein L33 1	core
	58	Complex 58	Function unknown	Mpn281	Mpn281	Uncharacterized lipoprotein Mpn281 precursor	core
				Mpn509	Mpn509	Uncharacterized protein Mpn509	core
	60	Complex 60	Function unknown	Mpn294	Mpn294	Uncharacterized protein Mpn294	core
				Mpn445	Mpn445	Uncharacterized lipoprotein MG309 homolog precursor	core
	61	Complex 61	Function unknown	Mpn328	Nfo	Probable endonuclease 4	core
				Mpn330	Mpn330	Uncharacterized protein MG237 homolog	core
	62	Complex 62	Function unknown	Mpn062	DeoD	Purine nucleoside phosphorylase DeoD-type	core
				Mpn079	FruK	Putative 1-phosphofructokinase	core
	63	Complex 63	Function unknown	Mpn323	NrdI	Protein NrdI	core
				Mpn407	Mpn407	Uncharacterized protein Mpn407	core
	64	Complex 64	Function unknown	Mpn029	Efp	Elongation factor P	core
				Mpn586	Mpn586	Uncharacterized protein Mpn586	core
	65	Complex 65	Function unknown	Mpn248	Mpn248	Putative serine/threonine-protein kinase	core
				Mpn504	Mpn504	UPF0134 protein Mpn504	core
	66	Complex 66	Function unknown	Mpn493	UlaD	Probable 3-keto-L-gulonate-6-phosphate decarboxylase	core
				Mpn596	Mpn596	Uncharacterized protein MG397 homolog	core
	67	Complex 67	Function unknown	Mpn350	Mpn350	UPF0078 membrane protein Mpn350	core
				Mpn401	GreA	Transcription elongation factor GreA	core
	68	Complex 68	Function unknown	Mpn329	Mpn329	Uncharacterized protein MG236 homolog	core
				Mpn440	Mpn440	Uncharacterized protein Mpn440	core
	69	Complex 69	Function unknown	Mpn302	PfkA	6-phosphofructokinase	core
				Mpn644	Mpn644	Uncharacterized lipoprotein MG439 homolog 3 precursor	core
	70	Complex 70	Function unknown	Mpn097	Mpn097	Uncharacterized lipoprotein Mpn097 precursor	core
				Mpn562	NadE	Probable NH_4 -dependent NAD^+ synthetase	core
	71	Complex 71	Function unknown	Mpn337	Mpn337	Uncharacterized protein MG241 homolog	core
				Mpn416	P29	Probable ABC transporter ATP-binding protein p29	core
	72	Complex 72	Function unknown	Mpn047	Mpn047	Uncharacterized protein MG037 homolog	core
				Mpn063	DeoC	Deoxyribose-phosphate aldolase	core
	73	Complex 73	Function unknown	Mpn529	Mpn529	Uncharacterized protein MG353 homolog	core
				Mpn646	Mpn646	Uncharacterized lipoprotein MG440 homolog 1 precursor	core
	74	Complex 74	Function unknown	Mpn051	Mpn051	Uncharacterized protein MG039 homolog	core
				Mpn627	PtsI	Phosphoenolpyruvate-protein phosphotransferase	core
	75	Complex 75	Function unknown	Mpn461	Mpn461	Uncharacterized protein MG323 homolog	core
				Mpn476	Cmk	Cytidylate kinase	core
	76	Complex 76	Function unknown	Mpn436	Mpn436	Uncharacterized lipoprotein MG307 homolog precursor	core
				Mpn492	UlaE	Probable L-ribulose-5-phosphate 3-epimerase	core
	77	Complex 77	Function unknown	Mpn126	Mpn126	Putative metallophosphoesterase MG207 homolog	core
				Mpn580	Mpn580	Uncharacterized protein Mpn580	core
	78	Complex 78	Function unknown	Mpn537	Mpn537	Uncharacterized protein MG360 homolog	core
				Mpn662	MsrB	Peptide methionine sulfoxide reductase	core
	79	Complex 79	Function unknown	Mpn307	Mpn307	Carbamate kinase-like protein	core
				Mpn590	Mpn590	Uncharacterized lipoprotein Mpn590 precursor	core
	80	Complex 80	Function unknown	Mpn081	Mpn081	Putative ABC transporter ATP-binding protein MG065 homolog	core
				Mpn505	Mpn505	Uncharacterized protein Mpn505	core
	83	Complex 83	Function unknown	Mpn005	SerS	Seryl-tRNA synthetase	core
				Mpn340	Mpn340	Probable DNA helicase	core
	85	Complex 85	Function unknown	Mpn027	Mpn027	Uncharacterized protein Mpn027	core
				Mpn628	GpmI	2,3-bisphosphoglycerate-independent phosphoglycerate mutase	core
	87	Complex 87	Function unknown	Mpn036	Mpn036	Uncharacterized protein Mpn036	core
				Mpn676	Mpn676	Uncharacterized protein Mpn676	core
	88	Complex 88	Function unknown	Mpn073	Prs	Ribose-phosphate pyrophosphokinase	core
				Mpn404	Mpn404	Uncharacterized protein MG285 homolog	core
	89	Complex 89	Function unknown	Mpn103	Mpn103	Uncharacterized protein Mpn103	core
				Mpn280	Mpn280	UPF0036 protein MG139 homolog	core
	91	Complex 91	Function unknown	Mpn210	SecA	Preprotein translocase subunit SecA	core
				Mpn379	PoiA	Probable 5'-3' exonuclease	core
	97	Complex 97	Function unknown	Mpn386	Mpn386	Uncharacterized protein MG268 homolog	core
				Mpn688	Mpn688	ParA family protein Mpn688	core
	98	Complex 98	Function unknown	Mpn413	Mpn413	Uncharacterized protein Mpn413	core
				Mpn475	EngA	GTP-binding protein EngA	core
	99	Complex 99	Function unknown	Mpn419	AlaS	Alanyl-tRNA synthetase	core
				Mpn582	Mpn582	Uncharacterized lipoprotein Mpn582 precursor	core
	106	Complex 106	Function unknown	Mpn025	Fba	Fructose-bisphosphate aldolase	core
				Mpn573	GroL	60 kDa chaperonin	core
	107	Complex 107	Function unknown	Mpn060	MetK	S-adenosylmethionine synthetase	core
				Mpn048	Mpn048	Uncharacterized protein Mpn048	core
	108	Complex 108	Function unknown	Mpn073	Prs	Ribose-phosphate pyrophosphokinase	core
				Mpn556	ArgS	Arginyl-tRNA synthetase	core
	110	Complex 110	Function unknown	Mpn189	RpsM	30S ribosomal protein S13	core
				Mpn629	TpiA	Triosephosphate isomerase	core
	111	Complex 111	Function unknown	Mpn623	Mpn623	Probable ATP-dependent RNA helicase MG425 homolog	core
				Mpn222	TilS	tRNA(Ile)-lysidine synthase	core
	112	Complex 112	Function unknown	Mpn340	Mpn340	Probable DNA helicase Mpn340	core
				Mpn521	Mpn521	Uncharacterized tRNA/rRNA methyltransferase MG346 homolog	core
	116	Complex 116	Function unknown	Mpn573	GroL	60 kDa chaperonin	core
				Mpn665	Tuf	Elongation factor Tu	core

Complex ID	Complex name	Systematic name	Protein name
ii) Homomultimeric complexes			
117	DNA polymerase III subunit β	Mpn001	DnaN
118	DNA gyrase subunit B	Mpn003	GyrB
119	Thymidylate kinase	Mpn006	Tmk
120	Uncharacterized protein MG007 homolog	Mpn007	Mpn007
121	Uncharacterized ATP-dependent helicase Mpn020	Mpn020	Mpn020
122	Putative proline iminopeptidase	Mpn022	Pip
123	Methionyl-tRNA synthetase	Mpn023	MetG
124	Fructose-biphosphate aldolase	Mpn025	Fba
125	DNA polymerase III PolC-type	Mpn034	PolC
126	Glycerol kinase	Mpn050	GlpK
127	Uncharacterized protein MG039 homolog	Mpn051	Mpn051
128	Deoxyribose-phosphate aldolase	Mpn063	DeoC
129	Phosphomannomutase	Mpn066	ManB
130	Putative 1-phosphofructokinase	Mpn079	FruK
131	Phenylalanyl-tRNA synthetase α chain	Mpn105	PheS
132	Phenylalanyl-tRNA synthetase β chain	Mpn106	PheT
133	Uncharacterized protein MG202 homolog	Mpn121	Mpn121
134	Putative Metallophosphoesterase	Mpn126	Mpn126
135	DNA-directed RNA polymerase subunit α	Mpn191	RpoA
136	HPr kinase/phosphorylase	Mpn223	HprK
137	Single stranded DNA binding protein	Mpn229	Ssb
138	Replicative DNA helicase	Mpn232	DnaB
139	Uncharacterized protein MG127 homolog	Mpn266	Mpn266
140	Probable inorganic polyphosphate/ATP-NAD kinase	Mpn267	PpnK
141	2',3'-cyclic-nucleotide 2'-phosphodiesterase	Mpn269	Mpn269
142	Uncharacterized protein MG134 homolog	Mpn275	Mpn275
143	6-phosphofructokinase /Phosphohexokinase	Mpn302	PfkA
144	Pyruvate kinase	Mpn303	Pyk
145	Protein MraZ	Mpn314	MraZ
146	Cell division protein	Mpn317	FtsZ
147	Thymidylate synthase	Mpn320	ThyA
148	Ribonucleoside-diphosphate reductase subunit β	Mpn322	NrdF
149	Ribonucleoside-diphosphate reductase subunit α	Mpn324	NrdE
150	Probable endonuclease 4	Mpn328	Nfo
151	Uncharacterized protein MG236 homolog	Mpn329	Mpn329
152	Uncharacterized protein MG246 homolog	Mpn349	Mpn349
153	Uncharacterized protein MG259 homolog	Mpn362	Mpn362
154	Uncharacterized protein MG263 homolog	Mpn381	Mpn381
155	Dihydrolipoyl lysine-residue acetyltransferase	Mpn391	PdhC
156	Pyruvate dehydrogenase E1 component subunit β	Mpn392	PdhB
157	Pyruvate dehydrogenase E1 component subunit α	Mpn393	PdhA
158	Alanyl-tRNA synthetase	Mpn419	AlaS
159	Uncharacterized protein MG293 homolog	Mpn420	Mpn420
160	Glyceraldehyde-3-phosphate-dehydrogenase	Mpn430	GapA

161	Putative Xaa-Pro aminopeptidase	Mpn470	PepP
162	FMN-dependent NADH-azoreductase	Mpn479	AzoR
163	Recombinase A	Mpn490	RecA
164	Probable L-ribulose-5-phosphate 3-epimerase UlaE	Mpn492	UlaE
165	Uncharacterized protein MG350 homolog	Mpn526	Mpn526
166	Chaperone protein ClpB	Mpn531	ClpB
167	Acetate kinase	Mpn533	AckA
168	Probable thiamine biosynthesis protein	Mpn550	ThiL
169	Uncharacterized protein MG377 homolog	Mpn555	Mpn555
170	Ribosomal RNA small subunit methyltransferase G	Mpn558	RsmG
171	Arginine deiminase-like protein	Mpn560	Mpn560
172	Probable NH ₃ -dependent NAD ⁺ synthetase	Mpn562	NadE
173	60 kDa chaperonin	Mpn573	GroL
174	Enolase	Mpn606	Eno
175	Putative type I restriction enzyme specificity protein	Mpn638	Mpn638
176	Elongation factor Tu	Mpn665	Tuf
177	L-lactate dehydrogenase	Mpn674	Ldh
178	Uncharacterized protein MG461 homolog	Mpn677	Mpn677

Table S5: List of literature curated protein complexes.

This table is a repository of literature evidence for i) 31 heteromultimeric and ii) 78 homomultimeric protein complexes that may exist in *M. pneumoniae* based on observed homologs in other species. In i) the first column states complex name. In columns 2-4 the table describes the protein components present in each complex, with their annotated systematic name, protein name and subunit/function, if known. In column 5, the PubMed ID or database evidence is highlighted. Column 6 specifies the organism for which the complex is observed. In ii) the first column provides the complex name. Columns 2 and 3 give the systematic and protein name of a homomultimer. Column 4 states the multimerization state of each complex. In column 5, the PubMed ID or database evidence is highlighted. Column 6 specifies the organism for which the complex is observed.

Complex name	Components			PubMed ID or source	Genus/ species
	Systematic name	Protein name	Subunit/function		
i) Heteromultimeric complexes					
DNA Gyrase complex	Mpn003	GyrB	B-subunit	Expasy	<i>M. pneumoniae</i>
	Mpn004	GyrA	A-subunit		
Phenylalanine-tRNA synthetase complex	Mpn105	SyfA	α -subunit	Expasy	<i>M. pneumoniae</i> by similarity
	Mpn106	SyfB	β -subunit		
RNA polymerase complex	Mpn024	RpoE	Probable δ -factor	Expasy	<i>M. pneumoniae</i> by similarity
	Mpn191	RpoA	α -subunit		
	Mpn352	RpoD	σ -factor		
	Mpn516	RpoB	β -subunit		
	Mpn515	RpoC	β' -subunit		
Transcription antitermination factor complex	Mpn067	NusG	Transcription antitermination	Expasy	by homology to <i>Bacillus subtilis</i>
	Mpn024	RpoE	Probable δ -factor		
	Mpn191	RpoA	α -subunit		
	Mpn352	RpoD	σ -factor		
	Mpn516	RpoB	β -subunit		
Pyruvate dehydrogenase complex	Mpn390	PdhD	E3 component Dihydrolipoyl dehydrogenase	Expasy	<i>M. pneumoniae</i> by similarity
	Mpn391	PdhC	E2 component Dihydrolipoamide acetyltransferase		
	Mpn392	PdhB	E1 component β -subunit		
	Mpn393	PdhA	E1 component α -subunit		
Ribonucleoside-diphosphate reductase complex	Mpn322	NrdF	α -subunit	Expasy	<i>M. pneumoniae</i> by similarity
	Mpn324	NrdE	β -subunit		
DNA Topoisomerase IV complex	Mpn122	ParE	b-subunit	Expasy	<i>M. pneumoniae</i>
	Mpn123	ParC	a-subunit		
DNA polymerase III complex	Mpn001	DnaN	β -subunit	Expasy	<i>M. pneumoniae</i> by similarity
	Mpn007	Mpn007	δ -subunit		
	Mpn034	PolC			
	Mpn378	DnaE	α -subunit		
	Mpn450	Mpn450	δ' -subunit		
	Mpn618	DnaX	γ/τ -subunit		
Translation elongation factor complex	Mpn631	Tsf		Expasy	<i>M. pneumoniae</i>
	Mpn665	Tuf			
GroEL-GroES complex	Mpn573	GroL		Expasy	<i>M. pneumoniae</i> by similarity
	Mpn574	GroS			
Restriction enzyme complexes	Mpn089	Mpn089	S protein	Expasy	<i>M. pneumoniae</i> by similarity
	Mpn201	Mpn201	S protein		
	Mpn285	Mpn285	S protein		
	Mpn289	Mpn289	S protein		
	Mpn290	Mpn290	S protein		
	Mpn342	Mpn342	M protein		
	Mpn343	Mpn343	S protein		
	Mpn345	Mpn345	R protein		
	Mpn347	Mpn347	R protein		
	Mpn365	Mpn365	S protein		
	Mpn507	Mpn507	S protein		
	Mpn615	Mpn615	S protein		
	Mpn638	Mpn638	S protein		

Ribosome complex	Mpn069	RpmG2	50S ribosomal protein L33 2	Expasy	<i>M. pneumoniae</i>
	Mpn116	RpmI	50S ribosomal protein L35		
	Mpn117	RplT	50S ribosomal protein L20		
	Mpn164	RpsJ	30S ribosomal protein S10		
	Mpn165	RplC	50S ribosomal protein L3		
	Mpn166	RplD	50S ribosomal protein L4		
	Mpn167	RplW	50S ribosomal protein L23		
	Mpn168	RplB	50S ribosomal protein L2		
	Mpn169	RpsS	30S ribosomal protein S19		
	Mpn170	RplV	50S ribosomal protein L22		
	Mpn171	RpsC	30S ribosomal protein S3		
	Mpn172	RplP	50S ribosomal protein L16		
	Mpn173	RpmC	50S ribosomal protein L29		
	Mpn174	RpsQ	30S ribosomal protein S17		
	Mpn175	RplN	50S ribosomal protein L14		
	Mpn176	RplX	50S ribosomal protein L24		
	Mpn177	RplE	50S ribosomal protein L5		
	Mpn178	RpsZ/RpsN	30S ribosomal protein S14 type Z		
	Mpn179	RpsH	30S ribosomal protein S8		
	Mpn180	RplF	50S ribosomal protein L6		
	Mpn181	RplR	50S ribosomal protein L18		
	Mpn182	RpsE	30S ribosomal protein S5		
	Mpn183	RplO	50S ribosomal protein L15		
	Mpn188	RpmJ	50S ribosomal protein L36		
	Mpn189	RpsM	30S ribosomal protein S13		
	Mpn190	RpsK	30S ribosomal protein S11		
	Mpn192	RplQ	50S ribosomal protein L17		
	Mpn208	RpsB	30S ribosomal protein S2		
	Mpn219	RplK	50S ribosomal protein L11		
	Mpn220	RplA	50S ribosomal protein L1		
	Mpn225	RpsL	30S ribosomal protein S12		
	Mpn226	RpsG	30S ribosomal protein S7		
	Mpn228	RpsF	30S ribosomal protein S6		
	Mpn230	RpsR	30S ribosomal protein S18		
	Mpn231	RplI	50S ribosomal protein L9		
	Mpn296	RpsU	30S ribosomal protein S21		
	Mpn325	RplU	50S ribosomal protein L21		
	Mpn327	RpmA	50S ribosomal protein L27		
	Mpn360	RpmE	50S ribosomal protein L31		
	Mpn446	RpsD	30S ribosomal protein S4		
	Mpn471	RpmG1	50S ribosomal protein L33 1		
	Mpn538	RplJ	50S ribosomal protein L10		
	Mpn539	RplL	50S ribosomal protein L7/L12		
	Mpn540	RpmF	50S ribosomal protein L32		
	Mpn541	RpsT	30S ribosomal protein S20		
	Mpn616	RpsI	30S ribosomal protein S9		
	Mpn617	RplM	50S ribosomal protein L13		
	Mpn622	RpsO	30S ribosomal protein S15		
	Mpn624	RpmB	50S ribosomal protein L28		
	Mpn658	RplS	50S ribosomal protein L19		
	Mpn660	RpsP	30S ribosomal protein S16		
	Mpn682	RpmH	50S ribosomal protein L34		
Translation initiation factor 1- Ribosome complex	Mpn187	InfA	Translation initiation factor IF-1	17329809, 11959450	by homology to <i>Mycobacterium tuberculosis</i>
	Ribosome	Ribosome			
Translation initiation factor 2- Ribosome complex	Mpn155	InfB	Translation initiation factor IF-2	Expasy	<i>M. pneumoniae</i> by similarity
	Ribosome	Ribosome			

Cohesin-like complex	Mpn300	ScpA		Expasy	<i>M. pneumoniae</i> by similarity
	Mpn301	ScpB			
	Mpn426	Smc			
RbfA-30S-ribosomal subunit complex	Mpn156	RbfA	Ribosome-binding factor A	15866174	by homology to <i>Thermus thermophilus</i>
	30S ribosomal subunit	30S ribosomal subunit			
Glycolytic enzyme complex 1	Mpn430	GapA	Glyceraldehyde-3-phosphate dehydrogenase	15701694	by homology to <i>Mus musculus</i>
	Mpn025	Fba	Fructose-bisphosphate aldolase		
	Mpn674	Ldh	L-lactate dehydrogenase		
	Mpn303	Pyk	Pyruvate kinase		
	Mpn302	PfkA	6-phosphofructokinase		
Dnak-GrpE-DnaJ complex	Mpn120	GrpE	Protein GrpE	15632130	By homology to <i>Escherichia coli</i>
	Mpn434	DnaK	Chaperone protein DnaK		
	Mpn021	DnaJ	Chaperone protein dnaJ		
Glycolytic enzyme complex 2	Mpn430	GapA	Glyceraldehyde-3-phosphate dehydrogenase	8613464	by homology to MCF-7 cells (eukaryotes)
	Mpn429	Fba	Phosphoglycerate kinase		
	Mpn606	Ldh	Enolase		
	Mpn303	Pyk	Pyruvate kinase		
Era-30S-ribosomal subunit complex	Mpn568	Era		15866174	by homology to <i>Thermus thermophilus</i>
	30S ribosomal subunit	30S ribosomal subunit			
GreA-RNA polymerase complex	Mpn401	GreA	Transcription elongation factor	17207814	<i>M. pneumoniae</i> by similarity
	Mpn024	RpoE	probable δ -factor		
	Mpn191	RpoA	α -subunit		
	Mpn352	RpoD	σ -factor		
	Mpn516	RpoB	β -subunit		
	Mpn515	RpoC	β' -subunit		
Dihydrofolate reductase-thymidylate synthase complex	Mpn321	FolA		Expasy	other
	Mpn320	ThyA			
Holliday junction ATP-dependent helicase complex	Mpn536	RuvB		Expasy	<i>M. pneumoniae</i>
	Mpn535	RuvA			
Aspartyl-glutamyl-tRNA(Asn/Gln) amidotransferase complex	Mpn238	GatB		Expasy	<i>M. pneumoniae</i> by similarity
	Mpn237	GatA			
	Mpn236	GatC			
UvrABC complex	Mpn125	UvrC		Expasy	<i>M. pneumoniae</i> by similarity
	Mpn619	UvrA			
	Mpn211	UvrB			
Phosphotransfer system maltose complex	Mpn053	PtsH		17803963	by homology to <i>Escherichia coli</i>
	Mpn651	MtlA			
	Mpn653	MtlF			
DnaB-DnaG complex	Mpn232	DnaB		17947583	by homology to <i>Escherichia coli</i>
	Mpn353	DnaG			
Phosphotransfer system glucose complex	Mpn053	PtsH		Expasy	<i>M. pneumoniae</i> by similarity
	Mpn207	PtsG			
Phosphotransfer system fructose complex	Mpn053	PtsH		Expasy	<i>M. pneumoniae</i> by similarity
	Mpn078	FruA			
Phosphotransfer system maltose complex	Mpn053	PtsH		Expasy	<i>M. pneumoniae</i> by similarity
	Mpn494	UlaC			
	Mpn495	UlaB			
	Mpn496	UlaA			
tRNA modification complex	Mpn557	MnmG		17062623	by homology to <i>Escherichia coli</i>
	Mpn008	TrmE			

Protein function	Systematic name	Protein name	Subunits per complex	PubMed ID or source	Genus/ species
ii) Homomultimeric complexes					
Cell division MraZ	Mpn314	MraZ	8	Expasy	<i>M. pneumoniae</i>
Pyruvate kinase	Mpn303	Pyk	4	Expasy	<i>M. pneumoniae</i> by similarity
Thymidylate synthase	Mpn320	ThyA	2	Expasy	<i>M. pneumoniae</i> by similarity
L-lactate dehydrogenase	Mpn674	Ldh	4	Expasy	<i>M. pneumoniae</i> by similarity
Fructose-biphosphate aldolase	Mpn025	Fba	2	Expasy	<i>M. pneumoniae</i> by similarity
Cell division FtsZ	Mpn317	FtsZ	multiple/ aggregates to form a ring-like structure	17977836	<i>M. pneumoniae</i> by similarity
HPr kinase/ phosphorylase	Mpn223	HprK	6	Expasy	<i>M. pneumoniae</i>
6-phosphofructokinase /Phosphohexokinase	Mpn302	PfkA	multiple	16289704	by homology to <i>Escherichia coli</i>
FMN-dependent NADH-azoreductase	Mpn479	AzoR	2	16684776	by homology to <i>Escherichia coli</i>
Glyceraldehyde-3-phosphate-dehydrogenase	Mpn430	GapA	4	Expasy	<i>M. pneumoniae</i> by similarity
Acetate kinase	Mpn533	AckA	2/4	216674, 29037,	by homology to <i>Veillonella alcalescens</i>
Alanyl-tRNA synthetase	Mpn419	AlaS	2	9207019	by homology to <i>Thermus thermophilus</i>
Probable inorganic polyphosphate/ATP-NAD kinase	Mpn267	PpnK	2	11006082	by homology to <i>Micrococcus flavus</i> & <i>Mycobacterium tuberculosis</i>
Recombinase A	Mpn490	RecA	6	17955055	by homology to <i>Escherichia coli</i>
Chaperone protein ClpB	Mpn531	ClpB	6	Expasy	<i>M. pneumoniae</i> by similarity
Replicative DNA helicase	Mpn232	DnaB	6	17947583	by homology to <i>Escherichia coli</i>
Glycerol kinase	Mpn050	GlpK	2/4	9930671	by homology to <i>Escherichia coli</i>
Putative proline iminopeptidase	Mpn022	Pip	multiple	8885412	by homology <i>Xanthomonas campestris pv. citri</i>
Methionyl-tRNA synthetase	Mpn023	MetG		2126467	by homology to <i>Escherichia coli</i>
Enolase	Mpn606	Eno	2	Expasy	<i>M. pneumoniae</i> by similarity
Putative Metallophosphoesterase	Mpn126	Mpn126	4	17586769	by homology to <i>Escherichia coli</i>
Single stranded DNA binding protein	Mpn229	Ssb	4	16041080	by homology to <i>Mycobacterium smegmatis</i>
Arginine deiminase-like protein	Mpn560	Mpn560	2/3	17223359, 11911465	by homology to <i>Lactococcus lactis</i>

DNA polymerase III subunit β	Mpn001	DnaN	2	Expasy	by homology to <i>Streptococcus pneumoniae</i>
DNA gyrase subunit B	Mpn003	GyrB	2	11850422	by homology to <i>Thermus thermophilus</i>
Thymidylate kinase	Mpn006	Tmk	2	11071809	by homology to <i>Homo sapiens</i>
Uncharacterized protein MG007 homolog	Mpn007	Mpn007	6	11545747	by homology to <i>Pyrococcus furiosus</i>
DNA polymerase III polC-type	Mpn034	PolC	2	Expasy	<i>M. pneumoniae</i> by similarity
Deoxyribose-phosphate aldolase	Mpn063	DeoC	1/2	7675789	by homology to <i>Escherichia coli</i>
Phosphomannomutase	Mpn066	ManB	2	16540464	by homology to <i>Sulfolobus Tokodaii</i>
Putative 1-phosphofructokinase	Mpn079	FruK	8	2527305	by homology to <i>Homo sapiens</i>
Phenylalanyl-tRNA synthetase α chain	Mpn105	PheS	2	7664121	by homology to <i>Thermus thermophilus</i>
Phenylalanyl-tRNA synthetase β chain	Mpn106	PheT	2	7664121	by homology to <i>Thermus thermophilus</i>
DNA-directed RNA polymerase subunit α	Mpn191	RpoA	2	Expasy	<i>M. pneumoniae</i> by similarity
Ribonucleoside-diphosphate reductase subunit β	Mpn322	NrdF	2	16301799	by homology to <i>Salmonella typhimurium</i>
Ribonucleoside-diphosphate reductase subunit α	Mpn324	NrdE	2	8052308	by homology to <i>Salmonella typhimurium</i>
Probable endonuclease 4	Mpn328	Nfo	2	10458614	by homology to <i>Escherichia coli</i>
Dihydrolipoyllysine-residue acetyltransferase	Mpn391	PdhC	60	9990008	<i>M. pneumoniae</i> by similarity
Pyruvate dehydrogenase E1 component subunit β	Mpn392	PdhB	2	Expasy	<i>M. pneumoniae</i> by similarity
Pyruvate dehydrogenase E1 component subunit α	Mpn393	PdhA	2	Expasy	<i>M. pneumoniae</i> by similarity
Putative Xaa-Pro aminopeptidase	Mpn470	PepP	2	12377124	by homology to <i>Pyrococcus horikoshii</i>
Probable L-ribulose-5-phosphate 3-epimerase UlaE	Mpn492	UlaE	4	11732895	by homology to <i>Pseudomonas cichorii</i>
Probable thiamine biosynthesis protein	Mpn550	ThiL	2	16343540	by homology to <i>Bacillus anthracis</i>
Ribosomal RNA small subunit methyltransferase G	Mpn558	RsmG	2	18667428	by homology to <i>Geobacter sulfurreducens</i>
Probable NH(3)-dependent NAD(+) synthetase	Mpn562	NadE	2	15645437	by homology to <i>Bacillus subtilis</i>
60 kDa chaperonin	Mpn573	GroL	2x7	Expasy	<i>M. pneumoniae</i> by similarity
Putative type I restriction enzyme specificity protein	Mpn638	Mpn638	2	16038930	by homology to <i>Methanocaldococcus jannaschii</i>
Elongation factor Tu	Mpn665	Tuf	2	16257965	by homology to <i>Escherichia coli</i>
Dihydrolipoyl dehydrogenase	Mpn390	PdhD	2	Expasy	<i>M. pneumoniae</i> by similarity
10 kDa chaperonin	Mpn574	GroS	7	Expasy	<i>M. pneumoniae</i> by similarity

DNA topoisomerase 4 subunit B	Mpn122	ParE	2	Expasy	<i>M. pneumoniae</i> by similarity
ATP-dependent protease Lon	Mpn332	Lon	4	Expasy	<i>M. pneumoniae</i> by similarity
Probable tRNA modification GTPase	Mpn008	TrmE	2	Expasy	by homology to <i>Thermotoga maritima</i>
UDP-galactopyranose mutase	Mpn278	GlF	2	10089346	by homology to <i>Escherichia coli</i>
Uridylate Kinase Mpn632	Mpn632	PyrH	6	17297917	by homology to <i>Sulfolobus solfataricus</i>
Ribose-5-phosphate-isomerase-B	Mpn595	RpiB	4	15162497	by homology to <i>Thermotoga maritima</i>
Methionine aminopeptidase, Peptidase M	Mpn186	Map	2	12297034	by homology to <i>Bacillus stearothermophilus</i>
FAD synthetase	Mpn158	RibF	2	15468322	by homology to <i>Thermotoga maritima</i>
GMP kinase	Mpn245	Def	2	17012781	by homology to <i>Staphylococcus aureus</i>
Acyl carrier protein (ACP) synthase	Mpn298	AcpS	2	16788183	<i>M. pneumoniae</i>
Adenine phosphoribosyltransferase	Mpn395	Apt	2	10393170, 11535055	by homology to <i>Leishmania donovani</i>
Threonyl-tRNA synthetase	Mpn553	ThrS	2	Expasy	<i>M. pneumoniae</i> by similarity
Thymidine phosphorylase	Mpn064	DeoA	multiple	Expasy	<i>M. pneumoniae</i> by similarity
Ribose-phosphate pyrophosphokinase	Mpn073	Prs	multiple	2169413	by homology to <i>Bacillus subtilis</i>
Protein GrpE	Mpn120	GrpE	2	Expasy	<i>M. pneumoniae</i> by similarity
UDP-glucose 4-epimerase	Mpn257	GalE	2	Expasy	<i>M. pneumoniae</i> by similarity
Tryptophanyl-tRNA synthetase	Mpn265	TrpS	2	Expasy	<i>M. pneumoniae</i> by similarity
Replication initiator protein DnaA	Mpn686	DnaA	multiple	16829961	by homology to <i>Aquifex aeolicus</i>
Triosephosphate isomerase	Mpn629	TpiA	2	Expasy	<i>M. pneumoniae</i> by similarity
Serine hydroxymethyltransferase	Mpn576	GlyA	4	Expasy	<i>M. pneumoniae</i> by similarity
Phosphoenolpyruvate-protein transferase	Mpn627	PtsL	2	Expasy	<i>M. pneumoniae</i> by similarity
Glycyl-tRNA synthetase	Mpn354	GlyqS	2	Expasy	<i>M. pneumoniae</i> by similarity
Seryl-tRNA synthetase	Mpn005	SerS	2	Expasy	<i>M. pneumoniae</i> by similarity
Bifunctional protein FoID	Mpn017	FoID	2	Expasy	<i>M. pneumoniae</i> by similarity
Lysyl-tRNA synthase	Mpn277	LysS	2	Expasy	<i>M. pneumoniae</i> by similarity
Histidyl-tRNA synthase	Mpn045	HisS	2	Expasy	<i>M. pneumoniae</i> by similarity
Thioredoxin reductase	Mpn240	TrxB	2	Expasy	<i>M. pneumoniae</i> by similarity
Chaperone protein dnaJ	Mpn021	DnaJ	2	Expasy	<i>M. pneumoniae</i> by similarity

Table S6: List of modeled protein complexes.

This table is a list of binary interactions that can be modeled for all protein complexes in this study. It is divided into i) a set of 29 heteromultimeric complexes and ii) a set of 57 homomultimeric complexes. In part i) the first column gives the unique complex ID for each protein complex modeled. Column 2 provides the complex name. In columns 3 and 4 give the systematic name and protein name of the first interactor modeled, columns 5 and 6 give the systematic name and protein name of the second interactor modeled. Columns 7 and 8 give the PDB ID and the species this information was obtained from. In part ii) the first column on the left shows the complex ID of the homomultimeric protein complexes modeled. The second column indicates the complex name. Columns 3 and 4 give the annotated systematic and protein name of each homomultimer. Columns 5 and 6 give the PDB ID and the species this information was obtained from. (* novel homomultimers supported by structural models)

Complex ID	Complex name	Interactor 1		Interactor 2		PDB ID	PDB species
		Systematic name	Protein name	Systematic name	Protein name		
i) Heteromultimeric complexes							
3	DNA Polymerase III γ complex	Mpn007	Mpn007	Mpn450	Mpn450	1sxj	<i>S. cerevisiae</i>
		Mpn007	Mpn007	Mpn618	DnaX	1njf	<i>E. coli</i>
		Mpn450	Mpn450	Mpn618	DnaX	1iqp	<i>P. furiosus</i>
8	Protein chaperone complex	Mpn002	Mpn002	Mpn434	DnaK	2qwq	<i>B. taurus</i>
		Mpn021	DnaJ	Mpn002	Mpn002	2och	<i>C. elegans</i>
		Mpn021	DnaJ	Mpn434	DnaK	2qwq	<i>B. taurus</i>
		Mpn120	GrpE	Mpn434	DnaK	1dkg	<i>E. coli</i>
		Mpn120	GrpE	Mpn531	ClpB	2tma	<i>O. cuniculus</i>
9	Dnak-GrpE complex	Mpn120	GrpE	Mpn434	DnaK	1dkg	<i>E. coli</i>
10	Aminoacyl-tRNA synthetase complex	Mpn023	MetG	Mpn678	GltX	1g59	<i>T. thermophilus</i>
		Mpn105	PheS	Mpn106	PheT	2rhs	<i>S. haemolyticus</i>
		Mpn105	PheS	Mpn553	ThrS	2zcd	<i>M. mazei</i>
		Mpn106	PheT	Mpn553	ThrS	2e3c	<i>M. mazei</i>
11	Phenylalanine-tRNA synthetase complex	Mpn105	PheS	Mpn106	PheT	2rhs	<i>S. haemolyticus</i>
15	DNA Recombination complex	Mpn122	ParE	Mpn123	ParC	1bjt	<i>S. cerevisiae</i>
17	DNA Topoisomerase IV complex	Mpn122	ParE	Mpn123	ParC	1bjt	<i>S. cerevisiae</i>
19	Complex 19	Mpn076	Mpn076	Mpn284	Mpn284	2r6g	<i>E. coli</i>
25	Complex 25	Mpn223	HprK	Mpn258	Mpn258	1kkm	<i>L. casei</i>
34	Restriction enzyme complex	Mpn185	Adk	Mpn671	FtsH	2rhm	<i>C. aurantiacus</i>
35	Restriction enzyme complex	Mpn089	Mpn089	Mpn507	Mpn507	1yf2	<i>M. jannaschii</i>
36	Complex 36	Mpn008	TrmE	Mpn426	P115	2ocy	<i>S. cerevisiae</i>
		Mpn194	CbiO2	Mpn426	P115	1mv5	<i>L. lactis</i>
		Mpn574	GroS	Mpn573	GroL	1gru	<i>E. coli</i>
37	GroEL-GroES complex	Mpn574	GroS	Mpn573	GroL	1gru	<i>E. coli</i>
38	Peptidase complex	Mpn218	OppF	Mpn609	PstB	2ouk	<i>G. stearothermophilus</i>
40	Cohesin like complex	Mpn426	P115	Mpn508	Mpn508	3bat	<i>A. irradians</i>
44	Pyruvate dehydrogenase complex	Mpn008	TrmE	Mpn426	P115	2ocy	<i>S. cerevisiae</i>
		Mpn392	PdhB	Mpn393	PdhA	1w85	<i>G. stearothermophilus</i>
		Mpn574	GroS	Mpn573	GroL	1gru	<i>E. coli</i>
45	Glycolytic enzyme complex 2	Mpn002	Mpn002	Mpn434	DnaK	2qwq	<i>B. taurus</i>
		Mpn120	GrpE	Mpn434	DnaK	1dkg	<i>E. coli</i>
		Mpn120	GrpE	Mpn531	ClpB	2tma	<i>O. cuniculus</i>

47	ATP synthase complex	Mpn598	AtpD	Mpn600	AtpA	2jj1	<i>B. taurus</i>
49	RNA polymerase complex	Mpn020	Mpn020	Mpn153	Mpn153	2qeq	<i>Kunjin virus</i>
		Mpn020	Mpn020	Mpn443	Mpn443	8ohm	<i>Hepatitis C virus</i>
		Mpn153	Mpn153	Mpn443	Mpn443	2qeq	<i>Kunjin virus</i>
		Mpn153	Mpn153	Mpn490	RecA	2gbl	<i>S. elongatus</i>
		Mpn191	RpoA	Mpn266	Mpn266	1z3e	<i>B. subtilis</i>
		Mpn191	RpoA	Mpn515	RpoC	2a69	<i>T. thermophilus</i>
		Mpn191	RpoA	Mpn516	RpoB	1ynj	<i>T. aquaticus</i>
		Mpn252	AsnS	Mpn354	GlyQS	3bju	<i>H. sapiens</i>
		Mpn261	TopA	Mpn515	RpoC	2nvq	<i>S. cerevisiae</i>
		Mpn261	TopA	Mpn516	RpoB	2r7z	<i>S. cerevisiae</i>
		Mpn352	RpoD	Mpn515	RpoC	2a69	<i>T. thermophilus</i>
		Mpn352	RpoD	Mpn516	RpoB	1smv	<i>T. thermophilus</i>
		Mpn515	RpoC	Mpn516	RpoB	1ynn	<i>T. aquaticus</i>
50	Ribosome complex	Mpn006	Tmk	Mpn386	Mpn386	2ccg	<i>S. aureus</i>
		Mpn006	Tmk	Mpn536	RuvB	2ar7	<i>H. sapiens</i>
		Mpn137	Mpn137	Mpn155	InfB	2ocy	<i>S. cerevisiae</i>
		Mpn137	Mpn137	Mpn475	EngA	2ocy	<i>S. cerevisiae</i>
		Mpn137	Mpn137	Mpn524	Mpn524	2ba2	<i>M. pneumoniae</i>
		Mpn155	InfB	Mpn475	EngA	2zej	<i>H. sapiens</i>
		Mpn155	InfB	Mpn524	Mpn524	2eqb	<i>S. cerevisiae</i>
		Mpn155	InfB	Mpn665	Tuf	2qmu	<i>S. solfataricus</i>
		Mpn165	RplC	Mpn658	RplS	1voy	<i>E. coli</i>
		Mpn167	RplW	Mpn174	RpsQ	1ml5	<i>E. coli</i>
		Mpn169	RpsS	Mpn189	RpsM	2vqe	<i>T. thermophilus</i>
		Mpn177	RplE	Mpn446	RpsD	1ml5	<i>E. coli</i>
		Mpn178	RpsZ	Mpn171	RpsC	2i2p	<i>E. coli</i>
		Mpn179	RpsH	Mpn182	RpsE	2avy	<i>E. coli</i>
		Mpn179	RpsH	Mpn208	RpsB	1s1h	<i>S. cerevisiae</i>
		Mpn182	RpsE	Mpn446	RpsD	1fjg	<i>T. thermophilus</i>
		Mpn230	RpsR	Mpn228	RpsF	2vho	<i>E. coli</i>
		Mpn386	Mpn386	Mpn536	RuvB	1sxj	<i>S. cerevisiae</i>
		Mpn475	EngA	Mpn524	Mpn524	2eqb	<i>S. cerevisiae</i>
		Mpn475	EngA	Mpn665	Tuf	2zej	<i>H. sapiens</i>
		Mpn536	RuvB	Mpn571	Mpn571	2chg	<i>A. fulgidus</i>
		Mpn571	Mpn571	Mpn685	Mpn685	2olk	<i>G. stearothermophilus</i>
52	Translation elongation factor complex	Mpn631	Tsf	Mpn665	Tuf	1xb2	<i>B. taurus</i>
54	Phosphotransfer system complex	Mpn207	PtsG	Mpn268	Mpn268	1o2f	<i>E. coli</i>
82	DNA Gyrase complex	Mpn003	GyrB	Mpn004	GyrA	1bjt	<i>S. cerevisiae</i>
92	Complex 92	Mpn258	Mpn258	Mpn259	Mpn259	2onk	<i>A. fulgidus</i>
93	ScpA-ThyA complex	Mpn300	ScpA	Mpn320	ThyA	2oip	<i>C. hominis</i>
94	Ribonucleoside-diphosphate reductase complex	Mpn322	NrdF	Mpn324	NrdE	2bq1	<i>S. typhimurium</i>
102	ParE-GyrA complex	Mpn004	GyrA	Mpn122	ParE	1bjt	<i>S. cerevisiae</i>
113	Restriction enzyme complex	Mpn343	Mpn343	Mpn365	Mpn365	1yf2	<i>M. jannaschii</i>
115	Restriction enzyme complex	Mpn365	Mpn365	Mpn615	Mpn615	1yf2	<i>M. jannaschii</i>

Complex ID	Complex name	Systematic name	Protein name	PDB ID	PDB species
ii) Homomultimeric complexes					
117	DNA polymerase III subunit β	Mpn001	DnaN	1mmi	<i>E. coli</i>
118	DNA gyrase subunit B	Mpn003	GyrB	1kij	<i>T. thermophilus</i>
119	Thymidylate kinase	Mpn006	Tmk	2uvq	<i>H. sapiens</i>
120	Uncharacterized protein MG007 homolog	Mpn007	Mpn007	1njf	<i>E. coli</i>
121*	Uncharacterized ATP-dependent helicase Mpn020	Mpn020	Mpn020	2qeq	<i>Kunjin virus</i>
122	Putative proline iminopeptidase	Mpn022	Pip	1xkt	<i>H. sapiens</i>
124	Fructose-biphosphate aldolase	Mpn025	Fba	1rv8	<i>T. aquaticus</i>
125	DNA polymerase III PolC-type	Mpn034	PolC	2p1j	<i>T. maritima</i>
126	Glycerol kinase	Mpn050	GlpK	2dpn	<i>T. thermophilus</i>
127*	Uncharacterized protein MG039 homolog	Mpn051	Mpn051	1ryi	<i>B. subtilis</i>
128	Deoxyribose-phosphate aldolase	Mpn063	DeoC	1jcj	<i>E. coli</i>
129	Phosphomannomutase	Mpn066	ManB	2f7l	<i>S. tokodaii</i>
130	Putative 1-phosphofructokinase	Mpn079	FruK	1gqt	<i>E. coli</i>
131	Phenylalanyl-tRNA synthetase α chain	Mpn105	PheS	1eqr	<i>E. coli</i>
132	Phenylalanyl-tRNA synthetase β chain	Mpn106	PheT	3bu2	<i>S. saprophyticus</i>
134	Putative Metallophosphoesterase	Mpn126	Mpn126	1z2w	<i>M. musculus</i>
135	DNA-directed RNA polymerase subunit α	Mpn191	RpoA	2cw0	<i>T. thermophilus</i>
136	HPr kinase/phosphorylase	Mpn223	HprK	1knx	<i>M. pneumoniae</i>
137	Single stranded DNA binding protein	Mpn229	Ssb	1s3o	<i>H. sapiens</i>
139*	Uncharacterized protein MG127 homolog	Mpn266	Mpn266	1r7h	<i>C. ammoniagenes</i>
140	Probable inorganic polyphosphate/ATP-NAD kinase	Mpn267	PpnK	1y3h	<i>M. tuberculosis</i>
141*	2',3'-cyclic-nucleotide 2'-phosphodiesterase	Mpn269	Mpn269	2pau	<i>E. coli</i>
142*	Uncharacterized protein MG134 homolog	Mpn275	Mpn275	1j8b	<i>H. Influenzae</i>
143	6-phosphofructokinase /Phosphohexokinase	Mpn302	PfkA	1pfk	<i>E. coli</i>
144	Pyruvate kinase	Mpn303	Pyk	2e28	<i>G. stearothermophilus</i>
146	Cell division protein	Mpn317	FtsZ	1w58	<i>M. jannaschii</i>
147	Thymidylate synthase	Mpn320	ThyA	1hw4	<i>H. sapiens</i>
148	Ribonucleoside-diphosphate reductase subunit β	Mpn322	NrdF	2bq1	<i>S. typhimurium</i>
149	Ribonucleoside-diphosphate reductase subunit α	Mpn324	NrdE	2bq1	<i>S. typhimurium</i>
150	Probable endonuclease 4	Mpn328	Nfo	1did	<i>Arthrobacter sp. nrml</i>
151*	Uncharacterized protein MG236 homolog	Mpn329	Mpn329	2fe3	<i>B. subtilis</i>
152*	Uncharacterized protein MG246 homolog	Mpn349	Mpn349	1t71	<i>M. pneumoniae</i>
153*	Uncharacterized protein MG259 homolog	Mpn362	Mpn362	2ipx	<i>H. sapiens</i>
154*	Uncharacterized protein MG263 homolog	Mpn381	Mpn381	2p9j	<i>A. aeolicus</i>
155	Dihydrolipoyl lysine-residue acetyltransferase	Mpn391	PdhC	1c4t	<i>E. coli</i>
156	Pyruvate dehydrogenase E1 component subunit β	Mpn392	PdhB	1w85	<i>G. stearothermophilus</i>
157	Pyruvate dehydrogenase E1 component subunit α	Mpn393	PdhA	2r8p	<i>E. coli</i>
158	Alanyl-tRNA synthetase	Mpn419	AlaS	1vhx	<i>B. subtilis</i>
159*	Uncharacterized protein MG293 homolog	Mpn420	Mpn420	2pz0	<i>T. tengcongensis</i>
160	Glyceraldehyde-3-phosphate-dehydrogenase	Mpn430	GapA	1nqa	<i>G. stearothermophilus</i>
161	Putative Xaa-Pro aminopeptidase	Mpn470	PepP	1wn1	<i>P. horikoshii</i>
162	FMN-dependent NADH-azoreductase	Mpn479	AzoR	2z9c	<i>E. coli</i>
163	Recombinase A	Mpn490	RecA	1oft	<i>P. aeruginosa</i>
164	Probable L-ribulose-5-phosphate 3-epimerase UlaE	Mpn492	UlaE	1k77	<i>E. coli</i>
166	Chaperone protein ClpB	Mpn531	ClpB	2tma	<i>O. cuniculus</i>
167	Acetate kinase	Mpn533	AckA	1g99	<i>M. thermophila</i>
168	Probable thiamine biosynthesis protein	Mpn550	Thil	1vbk	<i>P. horikoshii</i>
169*	Uncharacterized protein MG377 homolog	Mpn555	Mpn555	1zxj	<i>M. pneumoniae</i>
170	Ribosomal RNA small subunit methyltransferase G	Mpn558	GidB	1ne2	<i>T. acidophilum</i>
171	Arginine deiminase-like protein	Mpn560	Mpn560	2ci5	<i>B. taurus</i>
172	Probable NH ₃ -dependent NAD ⁺ synthetase	Mpn562	NadE	1wxg	<i>E. coli</i>
173	60 kDa chaperonin	Mpn573	GroL	2nwc	<i>E. coli</i>
174	Enolase	Mpn606	Eno	1pdz	<i>H. gammarus</i>
175	Putative type I restriction enzyme specificity protein	Mpn638	Mpn638	1yf2	<i>M. jannaschii</i>
176	Elongation factor Tu	Mpn665	Tuf	2bvn	<i>E. coli</i>
177	L-lactate dehydrogenase	Mpn674	Ldh	1uxg	<i>C. aurantiacus</i>
178*	Uncharacterized protein MG461 homolog	Mpn677	Mpn677	2q14	<i>B. thetaiotaomicron</i>

Table S7: List of multifunctional proteins.

This table gives a list of all multifunctional proteins. The first column indicates the annotated systematic name. The second column provides the protein name. The third column gives all complexes a multifunctional protein clusters to.

Systematic name	Protein name	Complexes
Mpn003	GyrB	35 83 58 59 112 99 34 46 56 101 96 29
Mpn004	GyrA	104 67 102 26 17 106 105 100 82 6 75 83 36 3 94 111 47 59 112 37 101 73
Mpn006	Tmk	90 17 15
Mpn007	Mpn007	28 3 10 105
Mpn011	Mpn011	6 4 99 3 5
Mpn014	Mpn014	84 103 106 97 10 12
Mpn020	Mpn020	104 11 26 17 48 93 106 3 97 61 94 108 98 101 43 10 86
Mpn025	Fba	28 106 97 9 105
Mpn027	Mpn027	109 85 79
Mpn034	PolC	112 90 21 83 86
Mpn042	Mpn042	74 60 1 24 14
Mpn048	Mpn048	14 107
Mpn060	MetK	16 17 15 107
Mpn061	Ffh	28 97 105 47
Mpn062	DeoD	92 11 56 97 10 62 54
Mpn063	DeoC	33 60 67 72 79
Mpn067	Mpn067	106 65
Mpn073	Prs	88 18 108 96 55
Mpn078	FruA	71 23 41
Mpn081	Mpn081	87 80 115
Mpn089	Mpn089	35 109
Mpn094	Mpn094	22 75
Mpn097	Mpn097	69 70
Mpn100	Mpn100	81 53 95
Mpn103	Mpn103	33 89 51
Mpn106	PheT	11 28 94 97 12 22 10 105 82
Mpn108	Mpn108	112 71 83 73
Mpn115	InfC	60 51 55 110
Mpn118	Mpn118	99 106 13 105 29 20
Mpn122	ParE	104 95 102 17 82
Mpn124	HrcA	73 13
Mpn141	MgpA	59 11 20
Mpn142	Mpn142	62 20
Mpn153	Mpn153	38 4 3 97 48
Mpn155	InfB	7 13 5 110
Mpn165	RplC	52 98 32 116 23 110
Mpn166	RplD	52 98 33 116 110
Mpn169	RpsS	51 110 55
Mpn177	RplE	111 79
Mpn189	RpsM	51 23 110
Mpn191	RpoA	67 53 51 18 106 10 105 110
Mpn192	RplQ	24 68
Mpn207	PtsG	3 64 68 12 47 79 48 98 18 93 86 54
Mpn210	SecA	74 67 33 32 90 91 98 103 56 24 43 86 105 100
Mpn211	UvrB	33 93
Mpn218	OppF	38 67 85
Mpn221	Pth	112 22 83
Mpn222	TilS	103 94 23 111
Mpn228	RpsF	33 53 57 51 110
Mpn230	RpsR	59 51 105
Mpn246	Def	6 4 34 3 100 5
Mpn257	GalE	112 27 109 28 83
Mpn258	Mpn258	25 92 3 68 86 47 48

Mpn264	Mpn264	36 75
Mpn269	Mpn269	25 77
Mpn277	LysS	25 53 76
Mpn280	Mpn280	53 64 89
Mpn284	Mpn284	19 86
Mpn294	Mpn294	60 57
Mpn300	ScpA	69 93 40 24
Mpn307	Mpn307	85 79
Mpn311	Mpn311	6 4 3 68 5
Mpn314	MraZ	42 71 66 68 43
Mpn315	MraW	43 105
Mpn316	Mpn316	46 68
Mpn320	ThyA	69 104 93 94 19
Mpn321	FolA	67 64 105
Mpn322	NrdF	11 63 94
Mpn324	NrdE	94 41
Mpn328	Nfo	61 24
Mpn329	Mpn329	11 67 68 79
Mpn330	Mpn330	60 61
Mpn331	Tig	32 90 13 65
Mpn340	Mpn340	112 27 28 83 101 102 82
Mpn341	UvrD	25 104 102 26 82
Mpn342	Mpn342	109 35 34 30 100 5
Mpn345	Mpn345	81 95 100 17
Mpn346	Mpn346	27 77 73 7 13 31 79
Mpn347	Mpn347	112 27 67 28 83 10 105
Mpn350	Mpn350	67 85
Mpn352	RpoD	84 103 67 37 61 51
Mpn354	GlyQS	60 28 97 105
Mpn361	PrfA	35 74 99 56 108 96
Mpn365	Mpn365	95 105 115 79 113 14
Mpn366	Mpn366	28 56 31
Mpn378	DnaE	105 20
Mpn379	PolA	116 91
Mpn387	Mpn387	76 16 17 15 107
Mpn388	Mpn388	1 24 105
Mpn390	PdhD	75 44
Mpn392	PdhB	106 36 71 66 37 97 17
Mpn393	PdhA	64 71 97 10 105
Mpn394	Nox	97 41 14
Mpn401	GreA	67 28 10 68 105 79
Mpn402	ProS	86 17
Mpn416	P29	71 100
Mpn419	AlaS	99 96
Mpn426	P115	93 86
Mpn430	GapA	33 85 53 56 106 12
Mpn434	DnaK	67 33 32 7 80 26 17 99 88 100 110 28 75 40 95 108 112 69 49 10 104 11
		53 78 107 93 106 46 23 96 65 50 3 51 111 9 114 58 12 15 8 52 81 34
		56 37 101 45 86 19 5
Mpn436	Mpn436	76 17 15
Mpn440	Mpn440	6 4 3 68
Mpn461	Mpn461	75 105
Mpn476	Cmk	75 58
Mpn480	ValS	18 30 51
Mpn486	Mpn486	90 101 82
Mpn487	Csd	98 76 111
Mpn497	Mpn497	109 85
Mpn509	Mpn509	81 58

Mpn514	Mpn514	1 39 21 2
Mpn515	RpoC	67 53 57 61 111 48 18 106 110
Mpn516	RpoB	53 57 61 111 51 17 47 107 106 10 105 110
Mpn520	IleS	11 97 108 51 87 106 43 44 96
Mpn526	Mpn526	112 27 28 21
Mpn531	ClpB	79 110
Mpn536	RuvB	95 71 100
Mpn547	Mpn547	81 25 23
Mpn550	ThiI	65 55
Mpn551	Mpn551	35 18 40 30 29 20
Mpn553	ThrS	11 97 51 12 38 106 37 44
Mpn556	ArgS	4 87 3 68 108 41 96
Mpn562	NadE	59 70 89
Mpn563	Mpn563	81 39 90 13
Mpn567	P200	6 24
Mpn568	Era	87 23 100
Mpn573	GroL	11 90 80 26 88 93 106 105 100 36 108 112 34 37 10 19 86 5
Mpn574	GroS	60 37 9
Mpn582	Mpn582	99 20
Mpn592	Mpn592	27 75
Mpn596	Mpn596	87 66 108 55
Mpn598	AtpD	3 97 12 47 48 38 4 106 105
Mpn599	AtpG	67 85 58
Mpn606	Eno	46 41 110
Mpn609	PstB	38 68
Mpn612	Mpn612	21 46 110
Mpn615	Mpn615	87 80 115 113
Mpn619	UvrA	75 105
Mpn622	RpsO	60 51
Mpn623	Mpn623	71 111
Mpn629	TpiA	3 46 110
Mpn635	Mpn635	7 55
Mpn644	Mpn644	69 93
Mpn650	Mpn650	81 90
Mpn656	Mpn656	109 71 100
Mpn658	RplS	33 23 26 100
Mpn662	MsrB	78 44
Mpn665	Tuf	104 33 32 90 53 48 116 18 106 23 96 110 50 36 51 111 47 52 49 37 54
Mpn668	Mpn668	39 90 41
Mpn669	TyrS	106 97 79
Mpn671	FtsH	38 92 34 100 54
Mpn673	Mpn673	67 61
Mpn676	Mpn676	87 80 115
Mpn677	Mpn677	21 26
Mpn678	GltX	22 103 23
Mpn685	Mpn685	56 37 51 110
Mpn686	DnaA	70 102 20 59 89 101 29 82

Table S8: List of literature curated multifunctional proteins.

This table is a repository of literature evidence for multifunctional proteins that may exist in *M. pneumoniae* based on observed homologs in other species. In total 50 multifunctional proteins are reported in this list. The first column on the left gives the annotated systematic name of each multifunctional protein. The second and third column provide the protein name and protein functions. The last column gives the PubMed ID or database, in which this evidence is reported.

Systematic name	Protein description	Functions, enzymatic activities	PMID or other source:
Mpn002	DnaJ-like protein	DnaJ has a function as co-chaperone of dnaK, but is also reported to be involved in DNA replication in <i>Escherichia coli</i>	17919282
Mpn003	DNA gyrase subunit B	DNA gyrase catalyzes the interconversion of various topological isomers of double-stranded DNA rings, including catenanes and knotted rings	Expasy
Mpn004	DNA gyrase subunit A	DNA gyrase catalyzes the interconversion of various topological isomers of double-stranded DNA rings, including catenanes and knotted rings	Expasy
Mpn006	Thymidylate kinase	Thymidylate kinase can use purine and pyrimidine nucleosides as substrates	18477629
Mpn008	tRNA modification GTPase mnmE	Has both GTPase and tRNA modification activities	12730230
Mpn025	Fructose-bisphosphate aldolase	Glycolytic enzyme. In eukaryotes associates with vacuolar H ⁺ -ATPase; contribute to its assembly, activity. Aldolase interacts with the endocytic proteins sorting nexin 9 (SNX9) Role in actin cytoskeleton assembly; bind F-actin. Also binds g-tubuline, heparin, phospholipase D and the glucose transporter Glut4.	18453690 15299020
Mpn045	Histidyl-tRNA synthetase	Is a class-II aminoacyl-tRNA synthetase and is reported to interact with translation elongation factors in mammals	12362348
Mpn046	Aspartyl-tRNA synthetase	Is a class-II aminoacyl-tRNA synthetase and is reported to interact with translation elongation factors in mammals	12362348
Mpn061	Signal recognition particle protein	Respond to acid stress Protein export and membrane biogenesis	11274114
Mpn105	Phenylalanyl-tRNA synthetase α-chain	Is a class-II aminoacyl-tRNA synthetase and is reported to interact with translation elongation factors in mammals	12362348
Mpn106	Phenylalanyl-tRNA synthetase β-chain	Is a aminoacyl-tRNA synthetase and is reported to interact with translation elongation factors in mammals	12362348
Mpn155	Translation initiation factor IF-2.	Is involved in the hydrolysis of GTP during the formation of the 70S ribosomal complex Protects formylmethionyl-tRNA from spontaneous hydrolysis and promotes its binding to the 30S ribosomal	Expasy
Mpn164	30S ribosomal protein S10	Involved in translation and NusB mediated antitermination of transcription	8871397
Mpn166	50S ribosomal protein L4	Suppresses translation of S10 operon mRNA also inhibits transcription of the s10 operon, binds to L4 mRNA and inhibits translation	8871397
Mpn171	30S ribosomal protein S3	Is involved in translation and has an endonuclease activity in <i>Homo sapiens</i> & <i>Escherichia coli</i> and a lyase function in <i>Drosophila melanogaster</i>	8871397
Mpn172	50S ribosomal protein L16	Is involved in translation; It is also seems to make contacts with the A and possibly P site tRNAs	Expasy
Mpn177	50S ribosomal protein L5	Is involved in translation and L5-5S rRNA complex stimulates aminoacyl-tRNA synthetases	8871397
Mpn178	50S ribosomal protein S14 type Z	Is involved in translation, binds to S14 gene and inhibits transcription	8871397
Mpn179	30S ribosomal protein S8	Is involved in translation and inhibits translation by binding to mRNAs in specific ribosomal cistrons	8871397
Mpn189	30S ribosomal protein S13	Is involved in translation and it associates with the ribosomal protein S19 and RNA fragments 16S rRNA recognition though its C-terminal domain	8193163
Mpn211	UvrABC system protein B	Helicase, nuclease and hydrolase activities	GO
Mpn220	50S ribosomal protein L1	Is involved in translation and inhibits translation by binding to mRNAs in specific ribosomal cistrons	8871397
Mpn221	Peptidyl-tRNA hydrolase	Releases tRNA from peptidyl-tRNAs by cleavage of ester bond Pth is also capable of hydrolysing an amide bond between the peptide and the 3'-amino group of the modified ribose at the end of the tRNA. It also associates with the 30S ribosomal subunit	16849786
Mpn226	30S ribosomal protein S7	Is involved in translation and inhibits translation by binding to mRNAs in specific ribosomal cistrons	8871397
Mpn235	Uracil-DNA glycosylase	Glyceraldehyde-3-phosphate dehydrogenase and transcriptional activation	1924305
Mpn263	Thioredoxin	Thioredoxin activity; synthesis of deoxyribonucleotides. Also binds to the <i>Escherichia coli</i> the T7 DNA polymerase	768986
Mpn300	Segregation and condensation protein A	Chromosome segregation and condensation DNA repair and transcription regulation	15009890
Mpn303	Pyruvate Kinase	Final step of glycolysis—the conversion of phosphoenolpyruvate to pyruvate. In mammalian muscle it binds thyroid hormone. It regulates the transcriptional responses of the thyroid-hormone receptor. In bacteria, it can synthesize nucleotide triphosphate (NTP) under anaerobic conditions	15877277
Mpn320	Thymidylate synthase	Pyrimidine metabolism. Translation inhibitor	1924359
Mpn332	ATP-dependent protease La	Chaperone	9149530
Mpn345	Putative type-1 restriction enzyme MpnORFDP R protein part 2	It has an ATP dependent nuclease as well as DNA translocation function	16126220
Mpn347	Putative type I restriction enzyme MpnORFDP R protein part 1	It has an ATP dependent nuclease as well as DNA translocation function	16126220
Mpn356	CysteinyI-tRNA synthetase	Is a class-I aminoacyl-tRNA synthetase and is reported to interact with translation elongation factors	12362348

Mpn379	DNA polymerase (PolA)	5'-3' exonuclease, but also reverse transcriptase activity in <i>Streptomyces</i> ; part of the <i>Streptomyces</i> telomere complex	15353591
Mpn384	Leucyl-tRNA synthetase	Is a class-I aminoacyl-tRNA synthetase and is reported to interact with translation elongation factors in mammals	12362348
Mpn430	Glyceraldehyde-3-phosphate dehydrogenase	Converts glyceraldehyde-3-phosphate to 1,3-diphosphoglycerate, but also binds RNA, RNA polymerase and HPr in <i>Bacillus subtilis</i>	17142398 15877277
Mpn434	Chaperone protein dnaK	Besides its role as protein chaperone dnaK was also shown to be involved in DNA replication <i>Escherichia coli</i>	17919282
Mpn446	30S ribosomal protein S4	Is involved in translation, nucleates assembly of the body of 30S ribosome by binding directly to the 16S rRNA and inhibits translation by binding to mRNAs in specific ribosomal cistrons in <i>Homo sapiens</i>	2470510 Uniprot
Mpn480	Valyl-tRNA synthetase	Is a class-I aminoacyl-tRNA synthetase and has two distinct active sites: one for aminoacylation and one for editing. Additionally it is reported to interact with translation elongation factors in mammals	12362348
Mpn520	Isoleucyl-tRNA synthetase	Is a class-I aminoacyl-tRNA synthetase and has two distinct active sites: one for aminoacylation and one for editing. Additionally it is reported to specifically interact with HSP90 in mammals	12362348
Mpn538	50S ribosomal protein L10	Is involved in translation and inhibits translation by binding to mRNAs in specific ribosomal cistrons in <i>Escherichia coli</i>	2470510
Mpn539	50S ribosomal protein L7/L12	Inhibits the translation of specific mRNAs including its own in <i>Homo sapiens</i>	8871397 2470510
Mpn541	30S ribosomal protein S20	Is involved in translation and polyamine biosynthesis in <i>Saccharomyces cerevisiae</i>	7539334
Mpn606	Enolase	Catalyzes the conversion of 2-phosphoglycerate to phosphoenolpyruvate in glycolysis, but further it is reported to function also as heat-shock protein, to bind cytoskeleton and chromatin structures	15877277
Mpn616	30S ribosomal protein S9	Is involved in translation and participates in the error prone SOS repair process in <i>Escherichia coli</i>	8871397
Mpn622	30S ribosomal protein S15	Is involved in translation and seems to selectively stabilize the temperature sensitive cap-binding subunit of eIF-4E in <i>Saccharomyces cerevisiae</i>	8871397
Mpn623	ATP-dependent RNA helicase	Some members of the same family are involved in RNA transcription, mRNA transport, translation initiation and cell cycle regulation in <i>Homo sapiens</i>	17979704
Mpn665	Elongation factor Tu	Elongation factor Tu promotes the GTP-dependent binding of aminoacyl-tRNA to the A-site of ribosomes during protein biosynthesis, in addition it has a direct function in host cell adherence in <i>M. pneumoniae</i>	17709513
Mpn671	Cell division protease ftsH homolog	Metalloprotease, but has also chaperone function in <i>Escherichia coli</i>	9149530
Mpn678	Glutamyl-tRNA synthetase	Is a class-I aminoacyl-tRNA synthetase and is reported to specifically interact with HSP90 in mammals	12362348

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